Community Detection on a Randomly Growing Network

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Abstract

We study community detection on Markovian random networks outside of the Stochastic Block Model (SBM) framework. Specifically, we consider a random network growth process which generates K separate preferential attachment trees and connects them with Erdős–Rényi edges, so that each tree represents a community and each node inherits the label of the tree to which it belongs. This model is able to produce many features of real-world networks that are improbable under SBM, such as power law degree distribution and the existence of chains and hubs. Given only the final graph, we seek to recover the unobserved community membership of the nodes. We first prove that it is impossible for any algorithm to consistently recover the community label of all the nodes. However, we design algorithms that are provably able to recover the community labels of subsets of central nodes, for several different notions of node centrality, such as arrival time or degree. Our procedure consists of two stages where we first classify high degree nodes and then extend the community assignments to the remaining vertices. Numerical experiments and a real data application on a coauthorship network demonstrate the effectiveness of our proposed approach.

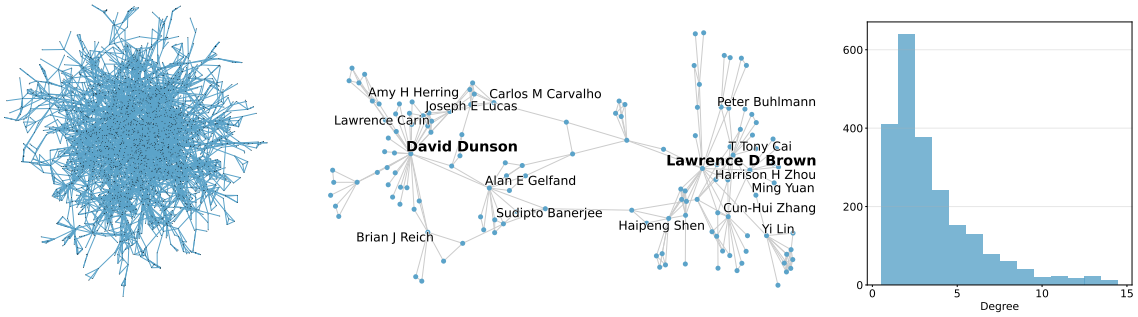
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1 Introduction

Community detection is a central problem in network data analysis. It seeks to cluster network nodes based on connectivity patterns, with applications in various domains such as biology (Krzakala et al., 2013; Traag et al., 2019; Yu et al., 2007), social science (Ji et al., 2022; Akbaritabar et al., 2020), computer science (Velickovic et al., 2018; Wang and Paschalidis, 2016), and more (de Berardinis et al., 2020). The predominant approach to community detection among statistical researchers has focused on the so-called Stochastic Block Model, which is a random graph model where edges are added independently, with probability dependent on whether the edge is between or within community. The order in which the edges are added is of no importance to the model.

In contrast, real-world networks are often formed from a growth process where vertices and edges are added sequentially. This explains the prevalence of certain network features, such as a power-law degree distribution (Barabási and Albert, 1999) and the existence of long chains and pendants (a node that connects to many degree 1 nodes). Examples of

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(a) Observed graph (b) Sub network (c) Degree distribution

Figure 1: Co-authorship network from Ji and Jin (2016). **(a)** Observed graph topology, where vertices correspond to authors and edges represent co-authorship relations. **(b)** Subnetwork consisting of all nodes within graph distance two of Lawrence D. Brown or David Dunson. **(c)** Degree distribution of the network. For readability, authors with degrees exceeding 15 are omitted.

graphs that exhibit such features include co-authorship networks (Ji and Jin, 2016) and gene co-expression networks (Cline et al., 2007); see Figure 1.

The growth process of a network leaves distinctive topological structures which contain information useful for estimating the communities. In the statistician co-authorship network from Ji and Jin (2016) for example, a research community tends to coalesce around a small number of influential researchers, creating a central core of high-degree nodes in that community. This is illustrated in Figure 1b, where we show the subnetwork of all the nodes within distance two of either Larry Brown, the renowned theoretical statistician, or David Dunson, the famous Bayesian statistician. A clear community of theoretical researchers forms around the former, and a separate community of Bayesian modelers around the latter. Identifying hub nodes could substantially aid community detection, but existing SBM-based approaches may overlook such growth-induced features due to the static nature of the block model. Our work addresses this gap.

To formalize the definition of our problem and to guide the development of our methodology, we consider a random graph model that captures both sequential growth and community structure in a simple way. Our random graph model, which we call the planted forest model, first generates K independent tree graphs using the preferential attachment process, where each node arrives sequentially and attaches to an existing node with probability dependent on the degree of the existing node. We let each tree represent a separate community so that the nodes in that tree take on the community label of that tree. We refer to this collection of disjoint trees as a forest graph. We then take the superposition of this forest graph with an independent Erdős–Rényi (ER) noise graph collapsing any multi-edges. Given an observation of the final graph, without any knowledge of the growth process or whether the edges belong to the forest graph or the noise graph, we seek to recover the unobserved community labels of the nodes.

Our first main contribution is an information-theoretic lower bound showing that no estimator can consistently recover the community labels of all the nodes even when the ER probability is at a relatively low level of $O(1/n)$. This is because the misclustering

error is driven by the later-arriving nodes, which cannot be classified correctly due to their peripheral location in the graph.

Despite the impossibility of global recovery, consistent recovery may still be possible *locally*, on a specific *subset* of the nodes. In our second main contribution, we propose a community recovery algorithm and prove that, asymptotically, it has either zero or vanishing misclustering errors on various subsets of central nodes, defined through three different but related notions of centrality: the arrival time of the node, the degree, and the distance of the node to the initial root node of the community. Our guarantees require only that community sizes be of comparable order and that the ER noise probability is not so high that the degrees of the nodes become uncorrelated with their arrival time.

Our estimation algorithm consists of two stages. We first iteratively remove low-degree nodes to reveal K disjoint connected components, which we refer to as anchor components. Intuitively, the anchor components capture the high-degree core of each of the communities and supply the initial seeds for our clustering. In the second stage, we propagate the community labels from the anchor components to the entire network using a range of recovery methods, the simplest of which is to label the nodes based on their shortest-path distance to the anchors.

Although our algorithm is motivated by the planted forest model, it depends on the model only loosely, mainly through the assumption that each community has a high-degree core that can be isolated by iterative degree-based pruning. In particular, it does not rely on the assumption that the connectivity structure within a community is tree-shaped. We demonstrate the practical effectiveness of our approach on a co-authorship network of statisticians, where we recursively apply our community detection procedure and recover a hierarchical clustering of research communities that largely aligns with recognized sub-fields of statistics.

The remainder of this paper is organized as follows. In Section 2, we introduce the planted forest model and present an impossibility result that frames the objective of our work. Section 3 details our SPAR (Selective Pruning with Anchor-based Recovery) algorithm. The theoretical guarantees of the algorithm are established in Section 4. Section 5 validates the method through simulation experiments under varied preferential attachment models and noise levels. Finally, Section 6 applies our approach to a co-authorship network, illustrating its practical performance.

1.1 Literature Review

Community detection in networks has been predominantly studied within the framework of SBM. Within this paradigm, widely used methods include spectral clustering (Amini et al., 2024; Li et al., 2022), likelihood-based inference (Zhen and Wang, 2023; Cerqueira and Levina, 2023), and hybrid approaches that combine spectral initialization with likelihood refinement (Xu et al., 2020; Deng et al., 2024). By contrast, relatively little work has addressed community detection in networks formed through sequential growth processes. Ben-Hamou and Velona (2025) and Hajek and Sankagiri (2019) extend classical preferential attachment models to multi-class settings in which incoming nodes connect to existing vertices according to both degree and latent class labels. Our setting differs in two respects: first, we consider an entirely different notion of community structure–they model

communities through class-dependent attachment probabilities, whereas our communities arise from disjoint growth processes–and second, most of their method and theory assume that the node arrival times are known, whereas we take the arrival ordering to be unobserved. Crane and Xu (2024) is most closely related to our work; they propose a Bayesian method for community detection on a similar model without any theoretical analysis. Our work differs in three crucial ways: first, our method comes with rigorous theoretical guarantees; second, our estimator is provably computationally efficient whereas Crane and Xu (2024) uses MCMC with no mixing time bounds; lastly, our approach depends only on the high-degree core structure within each community and may be less sensitive to deviation from the model.

Another related line of work in network data analysis considers inference on special vertices, which includes "root finding" (Bubeck et al., 2017; Banerjee and Bhamidi, 2022; Banerjee and Huang, 2023; Crane and Xu, 2021, 2024; Addario-Berry et al., 2025) and "diffusion source identification" (Dawkins et al., 2021; Li et al., 2021; Dong et al., 2022; Shah and Zaman, 2011). Most work in the root finding literature assumes that the observed network is generated entirely by a sequential growth mechanism, whereas most work in diffusion source identification literature assumes a fixed underlying network over which a stochastic diffusion process evolves. Despite these differing modeling assumptions, both lines of work focus on identifying or constructing confidence sets for a single distinguished vertex (the root node or the diffusion source) of a sequential growth process based on a single observed network snapshot. Among these works, Addario-Berry et al. (2025) employs a leaf-stripping procedure to construct confidence sets for root nodes, which closely parallels the graph pruning step of our algorithm. Our objective, however, is fundamentally different: rather than identifying a single special vertex, we aim to recover and cluster multiple planted components generated by distinct growth processes.

2 Model and Problem Formulation

In this section, we first introduce the notation and definition for the Planted Forest model. Then, we formalize the problem setup for community detection under the proposed model.

2.1 Notation

Throughout the paper, we will adopt the convention $0/0 := 0$. We define the natural numbers $\mathbb{N} = \{1, 2, 3, \ldots\}$. For an integer n, we write $[n] := \{1, 2, \ldots, n\}$ and let S_n denote the set of all permutations on $[n]$. For a discrete set A, we write $|A|$ as the cardinality of A.

We denote a graph by $\boldsymbol{g} = (V, E)$, where V and $E \subset V \times V$ represent the set of nodes and the set of undirected edges, respectively. For any two vertex sets $V_1, V_2 \subset V(\boldsymbol{g})$, we define the edge between them as $E_{\boldsymbol{g}}(V_1, V_2) := \{(u, v) \mid (u, v) \in E(\boldsymbol{g}), u \in V_1, v \in V_2\}$. When $V_1 = V_2$, we abbreviate $E_{\boldsymbol{g}}(V_1, V_1)$ as $E_{\boldsymbol{g}}(V_1)$. We write $\deg_{\boldsymbol{g}}(u)$ to denote u's degree in graph $\boldsymbol{g}$, i.e., the number of nodes directly connected to u via an edge. For a vertex set $V_1 \subset V$, we write $\deg_{\boldsymbol{g}}(V_1) := \max_{u \in V_1} \deg_{\boldsymbol{g}}(u)$.

For two labeled graphs $\boldsymbol{g}$ and $\boldsymbol{g}'$ defined on the same vertex set V, we write $\boldsymbol{g} + \boldsymbol{g}'$ to denote the graph obtained by taking the union of their edge sets, i.e., $E(\boldsymbol{g} + \boldsymbol{g}') = E(\boldsymbol{g}) \cup E(\boldsymbol{g}')$, with multi-edges collapsed. We write $\boldsymbol{g} \subset \boldsymbol{g}'$ if $\boldsymbol{g}$ is a subgraph of $\boldsymbol{g}'$, meaning $E(\boldsymbol{g}) \subset E(\boldsymbol{g}')$. When $\boldsymbol{g}$ and $\boldsymbol{g}'$ are two disjoint graphs (no shared vertices or

edges), we use $\boldsymbol{g} \oplus \boldsymbol{g}'$ to denote their disjoint union. In addition, for $v \in V(\boldsymbol{g})$, we write $\boldsymbol{g} \setminus v$ for the subgraph obtained by removing the vertex v together with all edges incident to v. Given a subset $\tilde{V} \subset V(\boldsymbol{g})$, we define $\boldsymbol{g} \cap \tilde{V}$ to be the subgraph induced by the nodes in $\tilde{V}$. Given a graph $\boldsymbol{g}$, we define $\mathcal{C}(\boldsymbol{g})$ to be the set of connected components of $\boldsymbol{g}$; if $\boldsymbol{g}$ is connected then $\mathcal{C}(\boldsymbol{g}) = \{\boldsymbol{g}\}$.

Throughout the paper, we use capital font (e.g., $\boldsymbol{G}$) to denote random objects and lowercase font to denote fixed objects. Graphs are represented via bold font.

2.2 Planted Forest Model

We first define the affine preferential attachment process for a randomly growing tree (Crane and Xu, 2021), which models the growth process of a single community.

Definition 1. *Let V_n be a set of n node labels and let $\pi : [n] \to V_n$ be the arrival ordering so that π_t denotes the t-th node to be added to the tree. Let $\alpha > -1$; we say that a random tree $\boldsymbol{T}$ with n nodes has the affine preferential attachment distribution* $\mathrm{APA}(\alpha, \pi)$ *if it is generated according to the following process: start with a single node π_1 as the root node and connect π_2 to π_1 deterministically; at every time step $t = 3, \dots, n$, add a new node π_t and connect it to an existing node $w \in \{\pi_1, \dots, \pi_{t-1}\}$ chosen with probability*

$$w \mapsto \frac{\deg_{\boldsymbol{T}_{t-1}}(w) + \alpha}{2(t-2) + (t-1)\alpha}, \tag{1}$$

where $\deg_{\boldsymbol{T}_{t-1}}(w)$ is the degree of node w at time $t-1$.

In affine preferential attachment, higher-degree nodes are more likely to attract new neighbors so that we get a rich-get-richer phenomenon. One important consequence of this is that the APA random graph tends to exhibit strong degree heterogeneity: asymptotically, the percentage of nodes with degree k is approximately proportional to $\frac{1}{k^{3+\alpha}}$ (Van Der Hofstad, 2024, Section 8.4). The parameter α controls the strength of the degree preference: larger values of α diminish the degree preference in the attachment probabilities, with the limiting case $\alpha = \infty$ corresponding to uniform attachment, in which each new node attaches to an existing node chosen uniformly at random.

To model multiple communities, we associate each community with an independent APA tree, producing a forest $\boldsymbol{F}_n$ of disjoint community-trees. To model between-community edges and to ensure that the overall graph is connected, we superpose $\boldsymbol{F}_n$ with an independent Erdős–Rényi random graph to obtain the final observed graph $\boldsymbol{G}_n$. The tree structure within each community is a simplification made for model parsimony. Definition 1 can be extended to allow for multiple edges per arrival, but doing so requires additional modeling choices, such as deciding how many edges to add at each step, that distract from our main focus on the interplay between the growth dynamics and community detection.

Definition 2. *Let $\alpha > -1$, V_n be a set of n node labels and let $\ell : V_n \to [K]$ be the community label function so that $\ell(v)$ is the community membership of node v. Let $\pi : [n] \to V_n$ be the arrival ordering of all n nodes. For each community $k \in [K]$, define $V^k := \{u \in V_n : \ell(u) = k\}$, $n_k := |V^k|$, and $\pi^k : [n_k] \to V^k$ as an ordering of V^k induced by π. For each $k \in [K]$, let $\boldsymbol{T}^k$ be an independent random tree generated according to the* $\mathrm{APA}(\alpha, \pi^k)$ *distribution. Let $\boldsymbol{F}_n = \oplus_{i=1}^K \boldsymbol{T}^i$ be the disjoint union of the K trees.*

Let $\theta \in [0,1]$ and let $\boldsymbol{R}_n$ be an Erdős–Rényi random graph with edge probability θ defined on V_n. Define

$$\boldsymbol{G}_n := \boldsymbol{F}_n + \boldsymbol{R}_n, \quad \text{collapsing multi-edges.} \tag{2}$$

We then say that $\boldsymbol{G}_n$ is distributed according to the planted forest distribution $\mathrm{PF}(\alpha, \theta, \ell, \pi)$. *As a short hand, we also write $\pi(\boldsymbol{T}^k) \equiv \pi^k$.*

The random tree $\boldsymbol{T}^k$ and the permutation π^k capture the growth process of community k. Since the model depends on the global arrival ordering π only through the community-specific orderings $\pi^1, \ldots, \pi^K$, the full π is not identifiable. It is however still convenient to have the notion of a global arrival ordering. We see that the forest graph $\boldsymbol{F}_n$ contains all the information about the community membership and that $\boldsymbol{R}_n$ may be regarded as noise.

Remark 3. *The planted forest model is basically equivalent to the multiple-roots PAPER (Preferential Attachment Plus Erös–Rényi) model proposed by Crane and Xu (2024). The main difference is that we take the community membership $\ell(\cdot)$ to be a fixed parameter. We use the name "planted forest" to emphasize the community structure induced by the disjoint collection of tree graphs.*

In our definition of the $\mathrm{PF}(\alpha, \theta, \ell, \pi)$ model, the community membership function $\ell(\cdot)$ is viewed as a fixed parameter; $\ell(u) = k$ means that the node u is in community-tree $\boldsymbol{T}^k$. However, for our lower bound analysis in Theorem 6 and algorithmic development in Section 5, it is useful to define an alternative model where $\ell(\cdot)$ is random. In this model, the nodes enter the network according the global ordering π and the t-th node π_t arrives, it chooses a parent-node at random from the set of all existing nodes $\{\pi_1, \ldots, \pi_{t-1}\}$. The new node π_t then joins the same community-tree as its parent. We refer to this model as the random community planted forest model (RC-PF) and define it formally below.

Definition 4. *Let $\alpha > -1$, $\theta \in [0,1]$, and $K \in \mathbb{N}$. Let $\pi : [n] \to V_n$ be an arrival ordering. For each $k \in [K]$, we initialize tree $\boldsymbol{T}^k$ as a graph with two nodes π_{2k-1}, π_{2k} and a single edge between them and assign $\ell(\pi_{2k-1}) = \ell(\pi_{2k}) = k$. We initialize $\boldsymbol{F}$ as the disjoint collection of $\boldsymbol{T}^1, \ldots, \boldsymbol{T}^K$.*

Then, for each $t = 2K+1, \ldots, n$, we connect the new node π_t to an existing node $w \in \{\pi_1, \ldots, \pi_{t-1}\}$ chosen with probability

$$w \mapsto \frac{\deg_{\boldsymbol{F}_{t-1}}(w) + \alpha}{2(t-K-1) + (t-1)\alpha}, \tag{3}$$

where $\deg_{\boldsymbol{F}_{t-1}}(w)$ is the degree of node w at time $t-1$. We set $\ell(\pi_t) = \ell(w)$.

Let $\boldsymbol{F}_n$ be the resulting random forest graph after all n nodes are added. Let $\boldsymbol{R}_n \sim$ Erdős–Rényi(θ) be an independent ER graph defined on the same set of vertices V_n and, as before, define

$$\boldsymbol{G}_n = \boldsymbol{F}_n + \boldsymbol{R}_n, \quad \text{collapsing multi-edges}$$

We then say that $\boldsymbol{G}_n$ is distributed according to the $\mathrm{PF}(\alpha, \theta, \pi, K)$ *model.*

We note that the RC-PF model produces a random community membership function ℓ so that the size of each community is random. A Pólya-urn argument (see e.g. Section 4.3.2 in Durrett (2019)), where we view each ball as one node and two edge-endpoints,

shows that the community size proportions $\left(\frac{n_1}{n}, \frac{n_2}{n}, \ldots, \frac{n_K}{n}\right)$ converge in distribution to Dirichlet($\frac{2+2\alpha}{2+\alpha}, \frac{2+2\alpha}{2+\alpha}, \ldots, \frac{2+2\alpha}{2+\alpha}$) supported on the $K-1$ dimensional simplex.

We also note that the RC-PF model is almost identical to the fixed K multiple roots PAPER model in Crane and Xu (2024). The only difference is that, in the latter, each tree is initialized with a single root node with a self-loop to guarantee positive degree.

2.3 Community Detection Problem

We observe the final snapshot $\boldsymbol{G}_n$ generated by the PF$(\alpha, \theta, \ell, \pi)$ model. The community detection task is to recover the community membership function ℓ.

To assess the performance of estimated community labels $\hat{\ell}$, we first define a discrepancy measure between two membership functions on the same graph. Since the community labels are arbitrary, the true membership function ℓ is identifiable only up to a permutation of the K community labels. Therefore, we do not compare $\hat{\ell}$ with the true membership ℓ directly but rather with $\sigma \circ \ell$ where $\sigma \in S_K$ is a permutation of the K labels. We choose σ to minimize the Hamming distance between the two membership functions. Finally, we allow the distance to be weighted where each node u is associated with a weight w_u so that the cost of misclustering u is large when w_u is large.

Definition 5. *Let $K \in \mathbb{N}$, let $\boldsymbol{g}$ be a graph, let $\ell_1, \ell_2 : V(\boldsymbol{g}) \to [K]$ be two labelings, and $\boldsymbol{w} = \{w_u\}_{u \in V(\boldsymbol{g})}$ be a set of non-negative numbers that sums to 1. We define the Hamming distance and the global misclustering error (viewing ℓ_1 as the true clustering) respectively as*

$$d^{\mathrm{Ham}}_{\boldsymbol{w}}(\ell_1, \ell_2) := \sum_{u \in V(g)} w_u \mathbb{1}\{\ell_1(u) \neq \ell_2(u)\}, \quad d_{\boldsymbol{w}}(\ell_1, \ell_2) := \min_{\sigma \in S_K} d^{\mathrm{Ham}}_{\boldsymbol{w}}(\ell_1, \sigma \circ \ell_2).$$

If the misclustering error is unweighted, for example when $w_u \propto 1$ for all u, we omit the subscript and simply write d^{Ham} and d. For a given subset of the nodes $V_0 \subseteq V(\boldsymbol{g})$, we also define the local *misclustering error with respect to V_0:*

$$d^{\mathrm{Ham}}_{V_0}(\ell_1, \ell_2) := \sum_{u \in V_0} \mathbb{1}\{\ell_1(u) \neq \ell_2(u)\}, \quad d_{V_0}(\ell_1, \ell_2) := \min_{\sigma \in S_K} d^{\mathrm{Ham}}_{V_0}(\ell_1, \sigma \circ \ell_2).$$

Equivalently, we may view d_{V_0} as a weighted error where the weights are $w_u \propto \mathbb{1}\{u \in V_0\}$.

Let us first consider the goal of designing an estimator $\hat{\ell}$ of the community membership labels so that the (unweighted) expected misclustering error rate $\frac{1}{n}\mathbb{E}[d(\hat{\ell}, \ell)]$ tends to 0 as n increases, which is known as weak recovery or consistent recovery. The difficulty of this naturally depends on the noise level θ since weak recovery is clearly impossible if θ is close to 1. Somewhat surprisingly, we prove that even when θ is at a relatively small level of $\Omega(1/n)$, weak recovery is still unattainable for any estimator.

Theorem 6. *Let $\boldsymbol{G}_n \sim$ PF$(\alpha, \theta, \pi, K = 2)$. Let $c > 0$ and suppose $\theta \geq \frac{c}{n}$. Then, for all sufficiently large n, there exists $\lambda := \lambda(c, \alpha) > 0$ such that*

$$\inf_{\hat{\ell}} \mathbb{E}\left[\frac{d\big(\hat{\ell}(\mathbf{G}_n), \ell\big)}{n}\right] \geq \lambda,$$

where the infimum is taken over all estimators of the community label, that is, over all function that on input $\boldsymbol{G}_n$ outputs a labeling function $\hat{\ell}(\mathbf{G}_n) : V(\mathbf{G}_n) \to \{1, 2\}$, and where d is the (unweighted) global misclustering error introduced in Definition 5.

We defer the proof of Theorem 6 to Section A of the appendix, but outline the high-level idea here. Suppose we fix a particular node u and know the community membership of all the nodes that arrived prior to u, then the task of identifying the community label of u is equivalent to that of looking at all the edges that connect u to the prior arriving nodes and picking out the single tree edge from the noise edges. We prove that if u is a late-arriving node, then it is impossible to identify the tree edge. Specifically, suppose (u, v) is a tree edge; we show that when the arrival time of u is greater than $n/2$, there exists a noise edge (u, v') such that v and v' have the same degree but belong to different communities with high probability. Swapping the roles (u, v) and (u, v') (so that (u, v') becomes a tree edge and (u, v) becomes a noise edge) produces an outcome with the same likelihood, making the two cases indistinguishable to any method.

We see from Theorem 6 that even when $\theta = \frac{0.1}{n}$ so that the expected number of noise edges is much smaller than $n - 2$, the number of edges in $\boldsymbol{F}_n$, the misclustering error rate of any estimator is still bounded away from 0. More generally, Theorem 6 implies that to achieve weak recovery, we would need the number of noise edges to be of smaller order than the number of tree edges, which is too restrictive for real-world data.

Rather than working in the unrealistic setting necessary to obtain weak recovery over all the nodes, our recovery results shift the focus from global recovery to recovery over a subset V_0 of nodes that are "informative" in that these nodes play important roles in the formation history of the network or if they carry strong structural signals that overcome the random perturbations introduced by the Erdős–Rényi noise. More precisely, we show that consistent local recovery is achievable for three classes of V_0: early arriving node, high-degree nodes, and nodes which are very close to the initial root node of any of the communities.

This fits with the proof of Theorem 6 where we see that the misclustering error is driven by nodes that are peripheral in the sense that they are late arriving and may have a long graph distance to the early arriving nodes. This conforms with our intuition that for real-world networks, nodes that are non-central and not well connected to the "core" of the graph may be much more difficult to cluster than centrally located nodes. The concept of graph core is studied in Zhang et al. (2015); Naik et al. (2021); Miao and Li (2023); Yanchenko et al. (2025), although the precise notion of graph core that they use is quite different and not based on growth dynamics.

Our community recovery method, described in detail in Section 3, first prunes the graph to reveal a set of K connected components (we call these anchor components) and then uses the anchor components to propagate the community labels to the remaining nodes to obtain our estimated community $\hat{\ell}$. For a graph $\boldsymbol{G}_n$ generated by the planted forest model $\mathrm{PF}(\alpha, \theta, \ell, \pi)$, supposing the noise level $\theta \leq O(n^{-\frac{1+\alpha}{2+\alpha}-\delta})$ for an arbitrarily small $\delta > 0$ and supposing that the community sizes are roughly balanced, we prove that with probability at least $1 - \varepsilon + o(1)$ the following stands (1) $d_{V_0^L}(\hat{\ell}, \ell) = 0$ for sufficiently large n where V_0^L is the set of L earliest arriving nodes, (2) $d_{V^{\tilde{C},\alpha}}(\hat{\ell}, \ell) = 0$ for sufficiently large n where $V^{\tilde{C},\alpha}$ is the set of nodes with degree at least $\tilde{C} n^{\frac{1}{2+\alpha}}$, and (3) that $d_{V_0^{(1)}}(\hat{\ell}, \ell) \leq O(n^{-\frac{1+\alpha}{2+\alpha}-\delta})|V_0^{(1)}|$

where $V_0^{(1)}$ is the set of nodes whose tree-distance to the root node is at most 1 (i.e. direct children). Based on these results, we also prove that the weighted misclustering error $d_{\boldsymbol{w}}(\hat{\ell}, \ell)$ goes to 0 for certain classes of weights $\boldsymbol{w}$ that decreases with the arrival time.

3 A Two-Stage Algorithm: SPAR

In this section, we propose a two–stage procedure for community detection in the network growth model, which we call Selective Pruning with Anchor-based Recovery (SPAR). In stage one, we iteratively prune the graph based on degree and extract K connected components which we refer to as anchor components. In stage two, we propagate the community labels to the entire network, either based on distance to the anchor components or on more complex model-based criteria.

The high-level rationale is as follows. If we knew the noise edges, we could remove them to immediately separate the communities, but since they are unknown, we aim to remove the noise edges indirectly by pruning the low-degree nodes. To ensure that we do not prune too many nodes and shatter the community structure, we proceed iteratively and at every step, try to preserve the core connected components formed by the earliest arriving nodes of each community, which we know will also contain the highest degree nodes of that community-tree (Sénizergues, 2021; Contat et al., 2024). In this way, we will eventually remove all the nodes that act as noise-induced bridges between the communities and reveal the K core connected components of the communities. We can then take these K connected components as anchor components to assign cluster memberships to all the nodes.

Our method requires knowledge of α and K. The α parameter can be estimated via the EM algorithm derived in Crane and Xu (2024) (see Section S3.1 therein). Estimation of K is much more difficult. Although we provide a theoretically valid method to estimate K on the planted forest model based on removing low degree nodes and examining the number of connected components that remain (c.f. Remark 8 and 15), it is difficult to use on real data. An alternative approach is to take $K = 2$ and perform hierarchical clustering. We adopted this second approach in our case study and were able to recover meaningful communities without extensive tuning.

3.1 Stage I: Selective Pruning

The goal of the first stage is to separate the graph into at least K connected components by pruning low-degree nodes. A simple way to do this is to set a threshold T and remove all nodes whose degree is less than T. The problem with this approach is that it is very sensitive to the choice of T and even if we have a good way of choosing T, we could end up with more than K components. This latter issue can be resolved by keeping only those components that contain at least one high-degree node, since we know that the earliest arriving nodes of each community must contain high-degree nodes.

To circumvent the need to choose T, which carries the risk that choosing too large a threshold may remove too many nodes and fragment the community structure, we propose an iterative graph pruning procedure in Algorithm 1. The algorithm has three integer input parameters K, τ, Q and either outputs a set of K anchor connected components, each of which has size at least Q and maximum-degree at least τ, or outputs failure if no qualified

anchor components can be found. In the failure case, we can decrease the input parameter; see Remarks 7 and 9.

The algorithm is iterative and performs three operations at each step: we first check if the current graph can already be separated into at least K connected components, each of which is of size at least Q and whose maximum degree is at least the anchor-degree threshold τ. If not, then we construct a set of candidate nodes for removal, remove the lowest degree nodes among the candidate set to get the pruned graph, and then go on to the next iteration. We note that the degree is always computed with respect to the input graph $\boldsymbol{G}_n$ and not updated as we prune the graph; this is mainly to make theoretical analysis more tractable – updating the degree at each iteration may also work well in practice.

The removal candidate set is designed to capture all the nodes that might have noise edges which connect different community-trees. It consists of all the nodes on all the cycles of the graph as well as any node whose removal would split the graph into a larger number of qualified connected components. Our specific definition of the removal candidate set endows Algorithm 1 the useful property that it is more conservative than simple degree thresholding in the sense that when both algorithms output K connected components, Algorithm 1 will never remove more nodes than simple degree thresholding; this property is deterministic and does not depend on the probabilistic model. We prove this property in Lemma 26 of Section B.2 of the appendix, where we also formalize the simple degree thresholding algorithm. A toy example illustrating the selective pruning stage and the remaining steps of SPAR is provided in Section C. We also note that the SPAR algorithm is computationally efficient. More precisely, $\tilde{V}'$ can be constructed by first computing a spanning tree/forest of $\bar{\boldsymbol{G}}$ (say via Kruskal's algorithm (Kruskal, 1956)) and then iterating over all edges not in the spanning tree/forest, with runtime of $O(|V|\cdot|E|)$. The set $\tilde{V}''$ can be constructed via depth-first-search on $\bar{\boldsymbol{G}}\backslash\{v\}$ with a runtime of $O(|V|^2)$. Moreover, by Lemma 26, the quantity D^* increases strictly at each iteration, and hence the number of iterations is bounded by $\deg(\boldsymbol{G}_n)$. Therefore, the overall runtime of Algorithm 1 is $O(|E||V|\deg(\boldsymbol{G}_n))$ in the worst case.

Remark 7. *The choice of the parameter τ in Algorithm 1 depends on the degree distribution of the input graph. If input graph is distributed according to the planted forest model, then one can determine the anchor-degree threshold τ via the asymptotic degree distribution described in Lemma 40. In practice, however, we give a heuristic for selecting τ through numerical approximation, as detailed in Algorithm 3 in Section B.1 of the appendix. We choose τ as the degree of the M-th highest-degree node, with M calibrated by Monte Carlo simulation. The method is robust to the choice of M.*

Remark 8. *For the most part, we assume that the number of communities K is known and does not increase with n. Although we do provide a method that can provably estimate K on the planted forest model (see Remark 15 and Algorithm 4 in Section B.2 of the appendix), the estimator is tailored to the specifics of the model and is in practice difficult to tune. Estimating the number of communities on large real-world network data is known to be a challenging problem (Le and Levina, 2022; Ma et al., 2021; Jin et al., 2023; Wang and Bickel, 2017); estimators that work in specific theoretical settings may not work well in practice. We defer a comprehensive study of how to best estimate K to future work.*

Remark 9. *The success of Algorithm 1 does not depend critically on the choice of the anchor-size threshold Q. We can take $Q = 1$ in all of our theoretical analysis although we find it still helpful to have an appropriately large Q to improve practical performance. If Q is chosen to be too large, Algorithm 1 may fail to give an output, which occurs when $\bar{\boldsymbol{G}}$ becomes a disjoint collection of trees such that no tree can be split, by the removal of a single node, into two trees both of size at least Q and with maximum degree at least τ. In such cases, we relax the size threshold by setting $Q \leftarrow Q - 1$ or $Q \leftarrow \lfloor \zeta Q \rfloor$ for some $\zeta \in (0, 1)$, and repeat until the procedure successfully outputs a set of K components.*

Remark 10. *Degree is not the only criterion one could use for pruning. Other centrality measures that correlate with node arrival order — eigenvector/betweenness centrality, or the posterior root probabilities of Crane and Xu (2024) may also work well in practice.*

Algorithm 1: Step I Selective Pruning

Input: Graph $\boldsymbol{G}_n = (V, E)$; community count K; anchor-degree threshold $\tau \in \mathbb{N}$; anchor-size threshold $Q \in \mathbb{N}$

Output: K community anchors $\{V_1, \ldots, V_K\}$ and D^*, or FAIL

Initialize: Let $\bar{\boldsymbol{G}} := \boldsymbol{G}_n$ and $D^* := 0$.
while *TRUE* **do**
  Set $K_c := |\{\boldsymbol{g} \in \mathcal{C}(\bar{\boldsymbol{G}}) : |V(\boldsymbol{g})| \geq Q,\ \deg_{\boldsymbol{G}_n}(V(\boldsymbol{g})) \geq \tau\}|$.
  if $K_c \geq K$ **then**
    return the K vertex sets of $\mathcal{C}(\bar{\boldsymbol{G}})$ with the largest maximum degree.
  end
  Set $\tilde{V}' := \{v \in V(\bar{\boldsymbol{G}}) : v \text{ belongs to at least one cycle of } \bar{\boldsymbol{G}}\}$.
  Set

$$\tilde{V}'' := \left\{ v \in V(\bar{\boldsymbol{G}}) : |\{\boldsymbol{g} \in \mathcal{C}(\bar{\boldsymbol{G}} \backslash \{v\}) : |V(\boldsymbol{g})| \geq Q,\ \deg_{\boldsymbol{G}_n}(V(\boldsymbol{g})) \geq \tau\}| > K_c \right\}.$$

  Set $\tilde{V} = \tilde{V}' \cup \tilde{V}''$.
  if $\tilde{V} = \emptyset$ **then**
    return FAIL.
  end
  Set $D^* := \min\{\deg_{\boldsymbol{G}_n}(v') : v' \in \tilde{V}\}$.
  Remove $\left\{v \in \tilde{V} : \deg_{\boldsymbol{G}_n}(v) = D^*\right\}$ and update $\bar{\boldsymbol{G}}$ accordingly.
end

3.2 Stage II: Anchor-based Recovery

The second stage of SPAR uses the anchor components $V_1, \ldots, V_K$ from the first stage to assign community labels to all nodes. Throughout the main text, we use the distance-based recovery method described in Algorithm 2. As an alternative, a model-based recovery method that assigns labels according to posterior probabilities under a specific model, computed using Monte Carlo approximations, is presented in Section B.3 of the Appendix.

Algorithm 2: Distance Recovery

Input: graph $\boldsymbol{G}_n = (V, E)$; community anchors $V_1, \cdots, V_K$; recovery threshold $\tau' \geq 0$.

Output: $\hat{\ell} : V \to \{1, \ldots, K\}$

Construct $\tilde{V}_i := \left\{u \in V_i : \deg_{\boldsymbol{G}_n}(u) \geq \tau'\right\}$ for all $i \in [K]$.

Labeling: for $u \in V$ **do**

  $D_k(u) = \mathrm{dist}_{\boldsymbol{G}_n}\left(u, \tilde{V}_k\right)$ for all $k \in [K]$; $\hat{\ell}(u) := \mathrm{argmin}_{k\in[K]} D_k(u)$ with uniform random tie–breaking if non-unique;

end

return $\hat{\ell}$;

Given two connected nodes u and v in a generic graph $\boldsymbol{g}$, we define the graph distance between u and v to be $\mathrm{dist}_{\boldsymbol{g}}(u, v) :=$ length of the shortest-path connecting u and v. If u and v are disconnected, we define $\mathrm{dist}_{\boldsymbol{g}}(u, v) = \infty$. For two subsets of nodes U, V, we define $\mathrm{dist}_{\boldsymbol{g}}(U, V) = \min\{\mathrm{dist}_{\boldsymbol{g}}(u, v) : u \in U, v \in V\}$.

One way to perform distance-based recovery is to assign each node u the community label of the closest anchor component. However, large anchor components may sometimes still contain misclassified nodes so that, to further improve the reliability of the recovery method, we instead assign each node u the community label of the anchor component whose high-degree nodes are nearest to u. To be specific, let $\tau' \geq 0$ be a tuning parameter and define the high-degree sub-component $\tilde{V}_k := \{u \in V_k : \deg_{\boldsymbol{G}_n}(u) \geq \tau'\}$. We then define a modified distance between a node u and a component V_k as $D_k(u) := \mathrm{dist}_{\boldsymbol{G}_n}(u, \tilde{V}_k)$, and assign u the label k' such that $D_{k'}(u)$ is minimized.

In practice, we can choose τ' by setting $\tau' = \tau/2$, where τ is the anchor-degree threshold in Algorithm 1, or let it be the degree of the last batch of removed nodes from the first stage, i.e. the value D^* at the end of Algorithm 1, which is what we use in Theorem 13.

For the theoretical results, set $\hat{\ell}(u) := 1$ for all $u \in V$ if Algorithm 1 returns FAIL.

4 Theoretical Analysis

In this section, we provide upper bounds on the misclustering error rate of our procedure local to specific subsets of nodes. We study three subsets of central nodes: earliest-arriving nodes, high-degree nodes, and nodes that are within tree-distance 1 or 2 of the root node, i.e. children or grandchildren of the root nodes. We give the proof of all the results in Section F of the appendix.

Before stating our results, we introduce some useful notation. For any pair of labeled graphs $\boldsymbol{g} \subset \boldsymbol{g}'$ and any $C \geq 0$, we define the following high degree subsets:

$$V^C_{\boldsymbol{g}'}(\boldsymbol{g}) := \{u \in V(\boldsymbol{g}) \,|\, \deg_{\boldsymbol{g}'}(u) \geq C\}, \qquad V^{C,\alpha}_{\boldsymbol{g}'}(\boldsymbol{g}) := V^{C|V(\boldsymbol{g}')|^{\frac{1}{2+\alpha}}}_{\boldsymbol{g}'}(\boldsymbol{g}). \tag{4}$$

That is, $V^C_{\boldsymbol{g}'}(\boldsymbol{g})$ is the set of nodes in $\boldsymbol{g}$ whose degree, computed with respect to $\boldsymbol{g}'$, is at least C. When $\boldsymbol{g} = \boldsymbol{g}'$, we write these more concisely as $V^C(\boldsymbol{g})$ and $V^{C,\alpha}(\boldsymbol{g})$ respectively. The scale factor of $|V(\boldsymbol{g}')|^{\frac{1}{2+\alpha}}$ in the definition of $V^{C,\alpha}_{\boldsymbol{g}'}(\boldsymbol{g})$ takes on that particular form

because, under affine preferential attachment mechanism, the maximum degree of a graph with n nodes grows at rate $n^{\frac{1}{2+\alpha}}$ (see Lemma 38 in Section F.7 of the appendix).

We now introduce the assumptions required for our theoretical results.

Assumption 11. *Define* $n_k := \left|\left\{u \mid u \in V(\boldsymbol{G}_n),\, \ell(u) = k\right\}\right|$ *as the size of community* k *and let* $H \in [K, \infty)$. *We assume that* $\frac{n_k}{n} \geq \frac{1}{H}$ *for every* $k \in [K]$.

Assumption 11 says that the K communities should be roughly balanced in the sense that the ratio of sizes between the largest and the smallest community should be less than a constant factor. Our numerical experiments (c.f. Section D.2 in the appendix) show that the recovery performance of SPAR deteriorates with the severity of community-size imbalance. This is potentially because very small communities may get accidentally removed in the first pruning stage of our algorithm.

Assumption 12. *We assume that there exist* $\delta > 0$ *and* $C_0 > 0$ *such that* $\theta \leq C_0 n^{-\frac{1+\alpha}{2+\alpha}-\delta}$.

Assumption 12 controls the magnitude of the Erdős–Rényi perturbation. To give an intuitive interpretation of the tolerance bound $n^{-\frac{1+\alpha}{2+\alpha}}$, we observe that if the noise probability $\theta = o(n^{-\frac{1+\alpha}{2+\alpha}})$, then the expected degree contributed by the Erdős–Rënyi random edges for a particular node is equal to $(n-1)\theta = o(n^{\frac{1}{2+\alpha}})$. For early arriving nodes, this is negligible compared to the degree generated by the preferential attachment mechanism, which is of order $n^{\frac{1}{2+\alpha}}$. We can therefore still use the degree to recover the early-arriving node of each of the communities. Based on our numerical experiments, we conjecture that $n^{-\frac{1+\alpha}{2+\alpha}}$ is a tight threshold in that if $\theta = \Omega(n^{-\frac{1+\alpha}{2+\alpha}+\delta})$ for some $\delta > 0$, then the planted forest graph is indistinguishable from pure ER graph. It is also worth noting that, in the linear preferential attachment case ($\alpha = 0$), our noise tolerance bound for θ coincides with that appearing in Theorem 11 of Crane and Xu (2024). In fact, our results suggest that the analysis in Crane and Xu (2024) may potentially be extended from the linear preferential attachment setting to the general affine regime $\alpha > -1$.

Our first result bounds the misclustering error rate on the first L earliest-arriving nodes in each of the communities, i.e., the subset $\cup_{i=1}^{K} \pi_{1:L}\left(\boldsymbol{T}^i\right)$. We define the constant $C_1(\varepsilon, H) := \gamma(\frac{\varepsilon}{3H}, 1)$ where $\gamma(\cdot,\cdot)$ is the function specified in Lemma 28 via equation (22).

Theorem 13. *Let* $\boldsymbol{G}_n \sim \mathrm{PF}(\alpha, \theta, \ell, \pi)$ *and suppose Assumptions 11 and 12 hold. Fix any* $\varepsilon > 0$ *and fixed* $L \in \mathbb{N}$. *Let* $\hat{\ell}(\cdot)$ *be the estimated community membership from Algorithm 1 followed by Algorithm 2 with* $\tau = C_1(\varepsilon, H) n^{\frac{1}{2+\alpha}}$, *any fixed* $Q \in \mathbb{N}$, *and* τ' *as any non-negative number at most* D^* *outputted by Algorithm 1. Then, with probability at least* $1 - \varepsilon + \eta_n$, *where the* $\eta_n = o(1)$ *as* $n \to \infty$ *and depending only on* $\varepsilon, L, Q, \alpha, H, \delta$,

$$d_{\cup_{i=1}^{K} \pi_{1:L}(\boldsymbol{T}^i)}(\hat{\ell}, \ell) = 0. \tag{5}$$

We note that Theorem 13 remains valid for any τ of order $n^{\frac{1}{2+\alpha}}$ that is bounded above by $C_1(\varepsilon, H) n^{\frac{1}{2+\alpha}}$. We fix a particular choice of τ only to simplify the statement of the theorem and avoid uninteresting technical complications in the proof.

Theorem 13 allows us to directly obtain a bound on a weighted misclustering error $d_{\boldsymbol{w}}(\hat{\ell}, \ell)$.

Corollary 14. *Under the same assumptions of Theorem 13, let $\{\omega_t\}_{t\geq 1}$ be a nonnegative sequence satisfying $\sum_{t=1}^{\infty}\omega_t \in (0,\infty)$. For each $i \in [K]$ and $t \in [n_i]$, assign $w_{\pi_t^i} \propto \omega_t$, where π_t^i is the t-th arriving vertex in $\boldsymbol{T}^i$. Let $\hat{\ell}$ be the estimator obtained from Algorithm 1, followed by Algorithm 2, with τ, τ', Q chosen as in Theorem 13. Then there exists a deterministic sequence $a_n \to 0$ (dependent only on $\varepsilon, Q, \alpha, H, \delta, \{\omega_t\}$) such that with probability at least $1-2\varepsilon$,*

$$d_{\boldsymbol{w}}(\ell, \hat{\ell}) \leq a_n, \tag{6}$$

where $d_{\boldsymbol{w}}(\cdot,\cdot)$ is from Definition 5.

Remark 15. *We state our results assuming that the number of communities K is given. In our theoretical setting where $\boldsymbol{G}_n$ is generated by the planted forest model, we can estimate K by carefully selecting two parameters $\tau_1 > \tau_2 > 0$, then removing all nodes of degree less than τ_2, and finally letting $\hat{K}$ be the number of remaining connected components that contain a node of degree at least τ_1; we describe this formally in Algorithm 4. By Lemma 30 in Section F.3 of the appendix, if we choose $\tau_1 = \gamma(\frac{\varepsilon}{3H}, 1)n^{\frac{1}{2+\alpha}}$ and $\tau_2 = \tilde{\gamma}(\varepsilon, 1, 1)n^{\frac{1}{2+\alpha}}$ (where $\tilde{\gamma}$ is defined in the proof of Lemma 30), then $\hat{K} = K$ with probability at least $1-\varepsilon+o(1)$. In practice however, the τ_2 parameter is critical to the estimation result and not much easier to tune than K.*

One somewhat restrictive aspect of Theorem 13 is that the target set $\cup_{i=1}^{K}\pi_{1:L}\left(\boldsymbol{T}^i\right)$ depends on the unobserved arrival ordering and therefore cannot be directly identified from the observed graph $\boldsymbol{G}_n$. However, we know that the early arriving nodes roughly coincide with the high-degree nodes of the graph (c.f. Lemma 28 in Section F.3 of the appendix) so that we may extend the recovery guarantee of Theorem 13 to the subset of high-degree vertex set $V^{\tilde{C},\alpha}(\boldsymbol{G}_n)$, which is readily computable from the observed graph.

Corollary 16. *Under the same setting as Theorem 13, let $V^{\tilde{C},\alpha}(\boldsymbol{G}_n)$ denote the set of high-degree nodes defined in (4) for any fixed $\tilde{C} > 0$. Then, the estimator $\hat{\ell}$ obtained using graph pruning through Algorithm 1 then with label recovery through Algorithm 2, with the same τ, τ', Q as specified in Theorem 13, satisfies, with probability at least $1-2\varepsilon+o(1)$ where the $o(1)$ term depends only on $\varepsilon, \tilde{C}, Q, \alpha, H, \delta$,*

$$d_{V^{\tilde{C},\alpha}(\boldsymbol{G}_n)}(\hat{\ell}, \ell) = 0. \tag{7}$$

By an argument identical to that of Remark 14, we can show that Corollary 16 holds for some sequence $\tilde{C} \equiv \tilde{C}_n$ that goes to 0 as $n \to \infty$.

The second part of our analysis considers vertex sets defined through distances from the root nodes. We define these by layers so that layer-1 nodes consist of all direct children of the K root nodes, layer-2 nodes consist of all grandchildren of the root nodes, etc. We formalize these definitions below:

Definition 17. *For $\boldsymbol{G}_n \sim \mathrm{PF}(\alpha, \theta, \ell, \pi)$ and $s \in \mathbb{N}$, we define $\mathcal{L}_s(\boldsymbol{F}_n)$ as the nodes in s layer of forest $\boldsymbol{F}_n$, that is*

$$\mathcal{L}_s(\boldsymbol{T}^i) := \left\{ v \in V(\boldsymbol{T}^i) \,\middle|\, \mathrm{dist}_{\boldsymbol{T}^i}\big(v, \pi_1(\boldsymbol{T}^i)\big) = s \right\}, \quad \mathcal{L}_s(\boldsymbol{F}_n) := \cup_{i=1}^{K}\mathcal{L}_s(\boldsymbol{T}^i).$$

These vertex sets have cardinalities that diverge with n. From Lemma 34 in the appendix, we have that if $\boldsymbol{G}_n$ is a planted forest random graph with linear preferential attachment, i.e $\alpha = 0$, then, for each $s \in \mathbb{N}$,

$$\mathbb{E}|\mathcal{L}_s(\boldsymbol{F}_n)| \asymp \sqrt{n}\log^{s-1} n.$$

For a general $\alpha > -1$, we have from Lemma 38 in the appendix that

$$|\mathcal{L}_1(\boldsymbol{F}_n)| \asymp_p n^{\frac{1}{2+\alpha}}. \tag{8}$$

Although the number of layer-s nodes is increasing with n, their proximity to the root node allows us to still bound the misclustering error rate on the high probability event that the K root nodes are correctly dispersed among the K anchor components.

Theorem 18. *For $\boldsymbol{G}_n \sim \mathrm{PF}(\alpha, \theta, \ell, \pi)$, and suppose Assumptions 11 and 12 hold. Fix any $\varepsilon > 0$ and $c_0 \in (0, 1]$. Let $\hat{\ell}(\cdot)$ be the estimated community membership from Algorithm 1 followed by Algorithm 2 with $\tau := C_1(\frac{\varepsilon}{6}, H)n^{\frac{1}{2+\alpha}}$, any fixed $Q > 0$, and any $\tau' \in [c_0\tau, \tau]$. Then, there exists a constant $C_2 \equiv C_2(\varepsilon, c_0) < \infty$, with probability at least $1-\varepsilon+o(1)$ (with the $o(1)$ term depending only on $\varepsilon, Q, \alpha, H, \delta, c_0$),*

$$\frac{d_{\mathcal{L}_1(\boldsymbol{F}_n)}(\hat{\ell}, \ell)}{\left|\mathcal{L}_1\left(\boldsymbol{F}_n\right)\right|} \leq C_2 n^{-\frac{1+\alpha}{2+\alpha}-\delta}. \tag{9}$$

Moreover, if $\alpha = 0$, then there exists a constant $C_3 \equiv C_3(\varepsilon, c_0) < \infty$ such that with probability at least $1 - \varepsilon + o(1)$ (with the $o(1)$ term depending only on $\varepsilon, Q, H, \delta, c_0$),

$$\frac{d_{\mathcal{L}_2(\boldsymbol{F}_n)}(\hat{\ell}, \ell)}{\left|\mathcal{L}_2\left(\boldsymbol{F}_n\right)\right|} \leq C_3 n^{-\delta}. \tag{10}$$

Remark 19. *It is worth noting that in the layer-1 bound (9), when $\alpha \geq 0$, we have by (8) that*

$$d_{\mathcal{L}_1(\boldsymbol{F}_n)}(\hat{\ell}, \ell) \leq C_2 n^{-\frac{1+\alpha}{2+\alpha}-\delta}\left|\mathcal{L}_1\left(\boldsymbol{F}_n\right)\right| = O_p(n^{-\frac{\alpha}{2+\alpha}-\delta})$$

so that, when n is large enough, we have $d_{\mathcal{L}_1(\boldsymbol{F}_n)}(\hat{\ell}, \ell) < 1$ and $d_{\mathcal{L}_1(\boldsymbol{F}_n)}(\hat{\ell}, \ell) = 0$ necessarily. In other words, when $\alpha \geq 0$, the SPAR algorithm can perfectly recover the community label of the layer-1 nodes with high probability. When $\alpha \in (-1, 0)$, the right-hand side of (9) appears to decrease as α increases, suggesting that larger α leads to better recovery. However, this interpretation is incorrect as, for larger α, the noise tolerance bound becomes smaller and the size $|\mathcal{L}_1(\boldsymbol{F}_n)|$ becomes smaller so that the recovery problem is substantially easier.

In Section F.1 of the appendix, we discuss the plausibility and challenges of extending Theorem 18 to more general settings such as when $s > 2$ or when the attachment mechanism is not necessarily an APA tree.

5 Simulation

We evaluate the performance of our proposed method with distance-based recovery and compare it with several benchmark methods on networks generated from the planted forest model under different attachment and noise regimes.

We consider the following four methods. The number of communities K is assumed to be known for all methods.

1. *Distance Recovery*: We apply the SPAR procedure in Algorithm 1, followed by Algorithm 2. We initialize the size threshold at $Q = n/10$ and use the shrinkage factor $\zeta = 0.8$ from Remark 9. The core degree threshold τ is the degree of the 10th highest-degree vertex in the observed graph. The recovery threshold is $\tau' = \min\{D^*, \tau/2\}$, where D^* is the degree at the last removal step. The calibration motivating the choice of τ is given in Section B.1. We denote this method by "Distance Recovery" in the plots.

2. *SCORE*: We use the spectral clustering method of Jin (2015), based on ratios of the leading eigenvectors of the adjacency matrix. We denote this method by "SCORE" in the plots.

3. *GMZZ*: We use the two-stage method of Gao et al. (2018), consisting of weighted spectral initialization followed by normalized neighbor voting. We denote this method by "GMZZ" in the plots.

4. *MCMC recovery*: We use the model-based recovery procedure in Algorithm 5, with the Gibbs sampler of Crane and Xu (2024) and the anchor components obtained from SPAR. The detail setting of Monte Carlo sampling method is explained in Section D.1. This method is denoted by "Monte Carlo" in the plots.

We simulate networks $\boldsymbol{G}_n \sim \mathrm{PF}(\alpha, \theta, \ell, \pi)$ over $B = 200$ independent repetitions, consider balanced component sizes $n_1 = n_2 \in \{500, 1000, 1500, 2000, 2500\}$ and attachment parameters $\alpha \in \{2, 0, -0.5\}$. The noise level is set to $\theta = cn^{-1}$, where c is chosen so that $\theta = 0.01$ when $n = 1000$. We repeat the same simulation design for noise level set as $\theta = c'n^{-3/4}$ and $\theta = c''n^{-1/2}$ where c' and c'' are again set so that $\theta = 0.01$ when $n = 1000$.

To facilitate comparisons of misclassification rates across different values of n and θ for a fixed α, we adopt the following construction. For each Monte Carlo repetition and each choice of component sizes $n_1 = n_2 \in \{500, 1000, 1500, 2000, 2500\}$, we first generate a single large preferential attachment forest $\boldsymbol{f}_{5000}$ consisting of two trees $\boldsymbol{t}^1_{2500}$ and $\boldsymbol{t}^2_{2500}$. We then extract the induced subgraph formed by the first n_i vertices in each tree to obtain the target forest $\boldsymbol{f}_{n_1+n_2} := \oplus_{i=1}^2 \boldsymbol{t}^i_{n_i}$. Finally, we superimpose an Erdős–Rényi random graph $\boldsymbol{r}_{n_1+n_2} \sim$ Erdős–Rényi(θ) with the corresponding noise level, yielding the observed network $\boldsymbol{g}_{n_1+n_2} := \boldsymbol{f}_{n_1+n_2} + \boldsymbol{r}_{n_1+n_2}$. Figure 2 compares the mean misclassification rates of the methods on the first 50 arriving vertices in each tree.

As expected, the near oracle method "Monte Carlo" achieves the lowest misclassification rates. The Distance Recovery method also consistently outperforms SCORE and GMZZ across all settings considered. In the regimes covered by our theoretical guarantees ($\theta = O(n^{-(1+\alpha)/(2+\alpha)-\delta})$), its misclassification rate decreases as the network size increases,

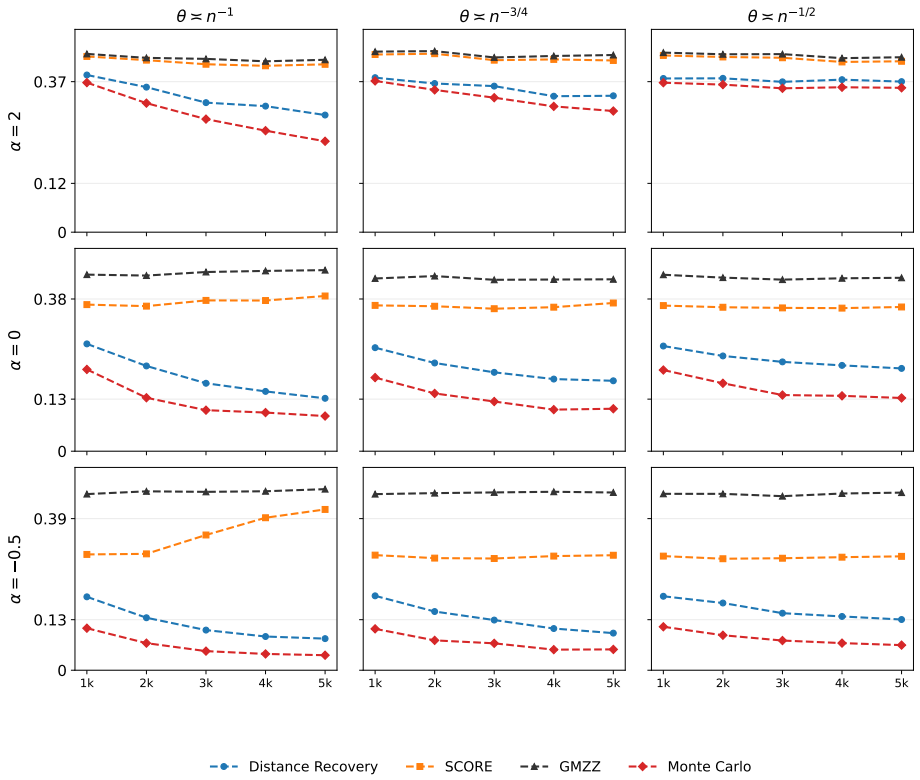


Figure 2: Mean misclassification rates for the first 50 arriving vertices in the network, computed 200 repetitions.

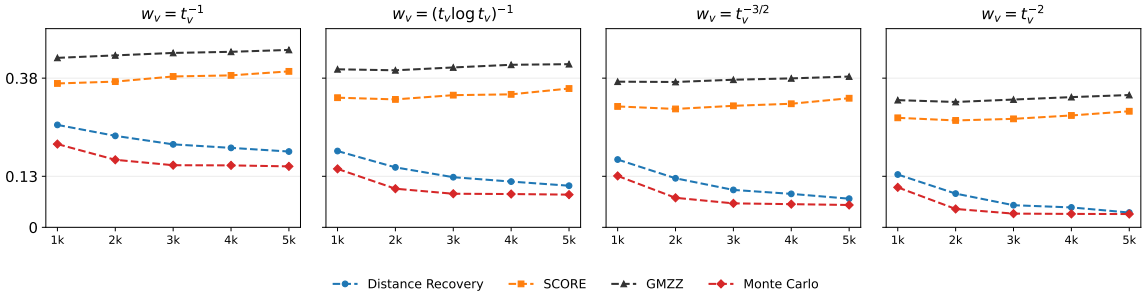


Figure 3: Mean weighted misclassification rates for $\alpha = 0$ and $\theta \asymp n^{-1}$, with weights t^{-1}, $(t \log t)^{-1}$, $t^{-3/2}$, and t^{-2} from left to right, where t is the arrival time within each tree.

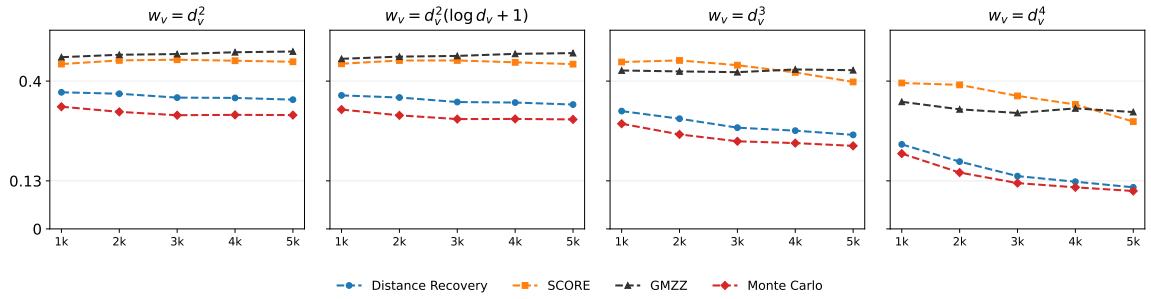


Figure 4: Mean weighted misclassification rates for $\alpha = 0$ and $\theta \asymp n^{-1}$, with weights d^2, $d^2(\log d+1)$, d^3, and d^4 from left to right, where d is the observed degree of each node.

approaching $\varepsilon = 0.05$ over the range of sample sizes considered. Sections D.2 in the appendix provide additional numerical results for multiple unbalanced trees setting.

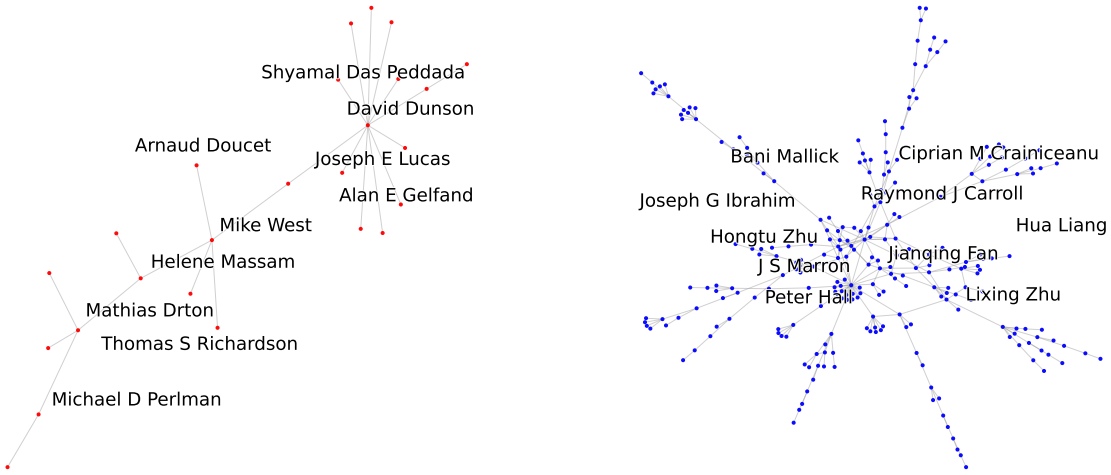
We further compared the weighted misclassification rate definied by weighted distance between $\hat{\ell}(\cdot)$ and $\ell(\cdot)$ in Definition (5). For the tth arriving vertex in each tree, we consider weights t^{-1}, $(t \log t)^{-1}$, $t^{-3/2}$, and t^{-2}. We set the logarithmic weight at $t = 1$ equal to its value at $t = 2$ to avoid division by zero.

Figure 3 shows that the weighted misclassification rate of Distance Recovery/Monte Carlo decreases toward zero for the weights $t^{-3/2}$ and t^{-2}. A decreasing trend is also apparent for the weights t^{-1} and $(t \log t)^{-1}$ whose summation does not converge, suggesting that vanishing weighted error may also be correct beyond the condition $\sum_{t=1}^{\infty} w_t = W \in (0, \infty)$ in Corollary 14.

We next consider data-dependent weights based on observed degree. For $\alpha = 0$, the degree growth scale $(n_k/t)^{1/2}$ for the tth arriving vertex in a tree of size n_k (equation (4.25) in Peköz et al. (2013)) motivates d^2, d^3, and d^4 as heuristic counterparts of the arrival-time weights t^{-1}, $t^{-3/2}$, and t^{-2} in the balanced setting. We also include $d^2(\log d + 1)$ as an intermediate choice between quadratic and cubic weighting. Figure 4 summarizes the results.

The degree-weighted results show a clearer contrast: the misclassification rates decrease substantially under d^3 and d^4 weighting, but only slightly under d^2 and $d^2(\log d+1)$ weighting. We leave the theoretical analysis of their convergence rates to future work.

A notable pattern in our simulation results is that the errors, even when they decrease rapidly, seem to level off around some small positive value of approximately 0.03; for example in the left bottom plot in Figure 2. The reason is that in about 5 percent of the simulations, the error is bounded away from 0. The average error thus converges to a value around 0.03 even though, in the other 95 percent of the simulations, the error actually does converge to 0. This is consistent with the theory and since the choice of core-degree threshold τ depends on the desired error probability ε in Theorem 13 and we choose τ by fixing ε in our simulations. Specifically, we fix τ as the 10-th largest observed degree. If all ten highest-degree vertices belong to the same planted tree, the other tree has no vertex meeting the threshold, so the procedure cannot identify a qualifying core from each tree. Choosing τ using a degree rank that slowly increases with n may reduce the frequency of this failure and allow the average error to go to 0.



(a) Community Core 1

(b) Community Core 2

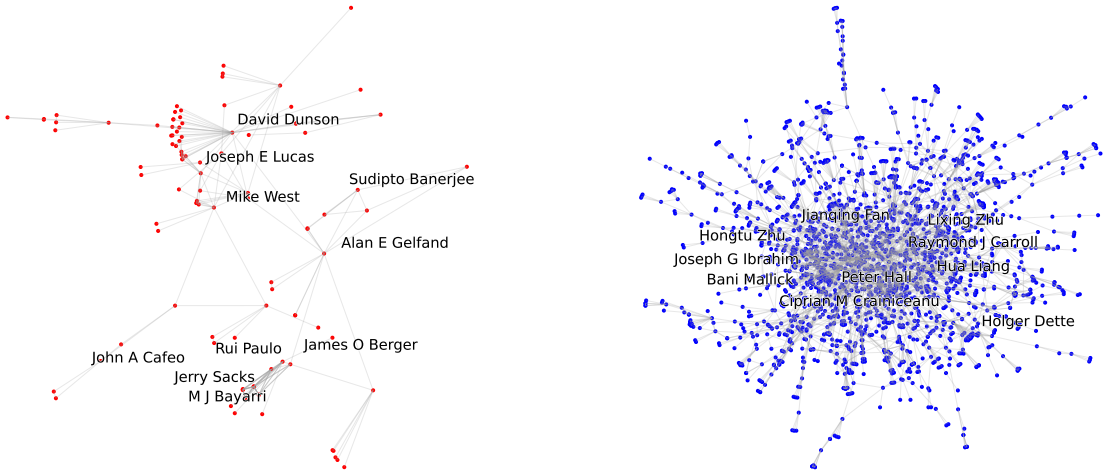
Figure 5: Community cores obtained from the co-authorship network of Ji and Jin (2016) using the graph pruning procedure in Algorithm 1. The parameters are set to $(K, \tau, Q, \zeta) = (2, 10\text{th highest degree}, 226, 0.8)$.

6 Case Study

We now apply our SPAR algorithm to perform an extensive analysis of a statistician co-authorship network constructed by Ji and Jin (2016). In this network, each node corresponds to a statistician and two nodes u and v have an edge between them if they have co-authored 1 or more publications in either Journal of Royal Statistical Society: Series B, Journal of the American Statistical Association, Annals of Statistics, or Biometrika from 2002 to 2013. We consider only the largest connected component which has $n = 2263$ nodes and $m = 4388$ edges. As a preliminary analysis of the dataset, we first apply Algorithm 1 to obtain the community anchors where the number of communities is set to be $K = 2$. Then, we apply the distance-based recovery method (Algorithm 2). To select the core degree threshold τ, we first estimate $\alpha = 0$ using the EM method described in Section S3.1 of the supplementary material of Crane and Xu (2024). Given this estimate, the core degree threshold is set to the degree of the 10th highest-degree node, following Algorithm 3. The size threshold is chosen as $\lfloor n/10 \rfloor = 226$, and the shrinkage parameter is set to $\zeta = 0.8$ (see Remark 9).

We give the intermediate estimate of community anchors produced by the graph pruning procedure (Algorithm 1) in Figure 5. In each graph, the ten highest-degree nodes are labeled by their name. The procedure reveals two prominent anchor components: The first core consists primarily of researchers specializing in Bayesian statistics, while the second core corresponds to a large group of high-dimensional statistics researchers.

We present the distance-based recovery results from Algorithm 2 in Figure 6. We use the same recovery threshold as in Section 5, where τ' is defined as the minimum of two quantities: first is the degree of the last removed node of Algorithm 1, that is, the value of D^* at the conclusion of Algorithm 1 and the second is $\tau/2$ where τ is the core degree



(a) Community 1 (b) Community 2

Figure 6: Communities obtained by the distance-based recovery procedure in Algorithm 2 with threshold $\tau' = \tau/2$, using the anchor components identified in Figure 5.

threshold from Algorithm 1. We also applied the model-based recovery method (Algorithm 5 in Section B.3 of the Appendix) to the dataset using the same community anchors shown in Figure 5. A comparison between the model-based recovery method and the distance-based recovery method is presented in Section E.1 of the Appendix.

While the classification results in Figure 5 suggest that our method produces clusters with coherent research themes, a direct observation is that some recovered cores or communities still contain multiple hidden clusters, particularly within Community 2 in Figure 6. This motivates further partitioning of the two detected communities. To avoid estimating the number of communities K and to capture actual hierarchical structure in research communities, we apply a hierarchical clustering procedure with tuning parameters $(K, \tau, Q) = (2, 10\text{th highest degree}, 25)$ for Algorithm 1 and recovery threshold $\tau' := \min\{\text{last removed node degree}, \tau/2\}$ for Algorithm 2.

Specifically, we take the two communities produced by the distance recovery stage, extract the largest connected component in each community, and reapply the community detection procedure using the same tuning parameters. We do this recursively and continue as long as the component can be separated into two cores, each containing at least 10 nodes. Once a cluster can no longer be further divided by the procedure, it becomes a terminal ("leaf") cluster in the hierarchical tree. We rank the resulting clusters from 1 to 23 according to the order in which they appear as leaf clusters during the process. The hierarchical structure of the clusters is illustrated by the dendrogram in Figure 7. Figure 8 presents six representative clusters. Within each cluster, the corresponding core is highlighted by coloring its edges in yellow, while all remaining edges are shown in gray.

Within most of the recovered communities, the researchers share similar research themes or institutional affiliations. For example, Community 1 can be interpreted as a "Bayesian modelling" cluster, while Community 4 appears to represent an "experimental design" clus-

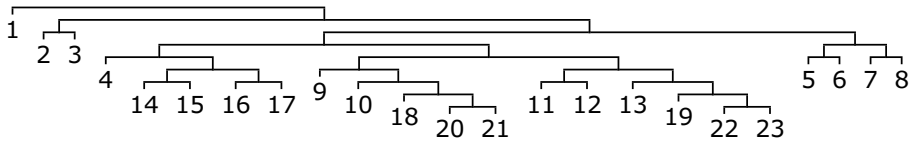
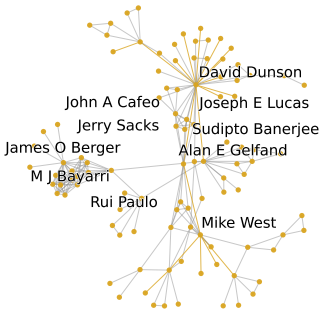
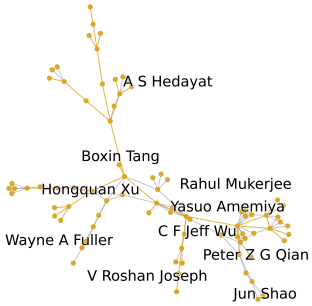


Figure 7: Hierarchical dendrogram from iterative core pruning

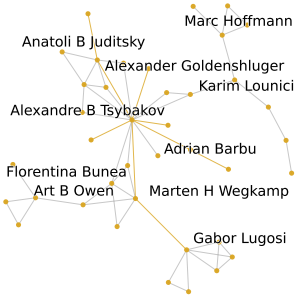
ter. The results are not perfect, as we can see from Community 23, which emerges last in the hierarchical clustering procedure and appears to pick up leftover nodes. Beyond similarities within individual communities, the dendrogram in Figure 7 also reveals structural relationships among communities that are close in the hierarchical tree. Communities 18, 20, and 21 emerge from the same broader cluster. Communities 18 and 20 mainly comprise European and U.S. theoretical statisticians, respectively, while Community 21 brings together theoreticians whose collaborations span both regions.



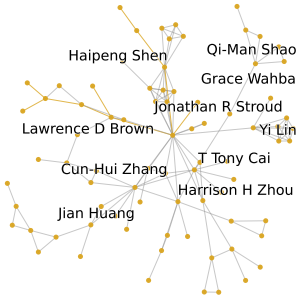
(a) Community 1
(Bayesian modelling)



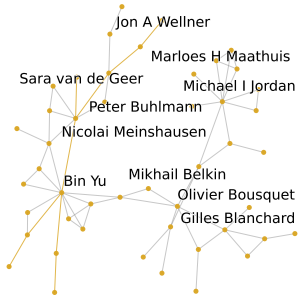
(b) Community 4
(Experimental design)



(c) Community 18
(European theoreticians)



(d) Community 20
(U.S. theoreticians)



(e) Community 21



(f) Community 23

Figure 8: Selected clusters obtained by distance recovery in the hierarchical procedure

7 Discussion

This paper studies the planted forest model for networks formed by the growth of K preferential attachment trees embedded in Erdös–Rényi noise and studies community detection on

these randomly growing networks. We propose a two-stage algorithm, SPAR, and establish local recovery guarantees with respect to several subsets of nodes, including early-arriving, high-degree, and layer-1 and layer-2 nodes, i.e. nodes within 1 or 2 distance away from the root nodes respectively. To the best of our knowledge, this is the first work on community detection with a rigorous theoretical guarantee for this class of randomly growing networks.

Several important questions remain open. From a methodological perspective, an important challenge is estimating the community counts K, which is mostly assumed to be known in this paper. Although Remark 15 provides a consistent estimator, it requires careful tuning that is difficult to perform in practice. Developing a more stable estimator of K would improve the practical applicability of the proposed method. Another direction is to explore alternative pruning criteria in the first stage of SPAR, as mentioned in Remark 10. From a theoretical perspective, an alternative to studying local community recovery is to replace the current definition of misclustering error with a criterion that assigns node-specific weights according to their structural importance in the network, such as arriving order. This may allow us to recover global consistency in community estimation, albeit with an alternative error metric.

Reproducibility. The code used to generate all simulation results and real-data analyses in this paper is publicly available at `https://github.com/JasonSmilingKnight/Forest-Community-Detection`. The repository also contains instructions and scripts for reproducing all numerical experiments and figures reported in the paper.

Acknowledgement:

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Supplementary Material for Community Detection on a Randomly Growing Network

Appendix A. Proofs for Section 2

Before proving Theorem 6, we introduce some notation to simplify the proof.

Let $\boldsymbol{G}_n := \boldsymbol{F}_n + \boldsymbol{R}_n$ generated from $\mathrm{PF}(\alpha, \theta, \pi, 2)$. Define the parent function $\mathrm{pa} : V(\boldsymbol{F}_n) \to V(\boldsymbol{F}_n)$ such that $\mathrm{pa}(v) = u$ indicates that node v is attached to node u during the generation process of $\boldsymbol{F}_n$, and, for the initial vertices, we set $\mathrm{pa}(\pi_{2i-1}) := \pi_{2i-1}$ and $\mathrm{pa}(\pi_{2i}) := \pi_{2i-1}$ for $i = 1, 2$.

For $t \in [n]$, we let $(e_1, \ldots, e_{h_t})$ be the set of noise edges incident between the new node π_t and previous nodes $\pi_{1:(t-1)}$; note that the number of such edges h_t is also random. Conditional on h_t, we define a random node $q(\pi_t)$ where we choose an edge e' in $(e_1, \ldots, e_{h_t})$ uniformly at random and let $q(\pi_t)$ be the endpoint of e' that is in $\pi_{1:(t-1)}$. If $h_t = 0$, then we let $q(\pi_t) = \emptyset$.

For an integer $t \in [n]$, denote by $\boldsymbol{T}_t^1$ and $\boldsymbol{T}_t^2$ the two tree components of the forest $\boldsymbol{F}_n$ at time t. With $v_1, v_2 \in \pi_{1:(t-1)}$, we define the "balance attachment" event $\mathcal{A}_t$ as follows:

$$\begin{aligned}\mathcal{A}_t\big(\{v_1, v_2\}\big) := \big\{ & v_1 \in V(\boldsymbol{T}_{t-1}^1), v_2 \in V(\boldsymbol{T}_{t-1}^2), \big\{\mathrm{pa}(\pi_t), q(\pi_t)\big\} = \{v_1, v_2\}, \\ & \{v_1, v_2\} \cap \pi_{1:4} = \emptyset, \quad \mathrm{pa}(\pi_s) \notin \{v_1, v_2\},\ \forall s \neq t \big\} \\ \mathcal{A}_t := \cup_{\{v_1, v_2\} \in \pi_{1:(t-1)}} & \mathcal{A}_t\big(\{v_1, v_2\}\big). \end{aligned} \tag{11}$$

The event $\mathcal{A}_t$ collects all growth histories for which the t-th node encounters an ambiguous attachment: the parent and a noise endpoint of π_t are two candidate nodes v_1 and v_2 from the two trees, respectively. Under this event, either node could serve as the parent of π_t, and neither v_1 nor v_2 receives any additional child nodes in the growth process. We note that if $\{v_1, v_2\} \neq \{v_1', v_2'\}$, then $\mathcal{A}_t(\{v_1, v_2\})$ and $\mathcal{A}_t(\{v_1', v_2'\})$ are disjoint.

Proof of Theorem 6. Let $\bar{\sigma} \in \mathrm{argmin}_{\sigma \in S_2} n^{-1} \sum_{t=1}^{\lceil n/2 \rceil} \mathbb{1}\left\{\hat{\ell}\left(\pi_t\right) \neq \sigma \circ \ell\left(\pi_t\right)\right\}$ (choose the identity when the argmin is non-unique). By Definition 5, we have

$$\begin{aligned}
\mathbb{E}\left[\frac{d\big(\hat{\ell}, \ell\big)}{n}\right] &= \mathbb{E}\left[\min_{\sigma \in S_2} n^{-1} \sum_{u \in \pi_{1:n}} \mathbb{1}\Big\{\hat{\ell}\big(u\big) \neq \sigma \circ \ell\big(u\big)\Big\}\right] \\
&\overset{(a)}{\geq} \mathbb{E}\left[\min\left\{n^{-1} \sum_{t=\lceil n/2 \rceil + 1}^{n} \mathbb{1}\Big\{\hat{\ell}\big(\pi_t\big) \neq \bar{\sigma} \circ \ell\big(\pi_t\big)\Big\}, \frac{1}{4}\right\}\right] \\
&\overset{(b)}{\geq} \frac{1}{4}\mathbb{E}\left[n^{-1} \sum_{t=\lceil n/2 \rceil + 1}^{n} \mathbb{1}\Big\{\hat{\ell}\big(\pi_t\big) \neq \bar{\sigma} \circ \ell\big(\pi_t\big)\Big\}\right] \min\left\{\mathbb{E}\left[n^{-1} \sum_{t=\lceil n/2 \rceil + 1}^{n} \mathbb{1}\Big\{\hat{\ell}\big(\pi_t\big) \neq \bar{\sigma} \circ \ell\big(\pi_t\big)\Big\}\right], \frac{1}{2}\right\} \\
&\overset{(c)}{\geq} \frac{1}{12} \min_{t > \lceil n/2 \rceil} \mathbb{P}\left(\hat{\ell}\left(\pi_t\right) \neq \bar{\sigma} \circ \ell\left(\pi_t\right)\right) \min\left\{\frac{1}{3} \min_{t > \lceil n/2 \rceil} \mathbb{P}\left(\hat{\ell}\left(\pi_t\right) \neq \bar{\sigma} \circ \ell\left(\pi_t\right)\right), \frac{1}{2}\right\} \\
&\overset{(d)}{\geq} \frac{1}{36}\Big(\min_{t > \lceil n/2 \rceil} \mathbb{P}\left(\hat{\ell}\left(\pi_t\right) \neq \bar{\sigma} \circ \ell\left(\pi_t\right)\right)\Big)^2,
\end{aligned} \tag{12}$$

where inequality (a) follows from Lemma 20, inequality (b) follows from Lemma 21, inequality (c) follows from the fact that the integer set $[\lceil n/2\rceil+1,\, n]$ contains more than $n/3$ integers when $n \geq 4$, and inequality (d) follows from the fact that $\frac{1}{3}\mathbb{P}\left(\hat{\ell}\left(\pi_t\right) \neq \bar{\sigma} \circ \ell\left(\pi_t\right)\right) \leq \frac{1}{3} < \frac{1}{2}$.

For a fixed integer t satisfying $t \geq \lceil n/2\rceil$, let the λ be chosen as Lemma 22. Recall the definition of a "balance attachment" event $\mathcal{A}_t$ in (11). Now, we lower bound $\mathbb{P}\left(\hat{\ell}\left(\pi_t\right) \neq \bar{\sigma} \circ l\left(\pi_t\right)\right)$ by restricting ourselves to the event $\mathcal{A}_t$:

$$
\begin{aligned}
\mathbb{P}(\hat{\ell}(\pi_t) \neq \bar{\sigma} \circ \ell(\pi_t)) &\geq \mathbb{P}\left(\left\{\hat{\ell}(\pi_t) \neq \bar{\sigma} \circ \ell(\pi_t)\right\} \cap \mathcal{A}_t\right) \\
&\overset{(a)}{=} \sum_{\{v_1,v_2\}\in\pi_{1:(t-1)}} \mathbb{P}\left(\left\{\hat{\ell}(\pi_t) \neq \bar{\sigma} \circ \ell(\pi_t)\right\} \cap \mathcal{A}_t\left(\{v_1,v_2\}\right)\right) \\
&\overset{(b)}{=} \sum_{\{v_1,v_2\}\in\pi_{1:(t-1)}} \mathbb{E}\left[\mathbb{E}\left(\mathbb{1}\{\hat{\ell}(\pi_t) \neq \bar{\sigma} \circ \ell(\pi_t)\}\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})} \,\middle|\, \left\{\mathrm{pa}(\pi_t), q(\pi_t)\right\},\, \{\mathrm{pa}(\pi_s)\}_{s\in[n]\setminus t}, \boldsymbol{G}_n\right)\right] \\
&\overset{(c)}{=} \sum_{\{v_1,v_2\}\in\pi_{1:(t-1)}} \mathbb{E}\left[\mathbb{P}\left(\hat{\ell}(\pi_t) \neq \bar{\sigma} \circ \ell(\pi_t) \,\middle|\, \left\{\mathrm{pa}(\pi_t), q(\pi_t)\right\} = \{v_1,v_2\},\, \{\mathrm{pa}(\pi_s)\}_{s\in[n]\setminus t}, \boldsymbol{G}_n\right)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}\right] \\
&\overset{(d)}{\geq} \sum_{\{v_1,v_2\}\in\pi_{1:(t-1)}} \mathbb{E}\left[\min_{i\in\{1,2\}} \mathbb{P}\left(\mathrm{pa}(\pi_t) = v_i \,\middle|\, \left\{\mathrm{pa}(\pi_t), q(\pi_t)\right\} = \{v_1,v_2\},\, \{\mathrm{pa}(\pi_s)\}_{s\in[n]\setminus t}, \boldsymbol{G}_n\right)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}\right] \\
&\overset{(e)}{=} \sum_{\{v_1,v_2\}\in\pi_{1:(t-1)}} \frac{1}{2}\mathbb{E}\left[\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}\right] \overset{(f)}{=} \frac{1}{2}\mathbb{P}(\mathcal{A}_t),
\end{aligned}
\tag{13}
$$

where the identities (a) and (f) follow from the fact that the events $\mathcal{A}_t(\{v_1,v_2\})$ are disjoint for different unordered pairs $\{v_1,v_2\}$. Step (b) follows from the tower property of conditional expectation, conditioning on the random unordered pair $\{\mathrm{pa}(\pi_t), q(\pi_t)\}$, the other parent choices, and $\boldsymbol{G}_n$. Step (c) follows because $\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}$ is fixed after we condition on $\left\{\mathrm{pa}(\pi_t), q(\pi_t)\right\} = \{v_1,v_2\}$, $\{\mathrm{pa}(\pi_s)\}_{s\in[n]\setminus t}, \boldsymbol{G}_n$. For (d), the estimator $\hat{\ell}(\pi_t)$ is fixed given the observed graph $\boldsymbol{G}_n$, whereas the random true label $l(\pi_t)$ depends only on whether $\mathrm{pa}(\pi_t) = v_1$ or $\mathrm{pa}(\pi_t) = v_2$, The random permutation $\bar{\sigma}$ is also fixed conditional on $\ell(\pi_{1:\lceil n/2\rceil})$ (which is fully determined by $\{\mathrm{pa}(\pi_t)\}_{t=1}^{\lceil n/2\rceil}$) and on $\boldsymbol{G}_n$ by its definition. For equality (e), we show that the two conditional probabilities corresponding to v_1 and v_2 are equal on the event $\mathcal{A}_t(\{v_1,v_2\})$. By Bayes' rule and the attachment probability in (3), we have

$$
\begin{aligned}
&\mathbb{P}\left(\mathrm{pa}(\pi_t) = v_1 \,\middle|\, \left\{\mathrm{pa}(\pi_t), q(\pi_t)\right\} = \{v_1,v_2\},\, \{\mathrm{pa}(\pi_s)\}_{s\in[n]\setminus t}, \boldsymbol{G}_n\right)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})} \qquad (14)\\
&= \frac{\mathbb{P}\left(\mathrm{pa}(\pi_t) = v_1,\, \left\{\mathrm{pa}(\pi_t), q(\pi_t)\right\} = \{v_1,v_2\},\, \{\mathrm{pa}(\pi_s)\}_{s\in[n]\setminus t}, \boldsymbol{G}_n\right)}{\mathbb{P}\left(\left\{\mathrm{pa}(\pi_t), q(\pi_t)\right\} = \{v_1,v_2\},\, \{\mathrm{pa}(\pi_s)\}_{s\in[n]\setminus t}, \boldsymbol{G}_n\right)}\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}
\end{aligned}
$$

$$= \frac{\mathbb{P}\begin{pmatrix}\mathrm{pa}(\pi_t) = v_1,\\ q(\pi_t) = v_2,\end{pmatrix}\mathbb{P}\left(\boldsymbol{G}_n, \{\mathrm{pa}(\pi_s)\}_{s>t} \middle| \begin{matrix}\mathrm{pa}(\pi_t) = v_1,\\ q(\pi_t) = v_2,\end{matrix} \{\mathrm{pa}(\pi_s)\}_{s<t}\right)}{\mathbb{P}\Big(\big\{\mathrm{pa}(\pi_t), q(\pi_t)\big\} = \{v_1, v_2\},\ \{\mathrm{pa}(\pi_s)\}_{s\in[n]\setminus t}, \boldsymbol{G}_n\Big)} \mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}$$

We first see that

$$\begin{aligned}&\mathbb{P}\big(\mathrm{pa}(\pi_t) = v_1, q(\pi_t) = v_2, \{\mathrm{pa}(\pi_s)\}_{s<t}\big)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}\\ &= \mathbb{P}\big(q(\pi_t) = v_2 \,\big|\, \mathrm{pa}(\pi_t) = v_1, \{\mathrm{pa}(\pi_s)\}_{s<t}\big)\mathbb{P}\big(\mathrm{pa}(\pi_t) = v_1 \,|\, \{\mathrm{pa}(\pi_s)\}_{s<t}\big)\\ &\qquad \mathbb{P}(\{\mathrm{pa}(\pi_s)\}_{s<t})\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})} \end{aligned} \tag{15}$$

By independence of the noise graph and the forest, and symmetry among the $t-1$ preceding vertices, the first term of (15) equals $[1-(1-\theta)^{t-1}]/(t-1)$. On $\mathcal{A}_t(\{v_1, v_2\})$, both candidate parents have degree one at time $t-1$, so the second term equals $(1+\alpha)/[2(t-3)+(t-1)\alpha]$. The third term is unchanged when v_1 and v_2 are exchanged. Therefore,

$$\begin{aligned}&\mathbb{P}\big(\mathrm{pa}(\pi_t) = v_1, q(\pi_t) = v_2, \{\mathrm{pa}(\pi_s)\}_{s<t}\big)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}\\ &\qquad = \mathbb{P}\big(\mathrm{pa}(\pi_t) = v_2, q(\pi_t) = v_1, \{\mathrm{pa}(\pi_s)\}_{s<t}\big)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}.\end{aligned}$$

Next, we observe that

$$\begin{aligned}&\mathbb{P}\left(\boldsymbol{G}_n, \{\mathrm{pa}(\pi_s)\}_{s>t} \middle| \mathrm{pa}(\pi_t) = v_1, q(\pi_t) = v_2,\ \{\mathrm{pa}(\pi_s)\}_{s<t}\right)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}\\ &= \mathbb{P}\left(\boldsymbol{G}_n \,\middle|\, \{\mathrm{pa}(\pi_s)\}_{s\in[n]\setminus\{t\}}, \begin{matrix}\mathrm{pa}(\pi_t) = v_1,\\ q(\pi_t) = v_2\end{matrix}\right)\mathbb{P}\big(\{\mathrm{pa}(\pi_s)\}_{s>t} \,\big|\, \mathrm{pa}(\pi_t) = v_1, \{\mathrm{pa}(\pi_s)\}_{s<t}\big)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}\end{aligned} \tag{16}$$

Here the future parent choices are independent of $q(\pi_t)$ given the forest history, since the noise graph is independent of the forest. For the first term of (16), we have

$$\begin{aligned}&\mathbb{P}\big(\boldsymbol{G}_n \mid \{\mathrm{pa}(\pi_s)\}_{s\neq t}, \mathrm{pa}(\pi_t) = v_1, q(\pi_t) = v_2\big)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}\\ &\qquad = \mathbb{P}\big(\boldsymbol{G}_n \mid \{\mathrm{pa}(\pi_s)\}_{s\neq t}, \mathrm{pa}(\pi_t) = v_2, q(\pi_t) = v_1\big)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}.\end{aligned}$$

Indeed, swapping the two candidate noise-edge indicators and the parent and selected endpoint preserves $\boldsymbol{G}_n$, the noise probability, and $1/h_t$. The endpoint events have equal probability, so the conditional likelihoods are equal.

Similarly, for the second term of (16), we also have

$$\begin{aligned}&\mathbb{P}\big(\{\mathrm{pa}(\pi_s)\}_{s>t} \,\big|\, \mathrm{pa}(\pi_t) = v_1, \{\mathrm{pa}(\pi_s)\}_{s<t}\big)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}\\ &\qquad = \mathbb{P}\big(\{\mathrm{pa}(\pi_s)\}_{s>t} \,\big|\, \mathrm{pa}(\pi_t) = v_2, \{\mathrm{pa}(\pi_s)\}_{s<t}\big)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}\end{aligned}$$

because, on the event $\mathcal{A}_t(\{v_1, v_2\})$, $\mathrm{pa}(\pi_s) \notin \{v_1, v_2\}$ for all $s > t$.

Returning to (14), we thus have that

$$\begin{aligned}&\mathbb{P}\left(\mathrm{pa}(\pi_t) = v_1 \,\middle|\, \big\{\mathrm{pa}(\pi_t), q(\pi_t)\big\} = \{v_1, v_2\},\ \{\mathrm{pa}(\pi_s)\}_{s\in[n]\setminus t}, \boldsymbol{G}_n\right)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}\\ &= \mathbb{P}\left(\mathrm{pa}(\pi_t) = v_2 \,\middle|\, \big\{\mathrm{pa}(\pi_t), q(\pi_t)\big\} = \{v_1, v_2\},\ \{\mathrm{pa}(\pi_s)\}_{s\in[n]\setminus t}, \boldsymbol{G}_n\right)\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}.\end{aligned}$$

Since these two probabilities sum to $\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}$, each must equal $\frac{1}{2}\mathbb{1}_{\mathcal{A}_t(\{v_1,v_2\})}$, which proves (e).

By Lemma 22, we already have $\mathbb{P}(\mathcal{A}_t) \geq 12\lambda^{1/2}$. Together with (12) and (13), we have

$$\begin{aligned}\mathbb{E}\left[\frac{d(\hat{\ell},\ell)}{n}\right] &\geq \frac{1}{36}\Big(\min_{t>\lceil n/2\rceil} \mathbb{P}\left(\hat{\ell}(\pi_t) \neq \bar{\sigma}\circ\ell(\pi_t)\right)\Big)^2 \\ &\geq \frac{1}{36}\Big(\min_{t>\lceil n/2\rceil} \mathbb{P}\left(\left\{\hat{\ell}(\pi_t) \neq \bar{\sigma}\circ\ell(\pi_t)\right\}\cap\mathcal{A}_t\right)\Big)^2 \\ &\geq \frac{1}{36}\left(6\sqrt{\lambda}\right)^2 = \lambda.\end{aligned}$$

The result follows as desired. ∎

Lemma 20. *Let* $(a_1,\ldots,a_n)$, $(b_1,\ldots,b_n) \in \{1,2\}^n$, *and* $\bar{\sigma} \in \arg\min_{\sigma\in S_2}\sum_{t=1}^{\lceil n/2\rceil}\mathbb{1}\{a_t \neq \sigma(b_t)\}$. *Then*

$$\min_{\sigma\in S_2}\frac{1}{n}\sum_{t=1}^{n}\mathbb{1}\{a_t\neq\sigma(b_t)\} \geq \min\left\{\frac{1}{n}\sum_{t=\lceil n/2\rceil+1}^{n}\mathbb{1}\{a_t\neq\bar{\sigma}(b_t)\},\,\frac{1}{4}\right\}.$$

Proof of Lemma 20. Let $m=\lceil n/2\rceil$, $\sigma^*\in\arg\min_{\sigma\in S_2}\sum_{t=1}^{n}\mathbb{1}\{a_t\neq\sigma(b_t)\}$. If $\sigma^*=\bar{\sigma}$, then

$$\sum_{t=1}^{n}\mathbb{1}\{a_t\neq\sigma^*(b_t)\} \geq \sum_{t=m+1}^{n}\mathbb{1}\{a_t\neq\bar{\sigma}(b_t)\},$$

and the conclusion of the lemma follows immediately. On the other hand, suppose $\sigma^*\neq\bar{\sigma}$. Since S_2 contains only two permutations, we have that $a_t\neq\sigma^*(b_t)$ if and only if $a_t=\bar{\sigma}(b_t)$ so that

$$\sum_{t=1}^{m}\mathbb{1}\{a_t\neq\sigma^*(b_t)\} = m-\sum_{t=1}^{m}\mathbb{1}\{a_t\neq\bar{\sigma}(b_t)\} \geq \frac{m}{2} \geq \frac{n}{4},$$

where the first inequality follows from the optimality of $\bar{\sigma}$. Therefore, we have

$$\frac{1}{n}\sum_{t=1}^{n}\mathbb{1}\{a_t\neq\sigma^*(b_t)\} \geq \min\left\{\frac{1}{n}\sum_{t=m+1}^{n}\mathbb{1}\{a_t\neq\bar{\sigma}(b_t)\},\frac{1}{4}\right\}.$$

The lemma follows as desired. ∎

Lemma 21. *Let* X *be a random variable taking value on* $[0,1]$. *Then*

$$\mathbb{E}\big[\min\{X,1/4\}\big] \geq \frac{1}{4}\min\big\{\mathbb{E}X,1/2\big\}\,\mathbb{E}X.$$

Proof of Lemma 21. Let $\mu:=\mathbb{E}X$ and set $a:=\mu/2$. Since $0\leq X\leq 1$,

$$\mu = \mathbb{E}\big[X\mathbb{1}\{X<a\}\big]+\mathbb{E}\big[X\mathbb{1}\{X\geq a\}\big] \leq a+\mathbb{P}(X\geq a).$$

Hence $\mathbb{P}(X\geq a)\geq\mu-a=\mu/2$. Therefore,

$$\mathbb{E}\big[\min\{X,1/4\}\big] \geq \min\{a,1/4\}\mathbb{P}(X\geq a) \geq \frac{\mu}{2}\min\left\{\frac{\mu}{2},\frac{1}{4}\right\} = \frac{1}{4}\mu\min\{\mu,\frac{1}{2}\}.$$

The desired bound follows. ∎

Lemma 22. *Let $\boldsymbol{G}_n \sim \mathrm{PF}(\alpha, \theta, \pi, 2)$ with $\theta \geq \frac{c}{n}$ for some constant $c > 0$. For any fixed $t \in \mathbb{N}$ satisfying $t \geq \lceil n/2 \rceil$, let $\mathcal{A}_t$ be defined in (11). Then there exists a constant $\lambda = \lambda(c, \alpha) > 0$ such that, for all sufficiently large n,*

$$\mathbb{P}(\mathcal{A}_t) \geq 12\lambda^{\frac{1}{2}}.$$

Proof of Lemma 22. To derive a lower bound for the probability of the event $\mathcal{A}_t$, we introduce a sequence of nested events $\Omega_3 \subset \Omega_2 \subset \Omega_1$, with $\Omega_3 \subset \mathcal{A}_t$, ensuring that all conditions in $\mathcal{A}_t$ defined in (11) are satisfied.

For each consecutive pair of events, we establish lower bounds for the corresponding conditional probabilities. By combining these bounds, we obtain

$$\mathbb{P}(\mathcal{A}_t) \geq \mathbb{P}(\Omega_3) = \mathbb{P}\left(\Omega_3 \mid \Omega_2\right) \mathbb{P}(\Omega_2) = \mathbb{P}\left(\Omega_3 \mid \Omega_2\right) \mathbb{P}\left(\Omega_2 \mid \Omega_1\right) \mathbb{P}(\Omega_1),$$

Then we bound $\mathbb{P}(\mathcal{A}_t)$ by bounding $\mathbb{P}\left(\Omega_3 \mid \Omega_2\right), \mathbb{P}\left(\Omega_2 \mid \Omega_1\right), \mathbb{P}(\Omega_1)$ separately. The precise definitions of the events Ω_i, $i \in [3]$, and their corresponding lower bounds will be introduced in the rest of the proof. Let

$$\Omega_1 := \left\{ \text{For each } i = 1, 2, \left|V\left(\boldsymbol{T}^i_{t-1}\right)\right| \geq \frac{n}{5}, \left|\left\{v : \deg_{\boldsymbol{T}^i_{t-1}}(v) = 1\right\}\right| \geq \frac{p(1)n}{10} \right\},$$

where $p(1)$ is defined in Lemma 36. A standard Pólya–urn argument implies that, as $t \to \infty$, the joint distribution of $\left(\frac{|V(\boldsymbol{T}^1_{t-1})|}{t-1}, \frac{|V(\boldsymbol{T}^2_{t-1})|}{t-1}\right)$ converges to Dirichlet$(\frac{2+2\alpha}{2+\alpha}, \frac{2+2\alpha}{2+\alpha})$. Consequently, by Lemma 36, the proportion of nodes with degree 1 in each tree converges to $p(1)$. Since we also have $t \geq n/2$, the two convergence statements imply that there exists $\lambda_1 > 0$ such that $\mathbb{P}(\Omega_1) > \lambda_1$ for all sufficiently large n.

For $i = 1, 2$, let $S_i := \{v \in V(\boldsymbol{T}^i_{t-1}) \setminus \pi_{1:4} : \deg_{\boldsymbol{T}^i_{t-1}}(v) = 1\}$. On Ω_1, we have $|S_i| \geq p(1)n/10 - 4 \geq p(1)n/20$ for all sufficiently large n. Define

$$\Omega_2 := \Omega_1 \cap \left\{\mathrm{pa}(\pi_t) \in S_1,\ q(\pi_t) \in S_2\right\}.$$

Since the vertices in $S_1 \cup S_2$ are noninitial degree-1 nodes, neither selected vertex has received a child before time t. By the attachment rule in (3), on Ω_1,

$$\mathbb{P}\left(\mathrm{pa}(\pi_t) \in S_1 \mid \boldsymbol{F}_{t-1}\right) = \frac{(1+\alpha)|S_1|}{2(t-3) + (t-1)\alpha} \geq \frac{p(1)(1+\alpha)}{20(2+\alpha)}.$$

By independence of the noise graph and the forest, and symmetry among the $t-1$ preceding vertices, on Ω_1,

$$\mathbb{P}\left(q(\pi_t) \in S_2 \mid \boldsymbol{F}_{t-1}, \mathrm{pa}(\pi_t)\right) = \frac{|S_2|}{t-1}\left\{1 - (1-\theta)^{t-1}\right\} \geq \frac{p(1)}{20}\left(1 - e^{-c/3}\right),$$

where we use $\theta \geq c/n$ and $t - 1 \geq n/3$ for all sufficiently large n. Consequently,

$$\mathbb{P}(\Omega_2 \mid \Omega_1) \geq \frac{p(1)^2(1+\alpha)}{400(2+\alpha)}\left(1 - e^{-c/3}\right) > 0.$$

Then we construct Ω_3 on the event of Ω_2 to capture the restriction of the graph for $\pi_{(t+1):n}$:

$$\Omega_3 := \Omega_2 \cap \left\{ \text{for each } i > t,\ \text{pa}\left(\pi_i\right) \notin \left\{ \text{pa}(\pi_t), q(\pi_t) \right\} \right\},$$

which represents the event that all nodes arriving after time t are not attached to $\text{pa}(\pi_t)$ or $q(\pi_t)$. Therefore, by the attachment rule in (3) and the fact that node $\deg_{\boldsymbol{F}_t}\left(\text{pa}(\pi_t)\right) = 2, \deg_{\boldsymbol{F}_t}\left(q(\pi_t)\right) = 1$ on the event Ω_2, we have, for all sufficiently large n,

$$\begin{aligned}
\mathbb{P}\left(\Omega_3 \mid \Omega_2\right) &= \prod_{i=t+1}^{n} \left(1 - \frac{2+\alpha+1+\alpha}{2(i-2-1)+(i-1)\alpha}\right) = \prod_{i=t+1}^{n} \left(1 - \frac{3+2\alpha}{2(i-3)+(i-1)\alpha}\right) \\
&\geq \left(1 - \frac{3+2\alpha}{2(n/2-3)+(n/2-1)\alpha}\right)^{n-t} \geq \left(1 - \frac{3+2\alpha}{n-6+(n/2-1)\alpha}\right)^{n-3} \\
&= \exp\left((n-3)\ln\left(1 - \frac{3+2\alpha}{n-6+(n/2-1)\alpha}\right)\right) \geq \exp\left(-\frac{(12+8\alpha)(n-3)}{2(n-6)+(n-2)\alpha}\right) \\
&= \exp\left(-\frac{(12+8\alpha)(n-3)}{(2+\alpha)(n-3)+\alpha-6}\right) \geq \exp\left(-\frac{24+16\alpha}{2+\alpha}\right).
\end{aligned}$$

The first inequality comes from the fact that $t \geq n/2$. In the third inequality, we apply the bound $\ln(1-x) \geq -2x$ for $0 \leq x \leq \frac{1}{2}$.

By the definition of $\mathcal{A}_t$ in (11) and the construction of the events Ω_i, $i \in [3]$, we observe that $\Omega_3 \subset \mathcal{A}_t$, since all the conditions required by $\mathcal{A}_t$ are satisfied on Ω_3. Combining the bounds above, define

$$\lambda^{\frac{1}{2}}(c,\alpha) := \frac{\lambda_1}{12}\exp\left(-\frac{24+16\alpha}{2+\alpha}\right)\frac{p(1)^2(1+\alpha)}{400(2+\alpha)}\left(1-e^{-c/3}\right) > 0.$$

Then, for all sufficiently large n,

$$\mathbb{P}(\mathcal{A}_t) \geq \mathbb{P}(\Omega_3 \mid \Omega_2)\mathbb{P}(\Omega_2 \mid \Omega_1)\mathbb{P}(\Omega_1) \geq \lambda_1 \frac{p(1)^2(1+\alpha)}{400(2+\alpha)}\left(1-e^{-c/3}\right)\exp\left(-\frac{24+16\alpha}{2+\alpha}\right) = 12\lambda^{\frac{1}{2}}.$$

The lemma follows as desired. ■

Appendix B. Appendix for Section 3

B.1 Choosing the Core Degree Threshold

We set the core degree threshold by matching the order of top degrees: We simulate the planted forest T times under the known parameters (n, K, α, θ) and a chosen minimum component proportion $1/H$ ($H \geq K$). In each run, we record the global degree rank of the weakest hub (i.e., the largest-degree node in each component; take the minimum across components) and take a high quantile (e.g., 95th percentile and above) of this rank, denoted r. On the observed graph, we sort vertices by degree and set the core degree threshold τ to the degree of the node at rank r. The procedure is summarized in Algorithm 3.

In the Section 5, we compute the 0.95 quantile of the maximum rank of the highest-degree vertex across planted components, using $H = 5$ and 200 Monte Carlo simulations. For $n = 1000$ and $\theta = 0.01$, this procedure yields thresholds of 5 for $\alpha = -0.5$ and 7 for both $\alpha = 0$ and $\alpha = 2$. As n increases, the corresponding 0.95 quantiles range from 5 to 9. For simplicity and consistency across all experimental settings, we therefore fix τ to be the degree of the 10th highest-degree vertex in the observed graph.

Remark 23. *Algorithm 3 assumes the graph parameters (α, θ) are available. If the graph parameters (α, θ) are assumed to be unknown, the noise level θ can still be easy to estimate, e.g. by $\hat{\theta} = \frac{|E(\boldsymbol{G}_n)|-(n-K)}{\binom{n}{2}-(n-K)}$. By contrast, α can be harder to estimate. A principled estimation procedure is given in the supplementary material of Crane and Xu (2024). In practice, one may use plug-in estimates or, for a conservative calibration in the simulations, a known upper bound for α.*

Remark 24. *With a chosen minimum proportion $1/H$. where $H \geq K$, the simulation setup we adopt here $\boldsymbol{G}_n^{(t)} \sim \mathrm{PF}(\alpha, \theta, \ell, \pi)$ with ℓ chosen as $|V^1| = \cdots = |V^{K-1}| = \frac{n}{H}$ and $|V^K| = n - \frac{n(K-1)}{H}$ represents the worst case: it maximizes imbalance across components and thus yields the largest (most conservative) values of $r^{(t)}$.*

Algorithm 3: Rank-based selection of core degree threshold τ

Input: Graph $\boldsymbol{G}_n = (V, E)$; community count $K \in \mathbb{N}$; minimum component proportion $1/H$; rank quantile $q = 0.95$; simulation times $B \in \mathbb{N}$.

Output: Core degree threshold τ.

for $t = 1$ **to** B **do**

  Simulate graph $\boldsymbol{G}_n^{(t)} \sim \mathrm{PF}(\alpha, \theta, \ell, \pi)$, where the community label function ℓ is chosen so that $|V^1| = \cdots = |V^{K-1}| = \frac{n}{H}$, and $|V^K| = n - \frac{n(K-1)}{H}$; find each planted tree's highest degree node; record $r^{(t)} = \max_k \mathrm{rank}^{(t)}(\text{highest degree node in comp } k)$.

end

$r := \mathrm{Quantile}_q\left(\{r^{(t)}\}_{t=1}^{B}\right)$;

return $\tau := \mathrm{rank}_r\left\{\deg(u) \mid u \in V(\boldsymbol{G}_n)\right\}$.

Algorithm 4: Simple Forest Community Detection Method

Input: Graph $\boldsymbol{G}_n = (V, E)$; core degree threshold $\tau_1 \in \mathbb{N}$; pruning degree threshold $\tau_2 \in \mathbb{N}$

Output: $\hat{K}$, core vertex set $\{V_1, \ldots, V_{\hat{K}}\}$, community label $\hat{\ell} : V \to \{1, \ldots, \hat{K}\}$

Remove $\left\{v \in V : \deg_{\boldsymbol{G}_n}(v) < \tau_2\right\}$ from $\boldsymbol{G}_n$ to get $\bar{\boldsymbol{G}}$;

Let $\hat{K} := \left|\left\{\boldsymbol{g} \in \mathcal{C}\left(\bar{\boldsymbol{G}}\right) : \deg_{\boldsymbol{G}_n}\left(V\left(\boldsymbol{g}\right)\right) \geq \tau_1\right\}\right|$;

Let $\left\{V_1, \ldots, V_{\hat{K}}\right\} := \left\{V\left(\boldsymbol{g}\right) : \boldsymbol{g} \in \mathcal{C}\left(\bar{\boldsymbol{G}}\right) \text{ and } \deg_{\boldsymbol{G}_n}\left(V\left(\boldsymbol{g}\right)\right) \geq \tau_1\right\}$;

Labeling: for $u \in V$ **do**

  $D_k(u) = \min_{v \in V_k} \mathrm{dist}_{\boldsymbol{G}_n}(u, v)$ for all $k \in [\hat{K}]$; $\hat{\ell}(u) := \mathrm{argmin}_{k \in [\hat{K}]} D_k(u)$ with uniform random tie–breaking if non-unique;

end

Remark 25. *A closely related alternative is to work with degrees directly instead of with degrees rank: in each simulation t, let $\tau^{(t)}$ be the smallest top–degree across components, and set $\tau := \mathrm{Quantile}_{1-q}\left(\{\tau^{(t)}\}_{t=1}^{T}\right)$.*

B.2 The Simple Degree Thresholding Algorithm

The toy example used to illustrate the selective pruning stage (Algorithm 1) is also used to illustrate the simple degree-thresholding algorithm in Section C.

Lemma 26. *Let $\hat{K}$ be the estimated number of clusters and $V_1^1, \ldots, V_{\hat{K}}^1$ be the core vertex sets outputted by Algorithm 4 with input parameters $\tau_1 \geq \tau_2 > 0$; assume $\hat{K} \geq 1$ and define $Q_{\max} := \min_{k \in [\hat{K}]} |V_k^1|$. Apply Algorithm 1 with input $K = \hat{K}$, $\tau = \tau_1$, and any $Q \in \mathbb{N}$ such that $Q \leq Q_{\max}$. Then the algorithm succeeds and returns $V_1^2, \ldots, V_{\hat{K}}^2$ such that, after suitable reindexing,*

$$V_i^1 \subseteq V_i^2 \quad \textit{for all } i \in \left[\hat{K}\right].$$

i.e., each core vertex set from Algorithm 4 is contained in the corresponding core vertex set returned by Algorithm 1. Moreover, D^* is strictly increasing over pruning iterations, and its final value is smaller than τ_2.

The following direct corollary of Lemma 26 will be useful for our theoretical analysis in Section F.

Corollary 27. *For any $\tau > 0$ and $Q, K \in \mathbb{N}$, let $\{V_1^2, \ldots, V_K^2\}$ and D^* be the output of Algorithm 1 with input τ, Q, and K. Let $0 < \tau' \leq \tau$ and define*

$$V_1^1, \ldots, V_M^1 = \mathcal{C}\big(\boldsymbol{G}_n \cap V^{\tau'}(\boldsymbol{G}_n)\big) \quad \textit{and}$$
$$\mathcal{K} = \left\{i \in [M] : \deg_{\boldsymbol{G}_n}(V_i^1) \geq \tau\right\}.$$

If $|\mathcal{K}| = K$ *and* $Q \le |V_i^1|$ *for all* $i \in \mathcal{K}$*, then Algorithm 1 succeeds,* $D^* < \tau'$*, and there exists a bijection* $\sigma : \mathcal{K} \to [K]$ *such that*

$$V_i^1 \subseteq V_{\sigma(i)}^2, \quad \forall i \in \mathcal{K}.$$

B.3 Model-based Recovery

By taking the random-community planted forest model specified in Definition 4 with a random arrival ordering π that is generated uniformly at random from the set of permutations of the n nodes, we can write down the joint probability

$$\mathbb{P}(\boldsymbol{G}_n, \boldsymbol{F}_n, \pi) = \mathbb{P}(\boldsymbol{G}_n \,|\, \boldsymbol{F}_n)\mathbb{P}(\boldsymbol{F}_n \,|\, \pi)\mathbb{P}(\pi)$$

where, on the right-hand side, the first term is determined by the Erdös–Rényi noise, the second term is determined by the affine preferential attachment mechanism, and the third term is $\frac{1}{n!}$.

For a fixed node u and a set of anchor components $\{V_1, \ldots, V_K\}$, we may then define the conditional probability that node u belongs to community k conditional on the observed graph $\boldsymbol{G}_n$ and the event that for every $i \in \{1, 2, \ldots, K\}$, each anchor component V_i contains the root node π_{2i-1} of community-tree $\boldsymbol{T}^i$:

$$\begin{aligned} p_k(u) &:= \mathbb{P}\big(\ell(u) = k \,\big|\, \pi_1 \in V_1, \ldots, \pi_{2K-1} \in V_K, \boldsymbol{G}_n\big) \\ &= \frac{\mathbb{P}\big(u \in V(\boldsymbol{T}^k),\, \pi_1 \in V_1, \ldots, \pi_{2K-1} \in V_K \,\big|\, \boldsymbol{G}_n\big)}{\mathbb{P}\big(\pi_1 \in V_1, \ldots, \pi_{2K-1} \in V_K \,\big|\, \boldsymbol{G}_n\big)}. \end{aligned}$$

The conditional probability $p_k(u)$ does not have a tractable analytic form but we can compute a Monte Carlo approximation using samples from the posterior distribution of the trees $\boldsymbol{T}^1, \ldots, \boldsymbol{T}^K$ and of the ordering π given the overall graph $\boldsymbol{G}_n$. Concretely, suppose we have M samples $\{(\boldsymbol{T}^{1,(j)}, \pi_1^{(j)}), \ldots, (\boldsymbol{T}^{K,(j)}, \pi_{2K-1}^{(j)})\}_{j=1}^M$ from the posterior distribution given $\boldsymbol{G}_n$, where $\boldsymbol{T}^{1,(j)}$ is the first community-tree and $\pi_1^{(j)}$ is the root node of the first community-tree in sample j. Then, we compute

$$\hat{p}_k(u) = \frac{\sum_{j=1}^M \mathbf{1}\left\{\pi_{2i-1}^{(j)} \in V_i \quad \text{for all } i \in [K]\right\} \mathbf{1}\left\{u \in \boldsymbol{T}^{k,(j)}\right\}}{\sum_{j=1}^M \mathbf{1}\left\{\pi_{2i-1}^{(j)} \in V_i \quad \text{for all } i \in [K]\right\}}.$$

We then assign node u the community label k which maximizes the posterior probability $\hat{p}_k(u)$. The overall procedure is summarized in Algorithm 5. It is worth noting that in addition to giving a hard community assignment $\hat{\ell}(u)$, we may also use the vector $(\hat{p}_1(u), \ldots, \hat{p}_K(u))$ to get a rough sense of community recovery uncertainty for node u.

To generate the posterior samples, we use the Gibbs sampler proposed in Crane and Xu (2024), with a slight modification to accommodate the fact that the model in Crane and Xu (2024) initializes each community-tree as a single node with a self-loop whereas we initialize each community-tree as two nodes connected by a single edge.

The toy example used to illustrate the selective pruning stage (Algorithm 1) is also used to illustrate the model-based recovery method in Section C.

Algorithm 5: Model Recovery

Input: Anchor connected components $V_1, \ldots, V_K$; M posterior samples $\{(\boldsymbol{T}^{1,(j)}, \pi_1^{(j)}), \ldots, (\boldsymbol{T}^{K,(j)}, \pi_{2K-1}^{(j)})\}_{j=1}^{M}$

Output: Soft assignment $\tilde{\ell} : V \to \Delta^{K-1}$ with $\tilde{\ell}(u) = \big(\hat{p}_u(1), \ldots, \hat{p}_u(K)\big)$; hard labels $\hat{\ell} : V \to [K]$

for $u \in V, k \in [K]$ **do**

Set

$$\hat{p}_k(u) = \frac{\sum_{j=1}^{M} \mathbf{1}\left\{\pi_{2i-1}^{(j)} \in V_i \quad \text{for all } i \in [K]\right\} \mathbf{1}\left\{u \in \boldsymbol{T}^{k,(j)}\right\}}{\sum_{j=1}^{M} \mathbf{1}\left\{\pi_{2i-1}^{(j)} \in V_i \quad \text{for all } i \in [K]\right\}}.$$

Set $\tilde{\ell}(u) := (\hat{p}_1(u), \ldots, \hat{p}_K(u))$ and $\hat{\ell}(u) := \arg\max_{k \in [K]} \hat{p}_k(u)$ (break ties uniformly at random).

end

return $\tilde{\ell}$, $\hat{\ell}$;

B.4 Proofs for Section 3

Proof of Lemma 26. Suppose the pruning procedure in Algorithm 1 produce a graph sequence such that $\boldsymbol{G}^{(T)} \subset \boldsymbol{G}^{(T-1)} \cdots \subset \boldsymbol{G}^{(0)} = \boldsymbol{G}_n$, where $\boldsymbol{G}^{(t)}$ denotes the graph after t-th pruning iteration. Denote the degree removed in pruning iteration t by $D^{(t)}$. If $T \geq 1$, the final value is $D^* = D^{(T)}$; otherwise $D^* = 0$. Let $\tilde{V}'^{(t)}, \tilde{V}''^{(t)}$ be the cycle-based and split-based target set of $\boldsymbol{G}^{(t)}$ from Algorithm 1 at iteration t, respectively. In the last iteration T, it must be either that we return $\hat{K}$ vertex sets $\{V_1^2, \ldots, V_{\hat{K}}^2\}$ such that $|V_k^2| \geq Q$ and $\deg_{\boldsymbol{G}_n}(V_k^2) \geq \tau_1$ for all $k \in [\hat{K}]$ or that $\tilde{V}'^{(T)} \cup \tilde{V}''^{(T)} = \emptyset$ and Algorithm 1 fails.

We first claim, for every $k \in [\hat{K}]$, that $V_k^1 \subset V(\boldsymbol{G}^{(t)})$ for every $t \in [T]$. We will then finish the proof by showing that failure is impossible and that each V_k^1 is contained in a distinct $V_{k'}^2$.

For the first claim, we use proof by induction on $t \leq T$. The induction hypothesis is $\cup_{i=1}^{\hat{K}} V_i^1 \subset V(\boldsymbol{G}^{(t)})$, which holds trivially for $t = 0$. Assume the hypothesis holds for some $t < T$.

We observe that, since Algorithm 1 did not terminate at step t, it must be that there exists $k \neq k'$ such that V_k^1 and $V_{k'}^1$ are connected in $\boldsymbol{G}^{(t)}$. Indeed, if $V_1^1, \ldots, V_{\hat{K}}^1$ are all disconnected in $\boldsymbol{G}^{(t)}$, then, writing $V_1^{(t)}, \ldots, V_{\hat{K}}^{(t)}$ as the connected components of $\boldsymbol{G}^{(t)}$ containing $V_1^1, \ldots, V_{\hat{K}}^1$, we have that $|V_k^{(t)}| \geq |V_k^1| \geq Q_{\max} \geq Q$ and that $\deg_{\boldsymbol{G}_n}(V_k^{(t)}) \geq \deg_{\boldsymbol{G}_n}(V_k^1) \geq \tau_1$ and thus, the condition in line 4 of Algorithm 1 would be satisfied and the algorithm would terminate, creating a contradiction.

Now we show that there exists a node $u \in \tilde{V}^{(t)} = \tilde{V}'^{(t)} \cup \tilde{V}''^{(t)}$ such that $\deg_{\boldsymbol{G}_n}(u) < \min\{\deg_{\boldsymbol{G}_n}(v) : v \in \cup_{k=1}^{\hat{K}} V_k^1\}$. To see this, consider any chain of nodes $\{u_1, \ldots, u_m\}$ connecting V_k^1 and $V_{k'}^1$ in $\boldsymbol{G}^{(t)}$. By line 1 of Algorithm 4, V_k^1 and $V_{k'}^1$ are disconnected after removing all nodes $u \in \boldsymbol{G}_n$ such that $\deg_{\boldsymbol{G}_n}(u) < \tau_2$. Thus, it must be that there is a node u_i on the chain has degree $\deg_{\boldsymbol{G}_n}(u_i) < \tau_2$. On the other hand, every node in $\cup_{k=1}^{\hat{K}} V_k^1$

must have degree (with respect to $\boldsymbol{G}_n$) at least τ_2 by construction. Moreover, we note that $u_i \in \tilde{V}^{(t)}$ because either u_i is in a cycle (and hence in $\tilde{V}'^{(t)}$) or if it is not in a cycle, then any path from a node in V_k^1 to a node in $V_{k'}^1$ must pass through u_i, and hence, removing u_i would disconnect V_k^1 and $V_{k'}^1$ in $\boldsymbol{G}^{(t)}$ so that u_i must be in $\tilde{V}''^{(t)}$. Thus,

$$\min_{v \in \tilde{V}^{(t)}} \deg_{\boldsymbol{G}_n}(v) < \tau_2, \tag{17}$$

so that the removed nodes in $(t+1)$th pruning iteration have degree smaller than τ_2 and cannot belong to $\cup_{i=1}^{\hat{K}} V_i^1$. This completes the induction and proves the claim that $\cup_{k=1}^{\hat{K}} V_k^1 \subset V(\boldsymbol{G}^{(t)})$ for each $t \in [T]$.

Suppose for contradiction that the algorithm fails. Then it must be that there exists $k \neq k'$ such that V_k^1 and $V_{k'}^1$ are connected in $\boldsymbol{G}^{(T)}$ and that $\tilde{V}^{(T)} = \emptyset$. Indeed, using the same argument as before, if $V_1^1, \ldots, V_{\hat{K}}^1$ are all disconnected in $\boldsymbol{G}^{(T)}$, then the connected components of $\boldsymbol{G}^{(T)}$ containing $V_1^1, \ldots, V_{\hat{K}}^1$ would have size at least $Q_{\max}$ and maximum degree at least τ_1 and satisfy condition in line 4 of Algorithm 1. However, if there exists a pair V_k^1 and $V_{k'}^1$ that are connected in $\boldsymbol{G}^{(T)}$, then at least one of the vertex on the chain connected V_k^1 and $V_{k'}^1$ would be in $\tilde{V}^{(T)}$ and thus creating a contradiction.

Since $\tau_1 \geq \tau_2$, every vertex of degree at least τ_1 belongs to some V_i^1. Each V_i^1 remains connected in $\boldsymbol{G}^{(T)}$, so every component containing such a vertex contains an entire V_i^1 and has size at least $Q_{\max} \geq Q$. Thus there are at most $\hat{K}$ qualifying components. The stopping condition gives at least $\hat{K}$, so each contains exactly one V_i^1. These are the components returned by Algorithm 1, and, after reindexing,

$$V_i^1 \subseteq V_i^2, \qquad i \in [\hat{K}].$$

We note that if $T \geq 1$, equation (17) at $t = T-1$ gives $D^* = D^{(T)} < \tau_2$. If $T = 0$, then $D^* = 0 < \tau_2$.

Now we show that $D^{(t)} > D^{(t-1)}$ for all $t \in \{2, 3, \ldots, T\}$. We note that the following must be true:

$$\tilde{V}'^{(t)} \subseteq \tilde{V}'^{(t-1)}, \quad \tilde{V}''^{(t)} \subseteq \tilde{V}'^{(t-1)} \cup \tilde{V}''^{(t-1)}, \tag{18}$$

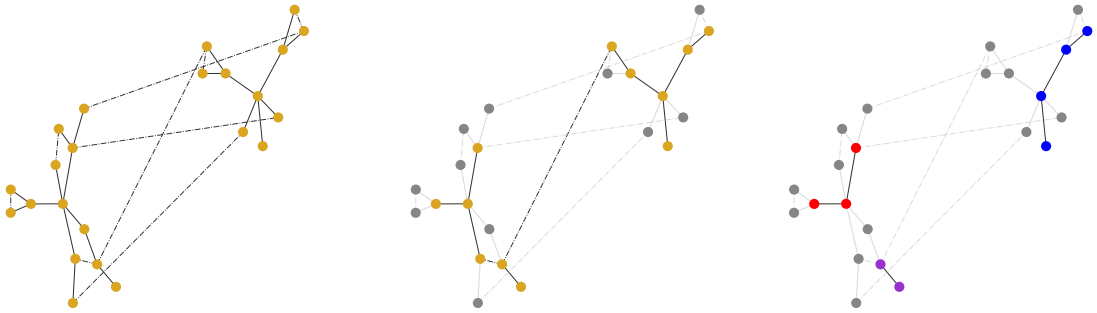
where the first inclusion holds because removing nodes from a graph cannot create new cycles. The second inclusion holds because if a node u in $\tilde{V}''^{(t)}$, then

$$\begin{aligned} |\{\boldsymbol{g} \in \mathcal{C}(\boldsymbol{G}^{(t)} \backslash \{u\}) \,:\, |V(\boldsymbol{g})| \geq Q, \deg_{\boldsymbol{G}_n}(V(\boldsymbol{g})) \geq \tau_1\}| > \\ |\{\boldsymbol{g} \in \mathcal{C}(\boldsymbol{G}^{(t)}) \,:\, |V(\boldsymbol{g})| \geq Q, \deg_{\boldsymbol{G}_n}(V(\boldsymbol{g})) \geq \tau_1\}|. \end{aligned} \tag{19}$$

Since $\boldsymbol{G}^{(t)}$ is a subgraph of $\boldsymbol{G}^{(t-1)}$, it must be that either u belongs in a cycle in $\boldsymbol{G}^{(t-1)}$ (so that $u \in \tilde{V}'^{(t-1)}$) or, if u does not belong in any cycle in $\boldsymbol{G}^{(t-1)}$, u must satisfy (19) with $\boldsymbol{G}^{(t-1)}$ in place of $\boldsymbol{G}^{(t)}$. Since we remove the nodes in $\tilde{V}'^{(t-1)} \cup \tilde{V}''^{(t-1)}$ with the smallest degree, we use (18) to conclude that for $1 \leq t < T$

$$D^{(t)} = \min_{v \in \tilde{V}'^{(t-1)} \cup \tilde{V}''^{(t-1)}} \deg_{\boldsymbol{G}_n}(v) < \min_{v \in \tilde{V}'^{(t)} \cup \tilde{V}''^{(t)}} \deg_{\boldsymbol{G}_n}(v) = D^{(t+1)}.$$

The lemma follows as desired. ∎



(a) Origin Graph (b) Graph after Iteration 1 (c) Graph after Iteration 2

Figure 9: Selective Pruning via Algorithm 1 on a toy example with $\tau = 5$, $K = 2$, and $Q = 3$. **(a)** Initial graph containing two planted tree structures. **(b)** Graph after the first pruning iteration. **(c)** Final graph after the second pruning iteration; red and blue anchors consist of nodes with degree ≥ 5 and size ≥ 3.

Appendix C. A Toy Example to Illustrate SPAR

C.1 Illustration of the Selective pruning stage

To illustrate the procedure of Algorithm 1, we consider a toy example shown in Figure 9a. The graph contains two planted trees of sizes 13 and 10, respectively, with 4 between-tree noise edges and 5 within-tree noise edges (drawn as dotted lines).

In the first iteration, all degree-2 nodes are removed, as they belong to the cycle-based candidate removal set $\tilde{V}'$ identified by Algorithm 1, yielding the graph in Figure 9b. In contrast, the split-based candidate removal set $\tilde{V}''$ selected by Algorithm 1 is empty in this iteration.

In the second iteration, no cycles remain in the graph, so the cycle-based candidate removal set $\tilde{V}'$ is empty. The split-based candidate removal set $\tilde{V}''$ contains 4 nodes, including 3 nodes of degree 3. Removing these three degree-3 nodes produces the graph shown in Figure 9c. The resulting red and blue anchors are the two components that satisfy both the minimum degree ($\tau = 5$) and minimum size ($Q = 3$) requirements.

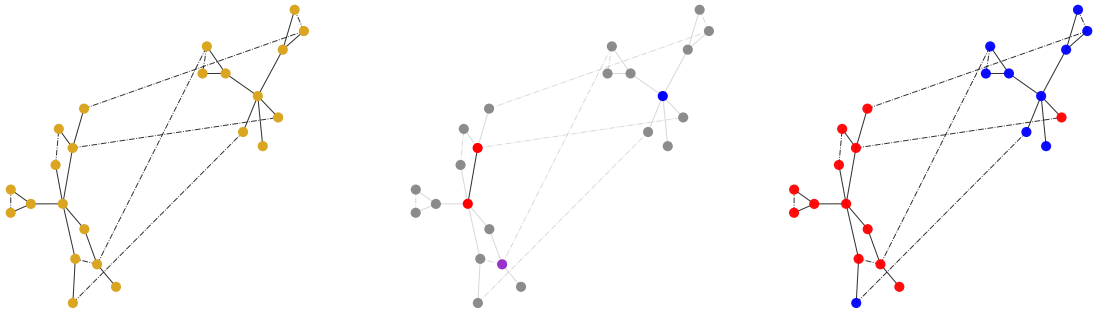
C.2 Illustration of the Simple Degree Thresholding Algorithm

We analyze the same synthetic example graph from Figure 9a to illustrate Algorithm 4. We set the core threshold to $\tau_1 = 5$ and the pruning threshold to $\tau_2 = 4$. After degree-based pruning (Figure 10b), the graph decomposes into $M = 3$ isolated cores. Among these, only two cores satisfy the highest-degree requirement of $\tau_1 = 5$, which includes the most "center" nodes in each planted tree.

The final distance recovery outcome, presented in Figure 10c shows that the procedure is effective for this example: only 2 nodes are misclassified in the entire graph.

C.3 Illustration of the Model-based Recovery

We evaluate the performance of Algorithm 5 using the anchors obtained from the example in Figure 9c. To generate posterior samples, we draw 500 posterior samples from the RC-

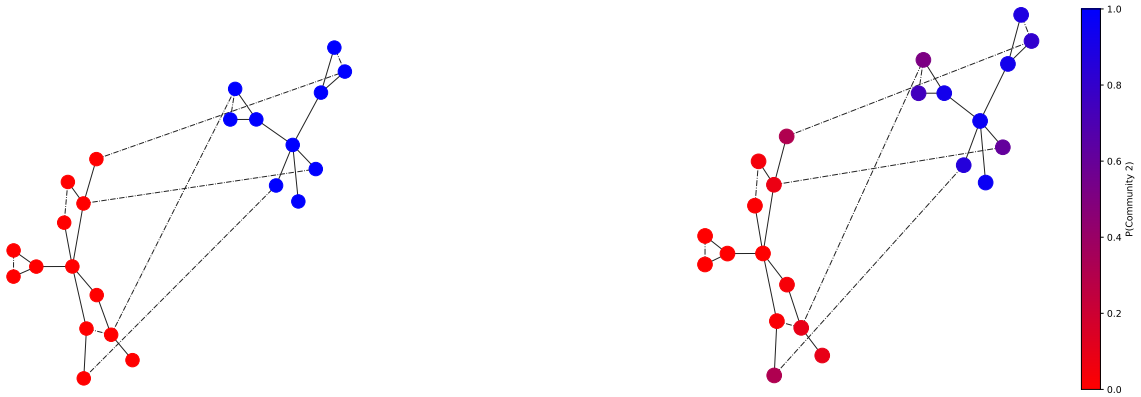


(a) Origin Graph (b) Graph after Degree Pruning (c) Final Classification

Figure 10: Forest community detection via Algorithm 4 on a toy example with thresholds $\tau_1 = 5$ and $\tau_2 = 4$. **(a)** Original graph containing two planted trees. **(b)** Three cores obtained after degree-based pruning; red/blue cores contain nodes of degree at least 5. **(c)** Final community assignment using red/blue cores for distance-based recovery.

PF model under the Gibbs sampling framework as described in Crane and Xu (2024), with parameters $\alpha = 0$, and $K = 2$. Among these,201 samples have roots falling within the identified anchors, indicating a moderate alignment between the core extraction method and the sampling mechanism. The hard classification recovers all 23 planted memberships correctly.

Figure 11a presents hard classification results, where each node is assigned to the class with the higher posterior probability. The soft classification results in Figure 11b encode the full posterior probability distribution into a color mapping, providing detailed insight into which nodes are confidently classified versus those with higher uncertainty. Boundary nodes connected by noisy edge to the opposite cluster have greater classification ambiguity, which is consistent with our intuition to prove Theorem 6.



(a) Hard classification results

(b) Soft classification results

Figure 11: Forest community detection results using Algorithm 5 on the toy example with input anchors from Figure 9c. **(a)** Hard classification outcomes, where each node is assigned to the community with the highest posterior probability: $\hat{\ell}(u) = \arg\max_{i\in[2]} \hat{p}_i(u)$. **(b)** Soft classification results, visualized through node colors corresponding to posterior probability vectors $\tilde{\ell}(u) = (\hat{p}_1(u), \hat{p}_2(u))$, with the colorbar indicating probability values.

Appendix D. Appendix for Section 5

D.1 Additional Details for Implementation of Monte Carlo Method

We provide additional comparisons across the simulation settings, including errors for depth-1 vertices, the 50 highest-degree vertices, and arrival-time and degree-weighted errors.

Here we provide some additional details of how we use Monte Carlo method in Section 5: we use the anchor components identified by SPAR to align the sampled communities. Each Gibbs chain produces 200 samples after 30 burn-in iterations. We retain samples with exactly one root in each anchor component and combine them across up to five chains, stopping once 200 valid samples are available. Each vertex is assigned to the community with the largest estimated posterior membership probability. When fewer than 50 valid samples are obtained or the posterior classification has fewer than K communities, we use the Distance Recovery labels.

D.2 Additional Numerical Results for Multiple Unbalanced Trees Setting

We use the simulation setup in Section 5, with three planted trees of sizes $n_1 = 3n_3$, $n_2 = 2n_3$, and $n_3 \in \{200, 400, 600, 800, 1000\}$. The noise probability is $\theta = 0.01(1200/n)^\gamma$, where $n = n_1 + n_2 + n_3$ and $\gamma \in \{1, 3/4, 1/2\}$.

We set τ to the degree of the 15th highest-degree vertex in the observed graph and retain the remaining distance-recovery parameters from the balanced setting. This choice follows the same calibration procedure: the estimated 0.95 quantiles range from 9 to 15 over the settings considered.

The figures below report the results for SPAR with distance recovery. The patterns are similar to those in the balanced two-tree setting, with generally higher misclassification rates when the trees have unequal sizes and the number of communities increases.

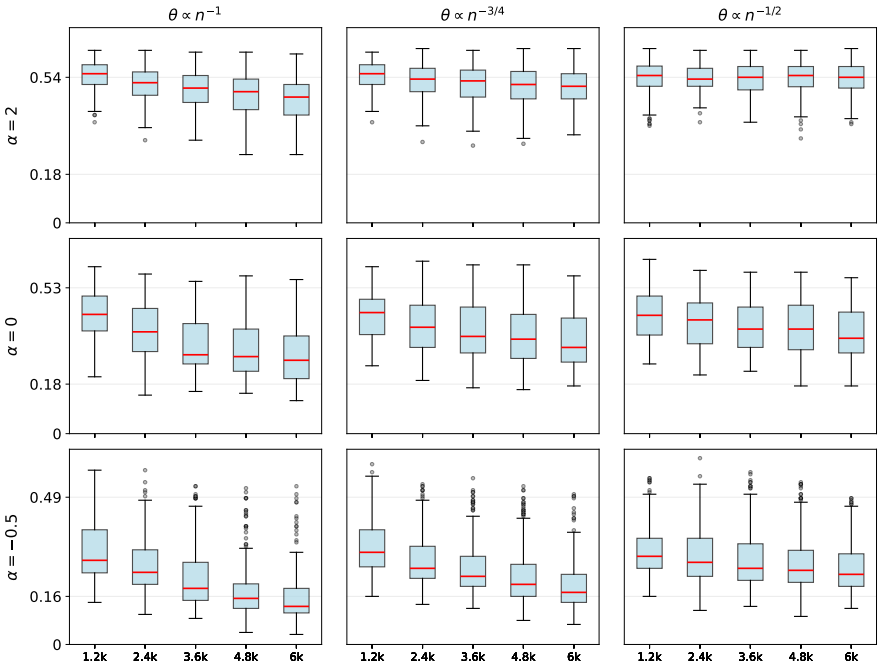


Figure 12: Boxplots of misclassification rates for the first 50 arriving vertices in each tree, computed over 200 simulated networks.

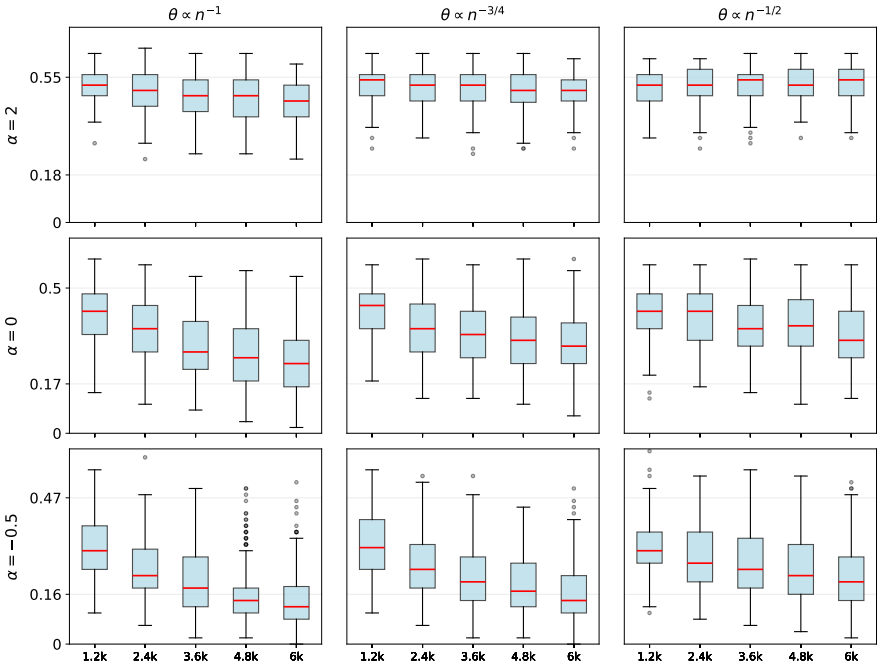


Figure 13: Boxplots of misclassification rates for the 50 highest degree vertices in the network, computed over 200 simulated networks.

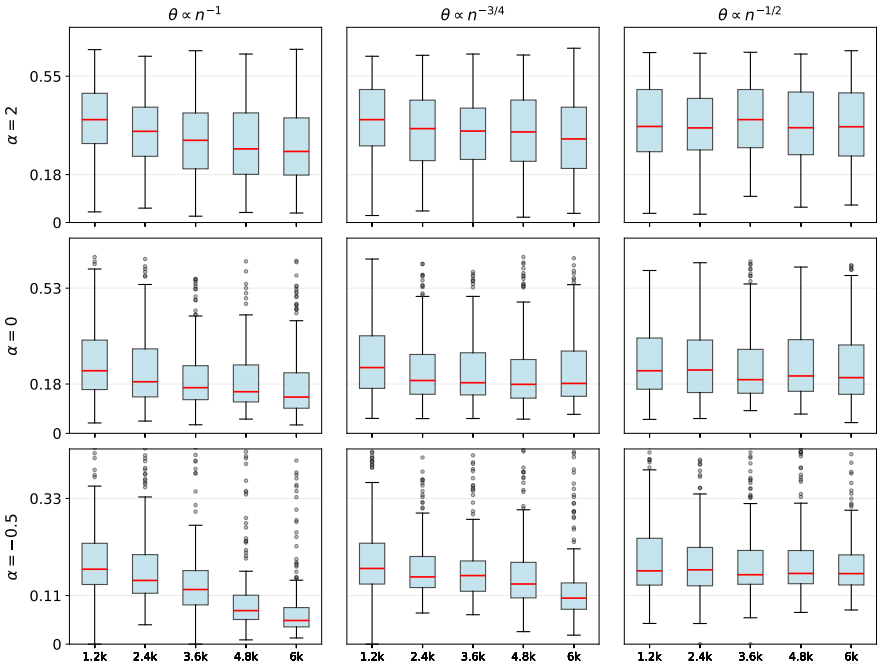


Figure 14: Boxplots of misclassification rates for the layer-1 vertices in the network, computed over 200 simulated networks.

Appendix E. Appendix for Section 6

E.1 Model-based Recovery Results for the Co-authorship Dataset

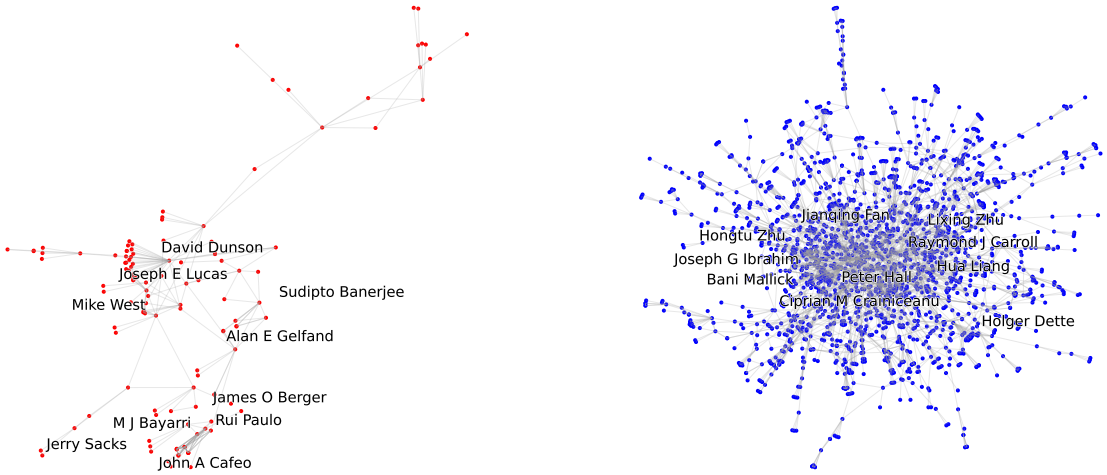
We report the model recovery results from Algorithm 5 in Figure 15, based on 1000 Monte Carlo samples generated from the implementation of Crane and Xu (2024). Among these samples, 572 root pairs are assigned to the two selected cores identified in Figure 5. For both figures, only the largest connected components of the recovered clusters are shown. The communities obtained from the two methods are highly consistent, with a 98.5% agreement in node classification.

Appendix F. Appendix for Section 4

F.1 Additional Discussion of Results in Section 4

Our layer-2 recovery result (10) currently pertains only to the linear preferential attachment case ($\alpha = 0$). The main obstacle to extending this to a general α is that we do not have bounds on the number of layer-2 and layer-3 nodes, i.e. $|\mathcal{L}_2(\boldsymbol{F}_n)|$ and $|\mathcal{L}_3(\boldsymbol{F}_n)|$, for a general value of α. Our proof technique for analyzing the number of layer-s nodes in the $\alpha = 0$ setting unfortunately does not directly generalize beyond linear preferential attachment. Nevertheless, we conjecture that $|\mathcal{L}_s(\boldsymbol{G}_n)|$ is of order $n^{\frac{1}{2+\alpha}} \log^{s-1} n$ for any $\alpha > -1$ in which case our bound (10) can be generalized.

Bounding the misclustering error rate for layer-s nodes where $s \geq 3$ becomes substantially more challenging, even in the linear preferential attachment case ($\alpha = 0$). The main



(a) Community 1

(b) Community 2

Figure 15: Communities obtained by the Model-based recovery procedure in Algorithm 5, based on 1000 Monte Carlo samples and using the anchor components identified in Figure 5.

difficulty is that the misclassification errors accumulate across layers. Indeed, if a parent node at layer s is misclassified, then we cannot meaningfully bound the misclassification error of all of its children at layer $s+1$. Our strong recovery guarantee for the layer-1 nodes allows us to obtain weaker but still non-trivial bounds for the layer-2 nodes, but these are not strong enough to yield a similar guarantee for the layer-3 nodes and beyond.

Our theoretical results are developed for the affine preferential–attachment regime, where the existing random graph theory is sufficiently well developed to support rigorous analysis. We expect these results to remain valid under mild deviations from this setting, such as when late–arriving nodes attach with a random but bounded number of edges, or when the attachment mechanism strengthens the advantage of early arrivals (e.g., exponential attachment). However, extending the theory beyond the affine case would require new technical tools and is left for future work.

F.2 Useful Intermediate Results

We first introduce several quantities that will be used repeatedly throughout the proof: For $L \in \mathbb{N}$, $\varepsilon > 0$ and $c \in (0, 1]$, we define

$$\begin{aligned} \tilde{L}(\varepsilon, c) &:= L\left(\frac{\varepsilon}{3H}, c\gamma\left(\frac{\varepsilon}{3H}, 1\right)\right), \\ \tilde{\gamma}(\varepsilon, L, c) &:= \gamma\left(\frac{\varepsilon}{3H},\ \tilde{L}(\varepsilon, c) \vee L\right). \end{aligned}$$

Here, the functions $\gamma(\varepsilon, L)$ and $L(\varepsilon, \gamma)$ are defined in Lemma 28.

F.3 Proof of Theorem 13

Proof. Let $\{V_1^2, \dots, V_K^2\}$ and D^* be the output on success of Algorithm 1 (and thus the input of Algorithm 2) with $\tau := C_1 n^{\frac{1}{2+\alpha}} = \gamma\left(\frac{\varepsilon}{3H}, 1\right) n^{\frac{1}{2+\alpha}}$. Let $\hat{\ell}$ be the estimated community labels outputted by Algorithm 2 with $0 \le \tau' \le D^*$ on success.

Let us define

$$V_1^1, \dots, V_M^1 = \mathcal{C}\big(\boldsymbol{G}_n \cap V^{\tilde{\gamma}(\varepsilon, L\vee Q, 1), \alpha}(\boldsymbol{G}_n)\big) \text{ and}$$
$$\mathcal{K} = \left\{ i \in [M] \,:\, \deg_{\boldsymbol{G}_n}(V_i^1) \ge \gamma\big(\frac{\varepsilon}{3H}, 1\big) n^{\frac{1}{2+\alpha}} \right\}.$$

Without loss of generality, we may suppose that $\mathcal{K} = \{1, 2, \dots, |\mathcal{K}|\}$. Define then the event

$$\Omega := \left\{ |\mathcal{K}| = K \text{ and } \exists \sigma \in S_K \text{ such that } \pi_{1:(L\vee Q)}(\boldsymbol{T}^{\sigma(i)}) \subset V_i^1, \text{ for all } i \in [K] \right\}.$$

We claim that $d_{\cup_{i=1}^K \pi_{1:(L\vee Q)}(\boldsymbol{T}^i)}(\hat{\ell}, \ell) = 0$ on the event Ω. To prove this, we work on the event Ω for the remainder of the proof. Since $|\mathcal{K}| = K$ and $Q \le |\pi_{1:(L\vee Q)}(\boldsymbol{T}^{\sigma(i)})| \le |V_i^1|$ for all $i \in \mathcal{K}$, we may conclude using Corollary 27 that there exists $\sigma' \in S_K$ such that

$$\forall i \in [K], \quad V_i^1 \subset V_{\sigma'(i)}^2. \tag{20}$$

Moreover, Corollary 27 gives $\tau' \le D^* < \tilde{\gamma}(\varepsilon, L \vee Q, 1) n^{\frac{1}{2+\alpha}}$, including the case $D^* = 0$. Therefore, via the definition of V_i^1, we have that

$$\forall i \in [K], \quad V_i^1 \subset V^{\tilde{\gamma}(\varepsilon, L\vee Q, 1), \alpha}(\boldsymbol{G}_n) \subset V^{\tau'}(\boldsymbol{G}_n). \tag{21}$$

From the definition of Algorithm 2, for any $i \in [K]$, we assign $\hat{\ell}(u) = i$ for all $u \in V_i^2 \cap V^{\tau'}(\boldsymbol{G}_n)$. Combining this with the definition of event Ω, with (20), and with (21), we have that

$$\forall u \in \pi_{1:(L\vee Q)}(\boldsymbol{T}^i), \quad \hat{\ell}(u) = \sigma'(\sigma^{-1}(i)) \text{ and } \ell(u) = i.$$

It holds therefore that

$$d_{\cup_{i=1}^K \pi_{1:L}(\boldsymbol{T}^i)}(\hat{\ell}, \ell) = \min_{\sigma'' \in S_K} d^{\mathrm{Ham}}_{\cup_{i=1}^K \pi_{1:L}(\boldsymbol{T}^i)}\left(\hat{\ell}, \sigma'' \circ \ell\right) \le d^{\mathrm{Ham}}_{\cup_{i=1}^K \pi_{1:L}(\boldsymbol{T}^i)}\left(\hat{\ell}, \sigma' \circ \sigma^{-1} \circ \ell\right) = 0.$$

It remains only to bound the probability of the event Ω. By Lemma 30, we have $\mathbb{P}(\Omega) \ge 1 - \varepsilon + \eta_n$ where $\eta_n = o(1)$ depends only on $\varepsilon, L, Q, H, \alpha, \delta$. The Theorem thus follows as desired.

■

Lemma 28. *For $\boldsymbol{G}_n \sim \mathrm{PF}(\alpha, \theta, \ell, \pi)$, under Assumption 11 and 12. For any $L \in \mathbb{N}$, and $\epsilon > 0$, there exists $\gamma(\varepsilon, L) \in (0, \infty)$ such that, for any $i \in [K]$,*

$$\liminf_{n\to\infty} \mathbb{P}\left\{ \pi_{1:L}(\boldsymbol{T}^i) \subset V_{\boldsymbol{G}_n}^{\gamma(\varepsilon, L), \alpha}(\boldsymbol{T}^i) \right\} \ge 1 - \varepsilon. \tag{22}$$

Similarly, for any $C > 0$, and $\epsilon > 0$, there exist an integer $L(\varepsilon, C)$ such that, for any $i \in [K]$,

$$\liminf_{n\to\infty} \mathbb{P}\left\{ V_{\boldsymbol{G}_n}^{C, \alpha}(\boldsymbol{T}^i) \subset \pi_{1:L(\varepsilon, C)}(\boldsymbol{T}^i) \right\} \ge 1 - \varepsilon. \tag{23}$$

Proof. We begin by proving the first claim (22). Let $i \in [K]$ and let $\gamma(\varepsilon, L)$ be a positive number whose definition we defer to later in (26). Recall the definition $\pi_{1:L}(\boldsymbol{T}^i) := \{\pi_t(\boldsymbol{T}^i) \mid t \in [L]\}$ from Definition 2 and $V_{\boldsymbol{G}_n}^{\gamma(\varepsilon,L),\alpha}(\boldsymbol{T}^i) := \{u \in V(\boldsymbol{T}^i) \mid \deg_{\boldsymbol{G}_n}(u) \geq \gamma(\varepsilon, L)\, n^{\frac{1}{2+\alpha}}\}$ from (4), we have

$$\mathbb{P}\left\{\pi_{1:L}(\boldsymbol{T}^i) \subset V_{\boldsymbol{G}_n}^{\gamma(\varepsilon,L),\alpha}\left(\boldsymbol{T}^i\right)\right\} \geq \mathbb{P}\left(\bigcap_{j=1}^{L}\left\{n^{-\frac{1}{2+\alpha}} \deg_{\boldsymbol{G}_n}(\pi_j(\boldsymbol{T}^i)) > \gamma(\varepsilon, L)\right\}\right). \tag{24}$$

Let $(Y_1, Y_2, \ldots)$ be a sequence of random variables specified in Lemma 40. By (24), the fact that $\frac{n}{n_i} \leq H$, and by applying Lemma 40 in conjunction with the Portmanteau lemma on the set $\prod_{j=1}^{L}\left(H^{\frac{1}{2+\alpha}}\gamma(\varepsilon, L), \infty\right) \times \prod_{j=L+1}^{\infty}\mathbb{R} \subset \mathbb{R}^{\mathbb{N}}$ which is open with respect to the ℓ_q topology for any $q \geq 2$, it follows that

$$\begin{aligned}
\liminf_{n\to\infty} \mathbb{P}\left(\pi_{1:L}\left(\boldsymbol{T}^i\right) \subset V_{\boldsymbol{G}_n}^{\gamma(\varepsilon,L),\alpha}\left(\boldsymbol{T}^i\right)\right) &\geq \liminf_{n\to\infty} \mathbb{P}\left(\bigcap_{j=1}^{L}\left\{n^{-\frac{1}{2+\alpha}} \deg_{\boldsymbol{G}_n}(\pi_j(\boldsymbol{T}^i)) > \gamma(\varepsilon, L)\right\}\right) \\
&\geq \liminf_{n\to\infty} \mathbb{P}\left(\bigcap_{j=1}^{L}\left\{n_i^{-\frac{1}{2+\alpha}} \deg_{\boldsymbol{G}_n}(\pi_j(\boldsymbol{T}^i)) > H^{\frac{1}{2+\alpha}}\gamma(\varepsilon, L)\right\}\right) \\
&\geq \mathbb{P}\left(\bigcap_{j=1}^{L}\left\{Y_{j,\alpha} > H^{\frac{1}{2+\alpha}}\gamma(\varepsilon, L)\right\}\right) \\
&= 1 - \mathbb{P}\left(\bigcup_{j=1}^{L}\left\{Y_{j,\alpha} \leq H^{\frac{1}{2+\alpha}}\gamma(\varepsilon, L)\right\}\right) \\
&\geq 1 - \sum_{j=1}^{L}\mathbb{P}\left(Y_{j,\alpha} \leq H^{\frac{1}{2+\alpha}}\gamma(\varepsilon, L)\right).
\end{aligned} \tag{25}$$

To bound the final term $\sum_{j=1}^{L}\mathbb{P}\left(Y_{j,\alpha} \leq H^{\frac{1}{2+\alpha}}\gamma(\varepsilon, L)\right)$. we note that, by Lemma 40, each random variable $Y_{j,\alpha}$ has a density $q_j(\cdot)$ supported on $[0, \infty)$ with respect to the Lebesgue measure. Thus, the function $t \mapsto \sum_{j=1}^{L}\mathbb{P}(Y_{j,\alpha} \leq t)$ is continuous on $[0, \infty)$ and hence surjective on its image $[0, L]$. Therefore, for any $\varepsilon > 0$, we choose $\gamma(\varepsilon, L)$ to be the unique positive number such that

$$\sum_{j=1}^{L}\mathbb{P}\left(Y_{j,\alpha} \leq H^{\frac{1}{2+\alpha}}\gamma(\varepsilon, L)\right) = \varepsilon. \tag{26}$$

Putting this together with inequality (25), we conclude that

$$\liminf_{n\to\infty} \mathbb{P}\left(\pi_{1:L}(\boldsymbol{T}^i) \subset V_{\boldsymbol{G}_n}^{\gamma(\varepsilon,L),\alpha}(\boldsymbol{T}^i)\right) \geq 1 - \sum_{j=1}^{L}\mathbb{P}\left(Y_{j,\alpha} \leq H^{\frac{1}{2+\alpha}}\gamma(\varepsilon, L)\right) = 1 - \varepsilon.$$

This completes the proof of the first claim (22) in the lemma. We now proceed to prove the second claim (23). We follow a similar argument. Let $L(\varepsilon, C)$ be a positive integer whose

definition we defer to later in (28). Recall the definition $\pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right) := \left\{\pi_t\left(\boldsymbol{T}^i\right) \mid t \in [L(\varepsilon,C)]\right\}$ from Definition 2 and $V^{C,\alpha}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^i\right) := \left\{u \in V\left(\boldsymbol{T}^i\right) \mid \deg_{\boldsymbol{G}_n}(u) \geq Cn^{\frac{1}{2+\alpha}}\right\}$ from (4), we have

$$\begin{aligned}
\mathbb{P}\left(V^{C,\alpha}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^i\right) \subset \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right) &= \mathbb{P}\left(\pi_{(L(\varepsilon,C)+1):n_i}\left(\boldsymbol{T}^i\right) \cap V^{C,\alpha}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^i\right) = \emptyset\right) \\
&= \mathbb{P}\left(\bigcap_{j=L(\varepsilon,C)+1}^{n_i}\left\{n^{-\frac{1}{2+\alpha}}\deg_{\boldsymbol{G}_n}\left(\pi_j\left(\boldsymbol{T}^i\right)\right) < C\right\}\right) \\
&\geq \mathbb{P}\left(\bigcap_{j=L(\varepsilon,C)+1}^{n_i}\left\{n_i^{-\frac{1}{2+\alpha}}\deg_{\boldsymbol{G}_n}\left(\pi_j\left(\boldsymbol{T}^i\right)\right) < C\right\}\right).
\end{aligned}$$

We fix $q > (2+\alpha) \vee \frac{1}{\delta}$, and again apply Lemma 40 in conjunction with the Portmanteau lemma, noting that the set $\prod_{j=1}^{L(\varepsilon,C)}\mathbb{R} \times \prod_{j=L(\varepsilon,C)+1}^{\infty}(-\infty, C) \subset \mathbb{R}^{\mathbb{N}}$ is open with respect to the ℓ_q topology, to obtain

$$\begin{aligned}
\liminf_{n\to\infty}\mathbb{P}\left(V^{C,\alpha}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^i\right) \subset \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right) &\geq \liminf_{n\to\infty}\mathbb{P}\left(\bigcap_{j=L(\varepsilon,C)+1}^{n_i}\left\{n_i^{-\frac{1}{2+\alpha}}\deg_{\boldsymbol{G}_n}\left(\pi_j\left(\boldsymbol{T}^i\right)\right) < C\right\}\right) \\
&\geq \mathbb{P}\left(\cap_{j=L(\varepsilon,C)+1}^{\infty}\left\{Y_{j,\alpha} < C\right\}\right) \\
&= 1 - \mathbb{P}\left(\cup_{j=L(\varepsilon,C)+1}^{\infty}\left\{Y_{j,\alpha} \geq C\right\}\right) \\
&= 1 - \mathbb{P}\left(\cup_{j=L(\varepsilon,C)+1}^{\infty}\left\{Y_j^q \geq C^q\right\}\right) \\
&\geq 1 - \mathbb{P}\left(\sum_{j=L(\varepsilon,C)+1}^{\infty} Y_j^q \geq C^q\right) \\
&\geq 1 - \frac{\sum_{j=L(\varepsilon,C)+1}^{\infty}\mathbb{E}Y_j^q}{C^q}, \qquad (27)
\end{aligned}$$

where the last inequality in (27) comes from Markov inequality. By Lemma 40, we have that $\sum_{j=1}^{\infty}\mathbb{E}\left(Y_j^q\right) < \infty$. Therefore, we choose $L(\varepsilon, C)$ to be the smallest positive integer such that

$$\sum_{j=L(\varepsilon,C)}^{\infty}\mathbb{E}Y_j^q \leq C^q\varepsilon. \qquad (28)$$

Putting this together with (27), we have that

$$\liminf_{n\to\infty}\mathbb{P}\left(V^{C,\alpha}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^i\right) \subset \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right) \geq 1 - \frac{\sum_{j=L(\varepsilon,C)+1}^{\infty}\mathbb{E}Y_j^q}{C^q} \geq 1 - \varepsilon.$$

The lemma follows as desired. ■

Lemma 29. *Let $\boldsymbol{G}_n \sim \mathrm{PF}(\alpha, \theta, \ell, \pi)$, and suppose Assumptions 11 and 12 hold. Then for any fixed $C > 0$, the following result holds:*

$$\lim_{n\to\infty} \mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(V^{C,\alpha}\left(\boldsymbol{G}_n\right)\right)\right| = 0\right) = 1.$$

Proof. Fix $\varepsilon > 0$ arbitrarily. Suppose $n \geq HL(\varepsilon, C)$ (where $L(\varepsilon, C)$ is defined in Lemma 28) so that $\min_{i\in[K]} n_i \geq L(\varepsilon, C)$ and the set $\pi_{1:L(\varepsilon,C)}(\boldsymbol{T}^i)$ is well-defined. Let $\Omega_i = \left\{V^{C,\alpha}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^i\right) \subset \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right\}$ and $\Omega := \cap_{i=1}^K \Omega_i$. Using the fact that $V^{C,\alpha}\left(\boldsymbol{G}_n\right) = \cup_{i=1}^K V^{C,\alpha}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^i\right)$, we have

$$\begin{aligned}
&\mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(V^{C,\alpha}\left(\boldsymbol{G}_n\right)\right)\right| = 0\right)\\
=\ &\mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^K V^{C,\alpha}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^i\right)\right)\right| = 0\right)\\
\geq\ &\mathbb{P}\left(\left\{\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^K V^{C,\alpha}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^i\right)\right)\right| = 0\right\} \cap \Omega\right)\\
\geq\ &\mathbb{P}\left(\left\{\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^K \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right)\right| = 0\right\} \cap \Omega\right)\\
=\ &\mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^K \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right)\right| = 0\right) - \mathbb{P}\left(\left\{\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^K \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right)\right| = 0\right\} \cap \Omega^c\right)\\
\geq\ &\mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^K \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right)\right| = 0\right) - \mathbb{P}\left(\Omega^c\right). \qquad (29)
\end{aligned}$$

We begin by bounding the first term of (29). Note that the set $\cup_{i=1}^K \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)$ is deterministic and not dependent on $\boldsymbol{R}_n$ by Definition 2. Therefore, the number of random edges of $\boldsymbol{R}_n$ with both endpoints in this set satisfies

$$\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^K \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right)\right| \sim \mathrm{Bin}\left(\frac{L(\varepsilon,C)K\left\{L(\varepsilon,C)K-1\right\}}{2}, \theta\right),$$

since each pair of nodes in the selected set contributes an edge independently with probability θ. Therefore

$$\begin{aligned}
\mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^K \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right)\right| = 0\right) &= 1 - \mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^K \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right)\right| \geq 1\right)\\
&\geq 1 - \mathbb{E}\left(\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^K \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right)\right|\right)\\
&= 1 - \frac{L(\varepsilon,C)K\left(L(\varepsilon,C)K-1\right)\theta}{2}\\
&= 1 - O\left(n^{-\frac{1+\alpha}{2+\alpha}-\delta}\right). \qquad (30)
\end{aligned}$$

The last equality follows from Assumption 12.

Now we bound the second term $\mathbb{P}(\Omega^c)$ of (29). By Lemma 28, we have

$$\mathbb{P}(\Omega^c) = \mathbb{P}\left(\cup_{i=1}^K \Omega_i^c\right) \le \sum_{i=1}^K \mathbb{P}(\Omega_i^c) = \sum_{i=1}^K \mathbb{P}\left(\left\{V_{\boldsymbol{G}_n}^{C,\alpha}(\boldsymbol{T}^i) \subset \pi_{1:L(\varepsilon,C)}(\boldsymbol{T}^i)\right\}^c\right) \le K\varepsilon + o(1). \tag{31}$$

Combining equations (29)–(31), we have shown that

$$\mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(V^{C,\alpha}(\boldsymbol{G}_n)\right)\right| = 0\right) \ge 1 - K\varepsilon - o(1).$$

Since $\varepsilon > 0$ can be arbitrary small, we have

$$\lim_{n\to\infty} \mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(V^{C,\alpha}(\boldsymbol{G}_n)\right)\right| = 0\right) = 1.$$

■

Lemma 30. *Let $L \in \mathbb{N}$, $\varepsilon > 0$ and $c \in (0,1]$. Let $\boldsymbol{G}_n \sim \mathrm{PF}(\alpha,\theta,\ell,\pi)$ and suppose Assumptions 11 and 12 hold. Define*

$$V_1,\dots,V_M = \mathcal{C}(\boldsymbol{G}_n \cap V^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{G}_n))$$
$$\mathcal{K} := \left\{i \in [M] : \deg_{\boldsymbol{G}_n}(V_i) \ge \gamma\left(\frac{\varepsilon}{3H}, 1\right) n^{\frac{1}{2+\alpha}}\right\}.$$

Then, with probability at least $1 - \varepsilon + \eta_n$ where $\eta_n = o(1)$ and depends only on $\varepsilon, L, \alpha, H, c, \delta$.

$$|\mathcal{K}| = K, \quad \text{and } \exists \text{ bijection } \sigma : [K] \to \mathcal{K} \text{ s.t.}$$
$$\pi_1(\boldsymbol{T}^i) \subset V_{\boldsymbol{G}_n}^{\gamma(\frac{\varepsilon}{3H},1),\alpha}(\boldsymbol{T}^i) \subset V_{\boldsymbol{G}_n}^{c\gamma(\frac{\varepsilon}{3H},1),\alpha}(\boldsymbol{T}^i) \subset \pi_{1:(L\vee\tilde{L}(\varepsilon,c))}(\boldsymbol{T}^i) \subset V_{\sigma(i)}, \text{ for all } i \in [K]. \tag{32}$$

Proof. Define

$$\tilde{\gamma}(\varepsilon,L,c) := \gamma\left(\frac{\varepsilon}{3H},\ \tilde{L}(\varepsilon,c)\vee L\right), \quad \text{where } \tilde{L}(\varepsilon,c) := L\left(\frac{\varepsilon}{3H}, c\gamma\left(\frac{\varepsilon}{3H},1\right)\right).$$

Let us define a "good" event $\Omega := \Omega_1 \cap \Omega_2 \cap \Omega_3$ where

$$\begin{aligned}
\Omega_1 &:= \left\{\pi_1(\boldsymbol{T}^i) \subset V_{\boldsymbol{G}_n}^{\gamma(\frac{\varepsilon}{3H},1),\alpha}(\boldsymbol{T}^i), \text{for all } i \in [K]\right\} \\
\Omega_2 &:= \left\{V_{\boldsymbol{G}_n}^{\gamma(\frac{\varepsilon}{3H},1),\alpha}(\boldsymbol{T}^i) \subseteq V_{\boldsymbol{G}_n}^{c\gamma(\frac{\varepsilon}{3H},1),\alpha}(\boldsymbol{T}^i) \subset \pi_{1:\tilde{L}(\varepsilon,c)}(\boldsymbol{T}^i), \text{for all } i \in [K]\right\} \\
\Omega_3 &:= \left\{\exists \text{ injective } \sigma : [K] \to [M] \text{ such that } \pi_{1:\tilde{L}(\varepsilon,c)\vee L}(\boldsymbol{T}^i) \subset V_{\sigma(i)}, \text{for all } i \in [K]\right\}.
\end{aligned}$$

By definition, on $\Omega \subset \Omega_3$, it holds that $\pi_{1:L}(\boldsymbol{T}^i) \subset V_{\sigma(i)}$ for some injective $\sigma : [K] \to [M]$. We now show that $|\mathcal{K}| = K$ on Ω and that σ is a bijection mapping $[K]$ to $\mathcal{K}$. To that end, suppose we are on Ω. Let $\sigma : [K] \to [M]$ be the injective mapping such

that $\pi_{1:\tilde{L}(\varepsilon,c)\vee L}(\boldsymbol{T}^i) \subset V_{\sigma(i)}$ for all $i \in [K]$ whose existence is guaranteed on Ω_3. Then, $\pi_1(\boldsymbol{T}^i) \in V_{\sigma(i)}$ and $\pi_1(\boldsymbol{T}^i) \in V_{\boldsymbol{G}_n}^{\gamma(\frac{\varepsilon}{3H},1),\alpha}(\boldsymbol{T}^i)$; hence, for all $i \in [K]$,

$$\deg_{\boldsymbol{G}_n}(V_{\sigma(i)}) \geq \deg_{\boldsymbol{G}_n}(\pi_1(\boldsymbol{T}^i)) \geq \gamma(\frac{\varepsilon}{3H},1)n^{\frac{1}{2+\alpha}}.$$

It thus holds that $\sigma([K]) \subset \mathcal{K}$ and so $|\mathcal{K}| \geq K$. To show that $|\mathcal{K}| \leq K$, first note that, by the fact that $V(\boldsymbol{G}_n) = \cup_{i=1}^K V(\boldsymbol{T}^i)$ and the definition of Ω_2 and Ω_3, we have

$$V^{\gamma(\frac{\varepsilon}{3H},1),\alpha}(\boldsymbol{G}_n) = \bigcup_{i=1}^K V_{\boldsymbol{G}_n}^{\gamma(\frac{\varepsilon}{3H},1),\alpha}(\boldsymbol{T}^i) \subset \bigcup_{i=1}^K V_{\boldsymbol{G}_n}^{c\gamma(\frac{\varepsilon}{3H},1),\alpha}(\boldsymbol{T}^i) \subset \bigcup_{i=1}^K \pi_{1:\tilde{L}(\varepsilon,c)}(\boldsymbol{T}^i) \subset \bigcup_{i=1}^K V_{\sigma(i)}.$$

Therefore, if $i \notin \sigma([K])$, then $V_i \cap V^{\gamma(\frac{\varepsilon}{3H},1),\alpha}(\boldsymbol{G}_n) = \emptyset$ and hence $i \notin \mathcal{K}$ as well. We thus have that $\mathcal{K} \subset \sigma([K])$. Therefore, we conclude that $|\mathcal{K}| = K$ and that σ is a bijection from $[K]$ to $\mathcal{K}$.

It remains to bound the probability of the event Ω. We have by Lemma 28 that for each $i \in [K]$, there exists $\eta_n^i = o(1)$ (dependent only on ε, L, H) such that

$$\mathbb{P}\big\{\pi_1(\boldsymbol{T}^i) \in V_{\boldsymbol{G}_n}^{\gamma(\frac{\varepsilon}{3H},1),\alpha}(\boldsymbol{T}^i)\big\} \geq 1 - \frac{\varepsilon}{3H} + \eta_n^i.$$

By a union bound and writing $\eta_n' = \sum_{i=1}^K \eta_n^i$, we have

$$\mathbb{P}\big\{\forall i \in [K],\, \pi_1(\boldsymbol{T}^i) \in V_{\boldsymbol{G}_n}^{\gamma(\frac{\varepsilon}{3H},1),\alpha}(\boldsymbol{T}^i)\big\} \geq 1 - \frac{K\varepsilon}{3H} + \eta_n' \geq 1 - \frac{\varepsilon}{3} + \eta_n'.$$

Thus, $\mathbb{P}(\Omega_1^c) \leq \frac{\varepsilon}{3} - \eta_n'$. Recall now that $\tilde{L}(\varepsilon,c) = L\big(\frac{\varepsilon}{3H}, c\gamma\big(\frac{\varepsilon}{3H},1\big)\big)$; via a similar argument, by using the second part of Lemma 28 and a union bound, we also obtain that $\mathbb{P}(\Omega_2^c) \leq \frac{\varepsilon}{3} - \eta_n''$ where $\eta_n'' \to 0$ (dependent only on ε, L, H) as $n \to \infty$.

To upper bound $\mathbb{P}(\Omega_3^c)$, we first define $\mathcal{A} := \mathcal{A}_1 \cap \mathcal{A}_2$, where

$$\begin{aligned}
\mathcal{A}_1 &:= \big\{\pi_{1:\tilde{L}(\varepsilon,c)\vee L}(\boldsymbol{T}^i) \subset V_{\boldsymbol{G}_n}^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{T}^i), \text{for all } i \in [K]\big\} \\
\mathcal{A}_2 &:= \Big\{\Big|E_{\boldsymbol{R}_n}\big(V^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{G}_n)\big)\Big| = 0\Big\}.
\end{aligned}$$

We now claim that $\mathcal{A} \subset \Omega_3$. We verify the claim by showing that on $\mathcal{A}$, there exist $\sigma : [K] \to [M]$ such that $\pi_{1:(\tilde{L}(\varepsilon,c)\vee L)}\left(\boldsymbol{T}^i\right) \subseteq V_{\sigma(i)}$ for all $i \in [K]$.

We note that since $V(\boldsymbol{G}_n) = \sqcup_{i=1}^K V(\boldsymbol{T}^i)$, we have $V^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{G}_n) = \sqcup_{i=1}^K V_{\boldsymbol{G}_n}^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{T}^i)$. Moreover, on $\mathcal{A}_2$, for any $i \neq j$, the two sets $V_{\boldsymbol{G}_n}^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{T}^i)$ and $V_{\boldsymbol{G}_n}^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{T}^j)$ are disconnected in $\boldsymbol{G}_n$; to see this, note that

$$\begin{aligned}
&E_{\boldsymbol{G}_n}\big(V_{\boldsymbol{G}_n}^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{T}^i), V_{\boldsymbol{G}_n}^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{T}^j)\big) \\
&= E_{\boldsymbol{R}_n}\big(V_{\boldsymbol{G}_n}^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{T}^i), V_{\boldsymbol{G}_n}^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{T}^j)\big) \subset E_{\boldsymbol{R}_n}\big(V_{\boldsymbol{G}_n}^{\tilde{\gamma}(\varepsilon,L,c),\alpha}\left(\boldsymbol{G}_n\right)\big) = \emptyset,
\end{aligned}$$

where the first equality follows because $\boldsymbol{T}^i$ and $\boldsymbol{T}^j$ are disconnected and the second equality follows from the definition of $\mathcal{A}_2$.

Therefore, each $V_{\boldsymbol{G}_n}^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{T}^i)$ must correspond to a subset of $V_1,\dots,V_M$; more precisely, there exists a mapping $\kappa : [M] \to [K]$ such that $V_{\boldsymbol{G}_n}^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{T}^i) = \sqcup_{j\in\kappa^{-1}(i)} V_j$. Since $\pi_{1:\tilde{L}(\varepsilon,c)\vee L}(\boldsymbol{T}^i)$ is connected in $\boldsymbol{T}^i$ and hence in $\boldsymbol{G}_n$, on event $\mathcal{A}_1$, it must be that there exists $j_i^* \in \kappa^{-1}(i)$ such that $\pi_{1:\tilde{L}(\varepsilon,c)\vee L}(\boldsymbol{T}^i) \subset V_{j_i^*}$. We may then define $\sigma(i) = j_i^*$ so that $\pi_{1:\tilde{L}(\varepsilon,c)\vee L}(\boldsymbol{T}^i) \subset V_{\sigma(i)}$; note that σ is injective by definition. The claim $\mathcal{A} \subset \Omega_3$ follows as desired.

It remains only then to upper bound the probability of $\mathbb{P}(\mathcal{A}^c)$. Recalling that $\tilde{\gamma}(\varepsilon,L,c) = \gamma(\frac{\varepsilon}{3H}, \tilde{L}(\varepsilon,c)\vee L)$, we have by Lemma 28 that, for each $i \in [K]$, there exists μ_n^i (dependent on ε, L) that is $o(1)$ as $n\to\infty$ such that

$$\mathbb{P}\big\{\pi_{1:\tilde{L}(\varepsilon,c)\vee L}(\boldsymbol{T}^i) \subset V_{\boldsymbol{G}_n}^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{T}^i)\big\} \geq 1 - \frac{\varepsilon}{3H} + \mu_n^i.$$

By an application of union bound, by defining $\mu_n' = \sum_{i=1}^K \mu_n^i$, and using the fact that $H \geq K$, we have that

$$\mathbb{P}(\mathcal{A}_1) = \mathbb{P}\big\{\forall i \in [K],\, \pi_{1:\tilde{L}(\varepsilon,c)\vee L}(\boldsymbol{T}^i) \subset V_{\boldsymbol{G}_n}^{\tilde{\gamma}(\varepsilon,L,c),\alpha}(\boldsymbol{T}^i)\big\} \geq 1 - \frac{K\varepsilon}{3H} + \mu_n' \geq 1 - \frac{\varepsilon}{3} + \mu_n'.$$

Now, by Lemma 29, there exists μ_n'' (dependent only on ε, L, H) that is $o(1)$ as $n\to\infty$ such that $\mathbb{P}(\mathcal{A}_2) \geq 1 + \mu_n''$. Writing $\mu_n = \mu_n' + \mu_n''$ and using another union bound, we then have that $\mathbb{P}(\Omega_3) \geq \mathbb{P}(\mathcal{A}) \geq 1 - \frac{\varepsilon}{3} + \mu_n$ as desired.

By another application of the union bound, we have that $\mathbb{P}(\Omega) \geq 1 - \varepsilon + \eta_n$ where $\eta_n = \eta_n' + \eta_n'' + \mu_n$. The proposition thus follows as desired.

■

F.4 Proof of Corollary 16

Proof. Define the event $\Omega = \Omega_1 \cap \Omega_2$ where

$$\begin{aligned}
\Omega_1 &:= \left\{ d_{\cup_{i=1}^K \pi_{1:L(K^{-1}\varepsilon,\tilde{C})}(\boldsymbol{T}^i)}(\hat{\ell},\ell) = 0 \right\}, \\
\Omega_2 &:= \left\{ V^{\tilde{C},\alpha}(\boldsymbol{G}_n) \subset \cup_{i=1}^K \pi_{1:L(K^{-1}\varepsilon,\tilde{C})}(\boldsymbol{T}^i) \right\}.
\end{aligned}$$

It is clear that on event Ω, the conclusion of the Corollary (7) holds; indeed, by Definition 5,

$$\begin{aligned}
d_{V^{\tilde{C},\alpha}(\boldsymbol{G}_n)}(\hat{\ell},\ell) &:= \min_{\sigma\in S_K} \sum_{u\in V^{\tilde{C},\alpha}(\boldsymbol{G}_n)} \mathbb{1}\left\{\ell(u) \neq \sigma\circ\hat{\ell}(u)\right\} \leq \min_{\sigma\in S_K} \sum_{u\in\cup_{i=1}^K \pi_{1:L(K^{-1}\varepsilon,\tilde{C})}(\boldsymbol{T}^i)} \mathbb{1}\left\{\ell(u) \neq \sigma\circ\hat{\ell}(u)\right\} \\
&= d_{\cup_{i=1}^K \pi_{1:L(K^{-1}\varepsilon,\tilde{C})}(\boldsymbol{T}^i)}(\hat{\ell},\ell) = 0.
\end{aligned}$$

We thus only need to lower bound the probability of Ω. By Theorem 13, we have $\mathbb{P}(\Omega_1^c) \leq \varepsilon + \eta_n$ where $\eta_n = o(1)$ and depends only on $\varepsilon, \tilde{C}, Q, \alpha, H, \delta$. By (23) in Lemma 28, we have

$$V^{\tilde{C},\alpha}(\boldsymbol{G}_n) := \cup_{i=1}^K V_{\boldsymbol{G}_n}^{\tilde{C},\alpha}\left(\boldsymbol{T}^i\right) \subset \cup_{i=1}^K \pi_{1:L\left(K^{-1}\varepsilon,\tilde{C}\right)}\left(\boldsymbol{T}^i\right).$$

with probability at least $1-\varepsilon-\eta_n'$ where $\eta_n'=o(1)$ and depends only on $\varepsilon,\tilde{C},Q,\alpha,H$. We thus have that $\mathbb{P}(\Omega_2^c)\le\varepsilon+\eta_n'$. Therefore, we have

$$\mathbb{P}(\Omega)\ge 1-2\varepsilon-\eta_n-\eta_n'.$$

The Corollary then follow as desired.

■

F.5 Proof of Corollary 14

Proof. Write the remainder term η_n in Theorem 13 as $\eta(L,n)$, so that $\eta(L,n)\to 0$ for each fixed L. Define L_n (dependent ε) as $L_n=\max\{L\in[n]\,:\,|\eta(L,n)|\le\varepsilon\}$ (where we let maximum over an empty set be 0). It holds that $L_n\to\infty$ as $n\to\infty$ since, for any arbitrary integer $L_0>0$, there exists an integer N_0 such that $|\eta(L_0,n)|\le\varepsilon$ for all $n\ge N_0$ so that $L_n\ge L_0$ for all $n\ge N_0$. By definition of L_n and Theorem 13, we have that the event $\Omega_n:=\big\{d_{\cup_{k=1}^K\pi_{1:L_n}(\boldsymbol{T}^k)}(\ell,\hat{\ell})=0\big\}$, has probability at least $1-2\varepsilon$.

On the event Ω_n, there exists a single permutation $\sigma_n\in S_K$ such that $\ell(\pi_t^k)=\sigma_n\circ\hat{\ell}(\pi_t^k)$ for every $k\in[K]$ and $t\le L_n$. Therefore we have

$$d^w(\ell,\hat{\ell})\le\sum_{k=1}^K\Big(\sum_{t=1}^{L_n}+\sum_{t=L_n+1}^{n_k}\Big)w_t\,\mathbb{1}\big\{\ell(\pi_t^k)\ne\sigma_n\circ\hat{\ell}(\pi_t^k)\big\}\le K\sum_{t=L_n+1}^{\infty}w_t:=a_n$$

Since $\sum_{t=1}^\infty w_t<\infty$ and $L_n\to\infty$, we have $a_n=K\sum_{t=L_n+1}^\infty w_t\to 0$, and claim follows as desired. ■

F.6 Proof of Theorem 18

Proof. Let $V_1^2,\dots,V_K^2$ be the output of Algorithm 1 (and thus the input of Algorithm 2). Recall that $\tau=\gamma(\frac{\varepsilon}{18H},1)n^{\frac{1}{2+\alpha}}$.

Let us define

$$V_1^1,\dots,V_M^1=\mathcal{C}\big(\boldsymbol{G}_n\cap V^{\tilde{\gamma}(\frac{\varepsilon}{6},Q,c_0),\alpha}(\boldsymbol{G}_n)\big)\ \text{ and}$$
$$\mathcal{K}=\Big\{i\in[M]\,:\,\deg_{\boldsymbol{G}_n}(V_i^1)\ge\gamma\big(\frac{\varepsilon}{18H},1\big)n^{\frac{1}{2+\alpha}}\Big\}.$$

It will be convenient to also define, for $j\in[K]$,

$$\tilde{V}_j^2:=V_j^2\cap V^{\tau'}(\boldsymbol{G}_n)\subset V_j^2\cap V^{c_0\gamma(\frac{\varepsilon}{18H},1),\alpha}(\boldsymbol{G}_n).$$

Without loss of generality, reindex the components so that $\mathcal{K}=[|\mathcal{K}|]$.

To prove the first claim of the Theorem, i.e. (9), we let C_2 be a positive constant whose definition will be specified later and also define the "good" event $\Omega=\Omega_1\cap\Omega_2$ where

$$\Omega_1:=\Big\{|\mathcal{K}|=K\text{ and }\exists\sigma\in S_K\text{ s.t.}\qquad(33)$$
$$\pi_1(\boldsymbol{T}^i)\subset V_{\boldsymbol{G}_n}^{\gamma(\frac{\varepsilon}{18H},1),\alpha}(\boldsymbol{T}^i)\subset V_{\boldsymbol{G}_n}^{c_0\gamma(\frac{\varepsilon}{18H},1),\alpha}(\boldsymbol{T}^i)\subset\pi_{1:(Q\vee\tilde{L}(\varepsilon/6,c_0))}(\boldsymbol{T}^i)\subset V_{\sigma(i)}^1,\quad\forall i\in[K]\Big\}$$

$$\Omega_2 := \left\{ \frac{\left| E_{\boldsymbol{R}_n}\left(V^{c_0\gamma(\frac{\varepsilon}{18H},1),\alpha}(\boldsymbol{G}_n),\, \mathcal{L}_1\left(\boldsymbol{F}_n\right)\right)\right|}{\left|\mathcal{L}_1\left(\boldsymbol{F}_n\right)\right|} \le \frac{C_2 n^{-\frac{1+\alpha}{2+\alpha}-\delta}}{2} \right\}.$$

Part A: We prove the first statement of the Theorem, i.e. (9). Let us first show that (9) holds on the event Ω. We proceed in 3 steps. Let $\sigma \in S_K$ be the permutation whose existence is guaranteed on Ω_1.

Step A1: We claim that on Ω, there exists $\sigma'' \in S_K$ such that

$$\forall i \in [K], \quad \pi_1(\boldsymbol{T}^i) \subset V^2_{\sigma''(i)} \cap V^{\tau'}(\boldsymbol{G}_n) = \tilde{V}^2_{\sigma''(i)} \tag{34}$$

To see this, we work on Ω and note that since $|\mathcal{K}| = K$ and $Q \le |\pi_{1:(Q\vee\tilde{L}(\varepsilon/6,c_0))}(\boldsymbol{T}^i)| \le |V^1_{\sigma(i)}|$ for all $i \in [K]$, we may apply Corollary 27 to conclude that there exists $\sigma' \in S_K$ such that

$$\forall i \in [K], \quad V^1_i \subset V^2_{\sigma'(i)}. \tag{35}$$

Since $\tau' \le \gamma(\frac{\varepsilon}{18H},1)n^{\frac{1}{2+\alpha}}$,

$$\begin{aligned}\pi_1(\boldsymbol{T}^i) &\subset V^{\gamma(\frac{\varepsilon}{18H},1),\alpha}_{\boldsymbol{G}_n}(\boldsymbol{T}^i) = V^{\gamma(\frac{\varepsilon}{18H},1),\alpha}_{\boldsymbol{G}_n}(\boldsymbol{T}^i) \cap V^1_{\sigma(i)} \\ &\subset V^{\gamma(\frac{\varepsilon}{18H},1),\alpha}_{\boldsymbol{G}_n}(\boldsymbol{T}^i) \cap V^2_{\sigma'(\sigma(i))} \subset V^{\tau'}(\boldsymbol{G}_n) \cap V^2_{\sigma'(\sigma(i))}.\end{aligned}$$

We conclude then that (34) holds with $\sigma'' = \sigma' \circ \sigma$.

Step A2: We next claim that, on Ω_1, for every $i, j \in [K]$ such that $j \neq \sigma''(i)$, $\tilde{V}^2_j \cap V(\boldsymbol{T}^i) = \emptyset$, i.e.,

$$\text{for all } j \in [K], \quad \tilde{V}^2_j \subseteq V(\boldsymbol{T}^{\sigma''^{-1}(j)}). \tag{36}$$

To see this, let $i, j \in [K]$ such that $j \neq \sigma''(i)$; since $\tilde{V}^2_j := V^2_j \cap V^{\tau'}(\boldsymbol{G}_n)$, we have

$$\begin{aligned}\tilde{V}^2_j \cap V(\boldsymbol{T}^i) &= \left(V^2_j \cap V^{\tau'}(\boldsymbol{G}_n)\right) \cap V(\boldsymbol{T}^i) = V^2_j \cap \left(V(\boldsymbol{T}^i) \cap V^{\tau'}(\boldsymbol{G}_n)\right) \\ &= V^2_j \cap V^{\tau'}_{\boldsymbol{G}_n}(\boldsymbol{T}^i) \overset{(a)}{\subseteq} V^2_j \cap V^{c_0\gamma(\frac{\varepsilon}{18H},1),\alpha}_{\boldsymbol{G}_n}(\boldsymbol{T}^i) \\ &\overset{(b)}{\subseteq} V^2_j \cap V^1_{\sigma(i)} \overset{(c)}{\subseteq} V^2_j \cap V^2_{\sigma''(i)} = \emptyset,\end{aligned} \tag{37}$$

where (a) follows since $\tau' \ge c_0\gamma(\frac{\varepsilon}{18H},1)n^{\frac{1}{2+\alpha}}$, where (b) follows from the definition of Ω_1, and where (c) follows by (35).

Step A3: We now proceed to bound $d_{\mathcal{L}_1(\boldsymbol{F}_n)}(\hat{\ell},\ell)$. Fix $i \in [K]$ and $u \in \mathcal{L}_1(\boldsymbol{T}^i)$. Then u is a neighbor of $\pi_1(\boldsymbol{T}^i)$ and hence, by (34),

$$D_{\sigma''(i)}(u) = \min_{v \in \tilde{V}^2_{\sigma''(i)}} \mathrm{dist}_{\boldsymbol{G}_n}(u,v) \le \mathrm{dist}_{\boldsymbol{G}_n}(u, \pi_1(\boldsymbol{T}^i)) = 1.$$

On the other hand, for any $j \neq \sigma''(i)$ and any $v \in \tilde{V}^2_j$, we know by (37) that u and v are disconnected in $\boldsymbol{F}_n$, so that $\mathrm{dist}_{\boldsymbol{G}_n}(u,v) = 1$ if and only if (u,v) is an edge in $\boldsymbol{R}_n$. Therefore,

$$D_j(u) = \min_{v \in \tilde{V}^2_j} \mathrm{dist}_{\boldsymbol{G}_n}(u,v) \begin{cases} = 1 & \text{if there exists } v \in \tilde{V}^2_j \text{ such that } (u,v) \in E(\boldsymbol{R}_n) \\ \ge 2 & \text{if } (u,v) \notin E(\boldsymbol{R}_n) \text{ for all } v \in \tilde{V}^2_j. \end{cases}$$

Via the definition of $\hat{\ell}(\cdot)$ in Algorithm 2 then,

$$\mathbb{1}\{\hat{\ell}(u) \neq \sigma''(i)\} \leq \mathbb{1}\left\{\exists j \neq \sigma''(i),\, D_{\sigma''(i)}(u) \geq D_j(u)\right\} \leq \mathbb{1}\left\{\exists j \neq \sigma''(i),\, D_j(u) = 1\right\}$$
$$\leq \mathbb{1}\left\{\exists v \in \cup_{j=1}^{K} \tilde{V}_j^2,\ (u,v) \in E(\boldsymbol{R}_n)\right\} \leq |E_{\boldsymbol{R}_n}(u, \cup_{j=1}^{K} \tilde{V}_j^2)|.$$

By the definition of $d(\cdot,\cdot)$ in (5), on Ω, we have by the definition of Ω_2 that

$$\frac{d_{\mathcal{L}_1(\boldsymbol{F}_n)}\left(\hat{\ell}, \ell\right)}{\left|\mathcal{L}_1\left(\boldsymbol{F}_n\right)\right|} = \min_{\sigma \in S_K} \frac{d^{\mathrm{Ham}}_{\mathcal{L}_1(\boldsymbol{F}_n)}\left(\hat{\ell}, \sigma \circ \ell\right)}{\left|\mathcal{L}_1\left(\boldsymbol{F}_n\right)\right|} \leq \frac{d^{\mathrm{Ham}}_{\mathcal{L}_1(\boldsymbol{F}_n)}\left(\hat{\ell}, \sigma'' \circ \ell\right)}{\left|\mathcal{L}_1\left(\boldsymbol{F}_n\right)\right|}$$
$$= \sum_{i=1}^{K} \sum_{u \in \mathcal{L}_1(\boldsymbol{T}^i)} \frac{\mathbb{1}\{\hat{\ell}(u) \neq \sigma''(i)\}}{\left|\mathcal{L}_1\left(\boldsymbol{F}_n\right)\right|} \leq \sum_{u \in \mathcal{L}_1(\boldsymbol{F}_n)} \frac{|E_{\boldsymbol{R}_n}(u, \cup_{j=1}^{K} \tilde{V}_j^2)|}{\left|\mathcal{L}_1\left(\boldsymbol{F}_n\right)\right|}$$
$$\overset{(a)}{\leq} \frac{2\left|E_{\boldsymbol{R}_n}\left(\cup_{j=1}^{K} \tilde{V}_j^2,\, \mathcal{L}_1\left(\boldsymbol{F}_n\right)\right)\right|}{\left|\mathcal{L}_1\left(\boldsymbol{F}_n\right)\right|} \leq \frac{2\left|E_{\boldsymbol{R}_n}\left(V^{c_0\gamma\left(\frac{\varepsilon}{18H},1\right),\alpha}(\boldsymbol{G}_n),\, \mathcal{L}_1\left(\boldsymbol{F}_n\right)\right)\right|}{\left|\mathcal{L}_1\left(\boldsymbol{F}_n\right)\right|} \leq C_2 n^{-\frac{1+\alpha}{2+\alpha}-\delta}.$$

where (a) follows from Lemma 42. We have thus shown that (9) holds on Ω.

It remains to bound the probability of the event Ω_1, Ω_2. By applying Lemma 30 (with $L \mapsto Q$, $\varepsilon \mapsto \varepsilon/6$, and $c \mapsto c_0$), we have

$$\mathbb{P}(\Omega_1) \geq 1 - \frac{\varepsilon}{6} + o(1) \tag{38}$$

where the $o(1)$ sequence depends only on $\varepsilon, Q, H, \alpha, \delta$. To bound $\mathbb{P}\left(\Omega_2^c\right)$, we have by Lemma 31 (with $C \mapsto c_0\gamma(\frac{\varepsilon}{18H}, 1)$ and $s \mapsto 1$) that there exists $C_2 > 0$ such that

$$\mathbb{P}\left(2n^{\frac{1+\alpha}{2+\alpha}+\delta} \frac{\left|E_{\boldsymbol{R}_n}\left(V^{c_0\gamma\left(\frac{\varepsilon}{18H},1\right),\alpha}(\boldsymbol{G}_n),\, \mathcal{L}_1\left(\boldsymbol{F}_n\right)\right)\right|}{\left|\mathcal{L}_1\left(\boldsymbol{F}_n\right)\right|} \geq C_2\right) \leq \frac{\varepsilon}{2}.$$

This immediately implies that $\mathbb{P}(\Omega_2) \geq 1 - \varepsilon/2$. Therefore, by an application of union bound,

$$\mathbb{P}\left(\Omega^c\right) \leq \mathbb{P}(\Omega_1^c) + \mathbb{P}(\Omega_2^c) \leq \frac{\varepsilon}{6} + \frac{\varepsilon}{2} + o(1).$$

Thus, we have have completed the proof of the first claim (9) of the theorem.

Part B: We now prove the second statement of the Theorem (10), in the linear preferential attachment regime where $\alpha = 0$. We let C_3 be a positive number whose definition will be specified later. We also define the "good" event $\tilde{\Omega} = \Omega_1 \cap \tilde{\Omega}_2 \cap \tilde{\Omega}_3$, where Ω_1 is defined in (33) and $\tilde{\Omega}_2$ and $\tilde{\Omega}_3$ are defined below:

$$\tilde{\Omega}_2 := \left\{\left|E_{\boldsymbol{R}_n}\left(V^{c_0\gamma\left(\frac{\varepsilon}{18H},1\right),0}\left(\boldsymbol{G}_n\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| = 0, \quad \text{for } s = 1, 3\right\}$$

$$\tilde{\Omega}_3 := \left\{ \frac{\left|E_{\boldsymbol{R}_n}\left(V_1^{c_0\gamma\left(\frac{\varepsilon}{18H},1\right),0}\left(\boldsymbol{G}_n\right),\mathcal{L}_2\left(\boldsymbol{F}_n\right)\right)\right| + \left|\bigcup_{\substack{i,j\in[K]\\ i\neq j}} \dot{E}_{\boldsymbol{R}_n}\left(V_{\boldsymbol{G}_n}^{c_0\gamma\left(\frac{\varepsilon}{18H},1\right),0}(\boldsymbol{T}^j),\mathcal{L}_2(\boldsymbol{T}^i)\right)\right|}{\left|\mathcal{L}_2\left(\boldsymbol{F}_n\right)\right|} \leq \frac{C_3 n^{-\delta}}{2} \right\},$$

where we define

$$V_1^{c_0\gamma\left(\frac{\varepsilon}{18H},1\right),0}\left(\boldsymbol{G}_n\right) := \left\{ v \in V\left(\boldsymbol{G}_n\right) \,\middle|\, \mathrm{dist}_{\boldsymbol{F}_n}\left(v, V^{c_0\gamma\left(\frac{\varepsilon}{18H},1\right),0}\left(\boldsymbol{G}_n\right)\right) \leq 1 \right\}$$

which consists of all nodes in $V^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{G}_n)$ as well as their $\boldsymbol{F}_n$–neighbors. We also define

$$\dot{E}_{\boldsymbol{R}_n}(A,B) := \left\{(u,v) \,\middle|\, u\in A,\, v\in B,\ \text{and } \exists w \in V(\boldsymbol{G}_n),\, (u,w),(w,v)\in E(\boldsymbol{R}_n)\right\},$$

as the set of pairs (u,v) with $u\in A$ and $v\in B$ such that u and v are connected by two noise edges.

We first show that (10) holds on $\tilde{\Omega}$. To that end, fix $i\in[K]$ and $u\in\mathcal{L}_2(\boldsymbol{T}^i)$. Since $\pi_1(\boldsymbol{T}^i)\in\tilde{V}^2_{\sigma''(i)}$ by (34), we have $D_{\sigma''(i)}(u)\leq \mathrm{dist}_{\boldsymbol{G}_n}(u,\pi_1(\boldsymbol{T}^i))\leq 2$. Let $j\neq\sigma''(i)$ and recall that on Ω_1, by (37), we have that $V(\boldsymbol{T}^i)\cap\tilde{V}^2_j=\emptyset$ so that

$$E_{\boldsymbol{G}_n}(u,\tilde{V}^2_j) = E_{\boldsymbol{R}_n}(u,\tilde{V}^2_j),$$

that is, all edges from u to $\tilde{V}^2_j$ must be noise edges. Moreover, we have that all the length-two path from u to $\tilde{V}^2_j$ can be decomposed as follows:

$$\begin{aligned}
\dot{E}(u,\tilde{V}^2_j) &:= \left\{(u,v) \,\middle|\, v\in\tilde{V}^2_j,\ \text{and } \exists w\in V(\boldsymbol{G}_n),\,(u,w),(w,v)\in E(\boldsymbol{G}_n)\right\}\\
&= \underbrace{\left\{(u,v) \,\middle|\, v\in\tilde{V}^2_j,\ \text{and } \exists w\in V(\boldsymbol{T}^i),\,(u,w)\in E(\boldsymbol{F}_n),(w,v)\in E(\boldsymbol{R}_n)\right\}}_{=:\dot{E}_{\mathrm{FR}}(u,\tilde{V}^2_j)} \cup\\
&\quad \underbrace{\left\{(u,v) \,\middle|\, v\in\tilde{V}^2_j,\ \text{and } \exists w\in V(\boldsymbol{G}_n),\,(u,w)\in E(\boldsymbol{R}_n),(w,v)\in E(\boldsymbol{R}_n)\right\}}_{=:\dot{E}_{\mathrm{RR}}(u,\tilde{V}^2_j)=\dot{E}_{\boldsymbol{R}_n}(u,\tilde{V}^2_j)} \cup\\
&\quad \underbrace{\left\{(u,v) \,\middle|\, v\in\tilde{V}^2_j,\ \text{and } \exists w\in V(\boldsymbol{G}_n),\,(u,w)\in E(\boldsymbol{R}_n),\,(w,v)\in E(\boldsymbol{F}_n)\right\}}_{=:\dot{E}_{\mathrm{RF}}(u,\tilde{V}^2_j)} \cup\\
&\quad \underbrace{\left\{(u,v) \,\middle|\, v\in\tilde{V}^2_j,\ \text{and } \exists w\in V(\boldsymbol{G}_n),\,(u,w)\in E(\boldsymbol{F}_n),\,(w,v)\in E(\boldsymbol{F}_n)\right\}}_{=:\dot{E}_{\mathrm{FF}}(u,\tilde{V}^2_j)}
\end{aligned}$$

Since u belongs to $\boldsymbol{T}^i$ and $\tilde{V}^2_j$ does not intersect $\boldsymbol{T}^i$ on Ω_1 by (37), there cannot exist a node w such that (u,w) and (w,v) are both edges in $\boldsymbol{F}_n$. Therefore,

$$\dot{E}_{\mathrm{FF}}(u,\tilde{V}^2_j)=\emptyset, \quad \text{on } \Omega_1. \tag{39}$$

For any $w\in V(\boldsymbol{T}^i)$ such that $(u,w)\in E(\boldsymbol{F}_n)$, it must be that $w\in\mathcal{L}_1(\boldsymbol{T}^i)\cup\mathcal{L}_3(\boldsymbol{T}^i)$. Since $\tilde{V}^2_j\subset V^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{G}_n)$, it must be that on $\tilde{\Omega}_2$, there are no edges in $E(\boldsymbol{R}_n)$ connecting such $w\in\mathcal{L}_1(\boldsymbol{T}^i)\cup\mathcal{L}_3(\boldsymbol{T}^i)$ in to $\tilde{V}^2_j$. Therefore,

$$\dot{E}_{\mathrm{FR}}(u,\tilde{V}^2_j)=\emptyset, \quad \text{on } \tilde{\Omega}_2. \tag{40}$$

We thus have that $D_j(u) = \min_{v\in\tilde{V}_j^2} \mathrm{dist}_{\boldsymbol{G}_n}(u,v) \le 2$ if and only if

$$E_{\boldsymbol{R}_n}(u,\tilde{V}_j^2) \cup \dot{E}_{\mathrm{RR}}(u,\tilde{V}_j^2) \cup \dot{E}_{\mathrm{RF}}(u,\tilde{V}_j^2) \neq \emptyset.$$

We thus have that, on the event $\tilde{\Omega}$,

$$\begin{aligned}
\mathbb{1}\{\hat{\ell}(u) \neq \sigma''(i)\} &\le \mathbb{1}\big\{\exists j \neq \sigma''(i),\, D_j(u) \le 2\big\} \\
&\le \mathbb{1}\big\{\exists j \neq \sigma''(i),\, E_{\boldsymbol{R}_n}(u,\tilde{V}_j^2) \cup \dot{E}_{\mathrm{RR}}(u,\tilde{V}_j^2) \cup \dot{E}_{\mathrm{RF}}(u,\tilde{V}_j^2) \neq \emptyset\big\} \\
&\overset{(a)}{\le} \mathbb{1}\Big\{E_{\boldsymbol{R}_n}\big(u, V_1^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{G}_n)\big) \neq \emptyset\Big\} + \mathbb{1}\big\{\exists j \neq \sigma''(i),\, \dot{E}_{\mathrm{RR}}(u,\tilde{V}_j^2) \neq \emptyset\big\} \\
&\le \Big|E_{\boldsymbol{R}_n}\big(u, V_1^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{G}_n)\big)\Big| + \sum_{j\neq\sigma''(i)} |\dot{E}_{\mathrm{RR}}(u,\tilde{V}_j^2)|.
\end{aligned}$$

where (a) follows because

$$E_{\boldsymbol{R}_n}\big(u, V_1^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{G}_n)\big) = \emptyset \implies E_{\boldsymbol{R}_n}(u,\tilde{V}_j^2) = \emptyset \text{ and } \dot{E}_{\mathrm{RF}}(u,\tilde{V}_j^2) = \emptyset.$$

To see this, note that $\tilde{V}_j^2 \subset V_1^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{G}_n)$ so that

$$E_{\boldsymbol{R}_n}(u,\tilde{V}_j^2) \subset E_{\boldsymbol{R}_n}\big(u, V_1^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{G}_n)\big)$$

. Note also that for any $(u,v) \in \dot{E}_{\mathrm{RF}}(u,\tilde{V}_j^2)$, there exists a node w such that $(w,v) \in E(\boldsymbol{F}_n)$ so that $w \in V_1^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{G}_n)$ and (u,w) is an element in $E_{\boldsymbol{R}_n}\big(u, V_1^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{G}_n)\big)$. Thus, we have

$$\begin{aligned}
\frac{d_{\mathcal{L}_2(\boldsymbol{F}_n)}(\hat{\ell},\ell)}{|\mathcal{L}_2(\boldsymbol{F}_n)|} &= \min_{\sigma\in S_K} \frac{d^{\mathrm{Ham}}_{\mathcal{L}_2(\boldsymbol{F}_n)}(\hat{\ell},\sigma\circ\ell)}{|\mathcal{L}_2(\boldsymbol{F}_n)|} \le \frac{d^{\mathrm{Ham}}_{\mathcal{L}_2(\boldsymbol{F}_n)}(\hat{\ell},\sigma''\circ\ell)}{|\mathcal{L}_2(\boldsymbol{F}_n)|} \\
&\le \sum_{i=1}^K \sum_{u\in\mathcal{L}_2(\boldsymbol{T}^i)} \frac{\mathbb{1}\{\hat{\ell}(u) \neq \sigma''(i)\}}{|\mathcal{L}_2(\boldsymbol{F}_n)|} \\
&\le \sum_{i=1}^K \sum_{u\in\mathcal{L}_2(\boldsymbol{T}^i)} \frac{\big|E_{\boldsymbol{R}_n}\big(u, V_1^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{G}_n)\big)\big| + \sum_{j\neq\sigma''(i)} |\dot{E}_{\mathrm{RR}}(u,\tilde{V}_j^2)|}{|\mathcal{L}_2(\boldsymbol{F}_n)|} \\
&\overset{(a)}{\le} \frac{1}{|\mathcal{L}_2(\boldsymbol{F}_n)|}\Big\{\sum_{u\in\mathcal{L}_2(\boldsymbol{F}_n)} \big|E_{\boldsymbol{R}_n}\big(u, V_1^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{G}_n)\big)\big| + \sum_{i\neq j}\sum_{u\in\mathcal{L}_2(\boldsymbol{T}^i)} |\dot{E}_{\boldsymbol{R}_n}(u, V_{\boldsymbol{G}_n}^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{T}^j))|\Big\} \\
&\overset{(b)}{\le} \frac{1}{|\mathcal{L}_2(\boldsymbol{F}_n)|}\Big\{2\big|E_{\boldsymbol{R}_n}\big(\mathcal{L}_2(\boldsymbol{F}_n), V_1^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{G}_n)\big)\big| + \sum_{i\neq j} 2\big|\dot{E}_{\boldsymbol{R}_n}\big(\mathcal{L}_2(\boldsymbol{T}^i), V_{\boldsymbol{G}_n}^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{T}^j)\big)\big|\Big\} \\
&\le C_3 n^{-\delta}, \quad \text{on } \tilde{\Omega}_3,
\end{aligned}$$

where inequality (a) follows from (36) and where inequality (b) follows from Lemma 42; to argue that

$$\sum_{u\in\mathcal{L}_2(\boldsymbol{T}^i)} |\dot{E}_{\boldsymbol{R}_n}(u, V_{\boldsymbol{G}_n}^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{T}^j))| \le 2\big|\dot{E}_{\boldsymbol{R}_n}\big(\mathcal{L}_2(\boldsymbol{T}^i), V_{\boldsymbol{G}_n}^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{T}^j)\big|,$$

we apply Lemma 42 with respect to a graph $\boldsymbol{g}'$ where (u,v) forms an edge if (u,v) belongs in the set $\dot{E}_{\boldsymbol{R}_n}\big(\mathcal{L}_2(\boldsymbol{T}^i), V_{\boldsymbol{G}_n}^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{T}^j)\big)$.

We have thus shown that (10) holds on the event $\tilde{\Omega}$. It remains to bound the probability of $\tilde{\Omega}$. Applying the first statement of Lemma 35 along with a union bound (with $C = c_0\gamma(\frac{\varepsilon}{18H},1)$ and $s=1$ and then $s=3$), we have that

$$\mathbb{P}(\tilde{\Omega}_2^c) \le 2o(1), \tag{41}$$

where the $o(1)$ sequence depends only on α, c_0, H, δ.

Noticing that $\big|\dot{E}_{\mathbb{R}_n}(A,B)\big| \le \big|E^2_{\mathbb{R}_n}(A,B)\big|$ defined in Lemma 35, and apply the second and third statement of the lemma (with $C = c_0\gamma(\frac{\varepsilon}{18H},1)$ and $s=2$), we may conclude that there exists $C_3 > 0$ such that

$$\mathbb{P}\bigg(4n^{2\delta}\frac{\big|\cup_{i,j\in[K],i\neq j}\dot{E}_{\mathbb{R}_n}\big(V_{\boldsymbol{G}_n}^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{T}^j), \mathcal{L}_2(\boldsymbol{T}^i)\big)\big|}{|\mathcal{L}_2(\boldsymbol{F}_n)} \ge C_3\bigg) \le \frac{\varepsilon}{6}$$

and that

$$\mathbb{P}\bigg(4n^{\delta}\frac{\big|E_{\boldsymbol{R}_n}\big(V_1^{c_0\gamma(\frac{\varepsilon}{18H},1),0}(\boldsymbol{G}_n), \mathcal{L}_2(\boldsymbol{F}_n)\big)\big|}{|\mathcal{L}_2(\boldsymbol{F}_n)|} \ge C_3\bigg) \le \frac{\varepsilon}{6}.$$

It holds therefore that $\mathbb{P}(\tilde{\Omega}_3^c) \le \frac{\varepsilon}{3}$. Combining these with (38) yields that $\mathbb{P}(\tilde{\Omega}^c) \le \varepsilon + o(1)$ and the Theorem thus follows.

We have proved that $\mathbb{P}\left(\Omega_1^c\right) \le \varepsilon/3$ in the above proof of (9). ∎

Lemma 31. *Let $\boldsymbol{G}_n \sim \mathrm{PF}(\alpha,\theta,\ell,\pi)$, and suppose Assumptions 11 and 12 hold. Then for any fixed $C>0$ and $s\in\mathbb{N}$, the following result holds:*

$$\frac{\left|E_{\boldsymbol{R}_n}\left(V^{C,\alpha}\left(\boldsymbol{G}_n\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right|}{\left|\mathcal{L}_s\left(\boldsymbol{F}_n\right)\right|} = O_p\left(n^{-\frac{1+\alpha}{2+\alpha}-\delta}\right), \tag{42}$$

where $\mathcal{L}_s\left(\boldsymbol{F}_n\right)$ denotes all the nodes in layer s of $\boldsymbol{T}^i$.

Proof. We fix $\varepsilon>0$ and define the "good event" $\Omega\left(t\right) := \left(\cap_{i=1}^K \Omega_i\right) \cap \left(\cap_{i=1}^K \breve{\Omega}_i\left(t\right)\right)$, where

$$\Omega_i = \left\{V_{\boldsymbol{G}_n}^{C,\alpha}\left(\boldsymbol{T}^i\right) \subset \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right\}$$

and

$$\breve{\Omega}_i\left(t\right) := \left\{\frac{\left|E_{\boldsymbol{R}_n}\big(\pi_{1:L(\varepsilon,C)}(\boldsymbol{T}^i), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\big)\right|}{\left|\mathcal{L}_s\left(\boldsymbol{F}_n\right)\right|} \le t n^{-\frac{1+\alpha}{2+\alpha}-\delta}\right\}.$$

for any $t>0$. The function $L(\cdot)$ is defined in Lemma 28.

We can observed that for fixed t, on the event $\Omega(t)$

$$\begin{aligned}\frac{\left|E_{\boldsymbol{R}_n}\left(V^{C,\alpha}(\boldsymbol{G}_n),\mathcal{L}_s(\boldsymbol{F}_n)\right)\right|}{\left|\mathcal{L}_s(\boldsymbol{F}_n)\right|} &\le \sum_{i=1}^{K}\frac{\left|E_{\boldsymbol{R}_n}\left(V^{C,\alpha}_{\boldsymbol{G}_n}(\boldsymbol{T}^i),\mathcal{L}_s(\boldsymbol{F}_n)\right)\right|}{\left|\mathcal{L}_s(\boldsymbol{F}_n)\right|}\\ &\le \sum_{i=1}^{K}\frac{\left|E_{\boldsymbol{R}_n}\left(\pi_{1:L(\varepsilon,C)}(\boldsymbol{T}^i),\mathcal{L}_s(\boldsymbol{F}_n)\right)\right|}{\left|\mathcal{L}_s(\boldsymbol{F}_n)\right|}\\ &\le Ktn^{-\frac{1+\alpha}{2+\alpha}-\delta},\end{aligned}$$

where the first inequality follows from the fact that $V^{C,\alpha}(\boldsymbol{G}_n)=\cup_{i=1}^{K}V^{C,\alpha}_{\boldsymbol{G}_n}(\boldsymbol{T}^i)$, and the remaining two inequalities follow directly from the definition of Ω_i and $\breve{\Omega}_i(t)$.

We now bound the probability of the event $\Omega(t)$. Note that by Lemma 28, we have that

$$\forall i\in[K],\quad \mathbb{P}(\Omega_i^c)\le\varepsilon+o(1). \tag{43}$$

Our next step then is to show that there exists $t>0$ such that $\mathbb{P}\big(\breve{\Omega}_i(t)^c\big)\le\varepsilon$. Note that the top $L(\varepsilon,C)$ nodes of each tree $\left\{\pi_{1:L(\varepsilon,C)}(\boldsymbol{T}^i),i\in[K]\right\}$ are deterministic and that the layer s nodes $\mathcal{L}_s(\boldsymbol{F}_n)$ are determined entirely by $\boldsymbol{F}_n$ and hence independent of $\boldsymbol{R}_n$ by the Definition 2 of our model. Therefore, the random variable $|E_{\boldsymbol{R}_n}(\pi_{1:L(\varepsilon,C)}(\boldsymbol{T}^i),\mathcal{L}_s(\boldsymbol{F}_n))|$, conditioning on $\boldsymbol{F}_n$, is stochastic dominated by the binomial distribution $\mathrm{Bin}\left(\left|\pi_{1:L(\varepsilon,C)}(\boldsymbol{T}^i)\right|\left|\mathcal{L}_s(\boldsymbol{F}_n)\right|,\theta\right)$. Combining with the fact that $\left|\pi_{1:L(\varepsilon,C)}(\boldsymbol{T}^i)\right|=L(\varepsilon,C)$, we have

$$\mathbb{E}_{\boldsymbol{R}_n}\left(\frac{\left|E_{\boldsymbol{R}_n}\left(\pi_{1:L(\varepsilon,C)}(\boldsymbol{T}^i),\mathcal{L}_s(\boldsymbol{F}_n)\right)\right|}{\left|\mathcal{L}_s(\boldsymbol{F}_n)\right|}\right)\le\left|\pi_{1:L(\varepsilon,C)}(\boldsymbol{T}^i)\right|\theta\le L(\varepsilon,C)\,\theta.$$

By Markov's inequality and Assumption 12, for any $\varepsilon>0$, there exist an universal constant $C^1>0$ such that the following holds with probability at least $1-\varepsilon$:

$$\frac{\left|E_{\boldsymbol{R}_n}\left(\pi_{1:L(\varepsilon,C)}(\boldsymbol{T}^i),\mathcal{L}_s(\boldsymbol{F}_n)\right)\right|}{\left|\mathcal{L}_s(\boldsymbol{F}_n)\right|}\le\varepsilon^{-1}L(\varepsilon,C)\,\theta\le C^1\varepsilon^{-1}L(\varepsilon,C)\,n^{-\frac{1+\alpha}{2+\alpha}-\delta}.$$

Therefore, we may take $t:=C^1\varepsilon^{-1}L(\varepsilon,C)$ to obtain that $\mathbb{P}\left(\breve{\Omega}_i^c(t)\right)\le\varepsilon$.

Combining the preceding bounds with a union bound, for all sufficiently large n, we have, for $t=C^1\varepsilon^{-1}L(\varepsilon,C)$,

$$\mathbb{P}\left(\Omega(t)^c\right)\;\le\;\sum_{i=1}^{K}\mathbb{P}(\Omega_i^c)+\sum_{i=1}^{K}\mathbb{P}\left(\breve{\Omega}_i(t)^c\right)\le K\varepsilon+K\varepsilon+o(1)\le 2K\varepsilon+o(1).$$

This completes the proof of (42). ∎

Lemma 32 (Degree sum of $\mathcal{L}_s(\boldsymbol{T}_n)$)**.** *Let $\boldsymbol{T}_n \sim \mathrm{APA}(0, n)$. For $s \in \mathbb{N}$, let $\mathcal{L}_s(\boldsymbol{T}_n)$ denote the set of nodes at layer s. The sum of the degrees of all the nodes in $\mathcal{L}_s(\boldsymbol{T}_n)$ has the following form:*

$$\sum_{u \in \mathcal{L}_s(\boldsymbol{T}_n)} \deg_{\boldsymbol{T_n}}(u) = \left|\mathcal{L}_{s+1}(\boldsymbol{T}_n)\right| + \left|\mathcal{L}_s(\boldsymbol{T}_n)\right|.$$

Proof. This result can be directly observed from the network structure. For a node $u \in \mathcal{L}_s(\boldsymbol{T}_n)$, any edge incident on u must either connect u to one of its children or connect u to its parent; we thus have that $\deg_{\boldsymbol{T}_n}(u) = |\text{children of } u \text{ in } \boldsymbol{T}_n| + 1$. We also note that any child of u is in layer $s+1$. Therefore,

$$\begin{aligned}\sum_{u \in \mathcal{L}_s(\boldsymbol{T}_n)} \deg_{\boldsymbol{T_n}}(u) &= \sum_{u \in \mathcal{L}_s(\boldsymbol{T}_n)} \big\{|\text{children of } u \text{ in } \boldsymbol{T}_n| + 1\big\} \\ &= |\mathcal{L}_{s+1}(\boldsymbol{T}_n)| + |\mathcal{L}_s(\boldsymbol{T}_n)|.\end{aligned}$$

The Lemma follows as desired. ∎

Lemma 33 (Difference equation for $\mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right)$)**.** *Let $\boldsymbol{T}_n \sim \mathrm{APA}(0, n)$. Suppose $s, n \in \mathbb{N}$ are integers satisfying $n \geq 2$. let $\mathcal{L}_s(\boldsymbol{T}_n)$ be the nodes at layer s, if $s \geq 2$ then the expectation obeys the following difference equation*

$$\mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_{n+1})\right|\right) - \mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right) = \frac{\mathbb{E}\left(\left|\mathcal{L}_{s-1}(\boldsymbol{T}_n)\right|\right) + \mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right)}{2(n-1)}.$$

If $s = 1$ then the expectation obeys the following difference equation

$$\mathbb{E}\left(\left|\mathcal{L}_1(\boldsymbol{T}_{n+1})\right|\right) - \mathbb{E}\left(\left|\mathcal{L}_1(\boldsymbol{T}_n)\right|\right) = \frac{\mathbb{E}\left(\left|\mathcal{L}_1(\boldsymbol{T}_n)\right|\right)}{2(n-1)}.$$

Proof. To derive the difference equation for $s \geq 2$, we first consider the conditional distribution of $\left|\mathcal{L}_s(\boldsymbol{T}_{n+1})\right|$ conditioning on $\boldsymbol{T}_n$. A direct observation is that $\left|\mathcal{L}_s(\boldsymbol{T}_{n+1})\right| = \left|\mathcal{L}_s(\boldsymbol{T}_n)\right| + 1$ if and only if the new node is attached to a node in layer $s-1$. Therefore, by the generation mechanism of preferential attachment tree and Lemma 32

$$\mathbb{P}\left(\left|\mathcal{L}_s(\boldsymbol{T}_{n+1})\right| = \left|\mathcal{L}_s(\boldsymbol{T}_n)\right| + 1 \,\|\, \boldsymbol{T}_n\right) = \frac{\sum_{u \in \mathcal{L}_{s-1}(\boldsymbol{T}_n)} \deg_{\boldsymbol{T_n}}(u)}{\sum_{u \in V(\boldsymbol{T}_n)} \deg_{\boldsymbol{T_n}}(u)} = \frac{\left|\mathcal{L}_{s-1}(\boldsymbol{T}_n)\right| + \left|\mathcal{L}_s(\boldsymbol{T}_n)\right|}{2(n-1)},$$

the second identity comes from Lemma 32. Then we proceed to the difference equation

$$\begin{aligned}\mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_{n+1})\right|\right) - \mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right) &= \mathbb{E}\big\{\mathbb{E}(|\mathcal{L}_s(\boldsymbol{T}_{n+1})| \,\|\, \boldsymbol{T}_n)\big\} - \mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right) \\ &= \mathbb{E}\big\{\mathbb{P}\left(\left|\mathcal{L}_s(\boldsymbol{T}_{n+1})\right| = \left|\mathcal{L}_s(\boldsymbol{T}_n)\right| + 1 \,\|\, \boldsymbol{T}_n\right)\big\} \\ &= \frac{\mathbb{E}\left(\left|\mathcal{L}_{s-1}(\boldsymbol{T}_n)\right|\right) + \mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right)}{2(n-1)}.\end{aligned}$$

We now turn to the case $s = 1$. The only change is that $\sum_{u\in\mathcal{L}_0(\boldsymbol{T}_n)} \deg_{\boldsymbol{T_n}}(u) = \left|\mathcal{L}_1(\boldsymbol{T}_n)\right|$ replaces $\left|\mathcal{L}_0(\boldsymbol{T}_n)\right| + \left|\mathcal{L}_1(\boldsymbol{T}_n)\right|$, here $\mathcal{L}_0(\boldsymbol{T_n})$ denotes the root node of tree $\boldsymbol{T_n}$. Applying the same procedure as above, we obtain

$$\mathbb{E}\left(\left|\mathcal{L}_1(\boldsymbol{T}_{n+1})\right|\right) - \mathbb{E}\left(\left|\mathcal{L}_1(\boldsymbol{T}_n)\right|\right) = \frac{\mathbb{E}\left(\left|\mathcal{L}_1(\boldsymbol{T}_n)\right|\right)}{2(n-1)}.$$

The lemma follows as desired. ■

Lemma 34 (Rate for $\mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right)$)**.** *Let $\boldsymbol{T}_n \sim \mathrm{APA}(0,n)$ and let $\mathcal{L}_s(\boldsymbol{T}_n)$ denote the nodes at layer $s \in \mathbb{N}$. Then, for any $s \in \mathbb{N}$,*

$$\mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right) \asymp n^{1/2}(\log n)^{s-1}.$$

Proof. We use proof by induction on s. The first step is to show that $\mathbb{E}\left(\left|\mathcal{L}_1(\boldsymbol{T}_n)\right|\right) \asymp n^{1/2}$. Denote $m_n := \mathbb{E}\left(\left|\mathcal{L}_1(\boldsymbol{T}_n)\right|\right)$, by Lemma 33, for $n \geq 2$

$$m_{n+1} - m_n = \mathbb{E}\left(\left|\mathcal{L}_1(\boldsymbol{T}_{n+1})\right|\right) - \mathbb{E}\left(\left|\mathcal{L}_1(\boldsymbol{T}_n)\right|\right) = \frac{m_n}{2(n-1)}.$$

indicates that

$$m_{n+1} = \frac{2n-1}{2n-2} m_n. \tag{44}$$

Putting it together with the boundary condition $m_2 = 1$, we have

$$\begin{aligned} m_{n+1} &= \frac{(2n-1)(2n-3)\cdots 3}{(2n-2)(2n-4)\cdots 2} \times m_2 = \frac{(2n-1)!}{\left((2n-2)(2n-4)\cdots 2\right)^2} = \frac{(2n-1)!}{2^{2n-2}\left((n-1)!\right)^2} \\ &\asymp \frac{\sqrt{2\pi}\sqrt{2n-1}(2n-1)^{2n-1}/e^{2n-1}}{2^{2n-2}\left(\sqrt{2\pi}\sqrt{n-1}(n-1)^{n-1}\right)^2/e^{2n-2}} \asymp \sqrt{n}. \end{aligned}$$

The asymptotic equivalent comes from the Stirling approximation.

Thus we have shown that $m_n := \mathbb{E}\left(\left|\mathcal{L}_1(\boldsymbol{T}_n)\right|\right) \asymp \sqrt{n}$, which in particular implies that for $s = 1$ we have $\mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right)/m_n \asymp (\log n)^{s-1}$. For $s > 1$, assume as the induction hypothesis that $\mathbb{E}(|\mathcal{L}_{s-1}(\boldsymbol{T}_n)|)/m_n \asymp (\log n)^{s-2}$ for all sufficiently large n; we then proceed to establish the corresponding asymptotic equivalence for s.

By Lemma 33 we have the following equation for layer s

$$\mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_{n+1})\right|\right) = \mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right) + \frac{\mathbb{E}\left(\left|\mathcal{L}_{s-1}(\boldsymbol{T}_n)\right|\right) + \mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right)}{2(n-1)} = \frac{2n-1}{2n-2}\mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right) + \frac{\mathbb{E}\left(\left|\mathcal{L}_{s-1}(\boldsymbol{T}_n)\right|\right)}{2n-2}.$$

Divide both side with m_{n+1} with the result (44), we have

$$\begin{aligned} \frac{\mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_{n+1})\right|\right)}{m_{n+1}} &= \frac{\mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right)}{m_n} + \frac{\mathbb{E}\left(\left|\mathcal{L}_{s-1}(\boldsymbol{T}_n)\right|\right)}{(2n-2)m_{n+1}} = \frac{\mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right)}{m_n} + \frac{\mathbb{E}\left(\left|\mathcal{L}_{s-1}(\boldsymbol{T}_n)\right|\right)}{(2n-1)m_n} \\ &\asymp \frac{\mathbb{E}\left(\left|\mathcal{L}_s(\boldsymbol{T}_n)\right|\right)}{m_n} + \frac{(\log n)^{s-2}}{2n-1}. \end{aligned}$$

Applying the above inequality repeatedly starting from $n = 2$ and noting that $\frac{\mathbb{E}(|\mathcal{L}_s(\boldsymbol{T}_2)|)}{m_2} = 0$ since $s > 1$, we obtain

$$\frac{\mathbb{E}\left(\left|\mathcal{L}_s\left(\boldsymbol{T}_n\right)\right|\right)}{m_n} \asymp \sum_{i=2}^{n-1} \frac{(\log i)^{s-2}}{2i-1} \asymp (\log n)^{s-1},$$

where the last inequality follows from Lemma 43. Therefore, $\mathbb{E}\left(\left|\mathcal{L}_s\left(\boldsymbol{T}_n\right)\right|\right)/m_n \asymp (\log n)^{s-1}$. We have shown that the inductive hypothesis holds and the proof is thus complete.

■

Lemma 35. *Let $\boldsymbol{G}_n \sim \mathrm{PF}(0, \theta, \ell, \pi)$, and suppose Assumptions 11 and 12 hold. Then for any fixed $C > 0$ and $s \in \mathbb{N}$, the following result holds:*

$$\lim_{n\to\infty} \mathbb{P}\left\{\left|E_{\boldsymbol{R}_n}(V^{C,0}(\boldsymbol{G}_n), \mathcal{L}_s(\boldsymbol{F}_n))\right| = 0\right\} = 1, \tag{45}$$

where $\mathcal{L}_s\left(\boldsymbol{F}_n\right)$ denotes all the nodes in layer s of $\boldsymbol{T}^i$. Furthermore, letting

$$E^2_{\boldsymbol{R}_n}\left(V^{C,0}_{\boldsymbol{G}_n}(\boldsymbol{T}^j), \mathcal{L}_s(\boldsymbol{T}^i)\right) = \left\{(v_1, v_2, v_3) \,\middle|\, (v_1, v_2) \in E_{\boldsymbol{R}_n}\left(V^{C,0}_{\boldsymbol{G}_n}(\boldsymbol{T}^j), V\left(\boldsymbol{G}_n\right)\right), \right.$$
$$\left. (v_2, v_3) \in E_{\boldsymbol{R}_n}\left(V\left(\boldsymbol{G}_n\right), \mathcal{L}_s(\boldsymbol{T}^i)\right), \right\}, \tag{46}$$

we have

$$\frac{\left|\bigcup_{\substack{i,j\in[K]\\ i\neq j}} E^2_{\boldsymbol{R}_n}\left(V^{C,0}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^j\right), \mathcal{L}_s\left(\boldsymbol{T}^i\right)\right)\right|}{\left|\mathcal{L}_s\left(\boldsymbol{F}_n\right)\right|} = O_p\left(n^{-2\delta}\right). \tag{47}$$

And, letting

$$V_1^C\left(\boldsymbol{G}_n\right) := \left\{v \in V\left(\boldsymbol{G}_n\right) \,\middle|\, \mathrm{dist}_{\boldsymbol{F}_n}\left(v, V^{C,0}\left(\boldsymbol{G}_n\right)\right) \leq 1\right\}, \tag{48}$$

we have

$$\frac{\left|E_{\boldsymbol{R}_n}\left(V_1^C\left(\boldsymbol{G}_n\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right|}{\left|\mathcal{L}_s\left(\boldsymbol{F}_n\right)\right|} = O_p\left(n^{-\delta}\right). \tag{49}$$

Proof. We first show (45). We fix $\varepsilon > 0$. Since $V^{C,0}\left(\boldsymbol{G}_n\right) = \cup_{i=1}^K V^{C,0}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^i\right)$, it follows that

$$\mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(V^{C,0}\left(\boldsymbol{G}_n\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| = 0\right) = \mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^K V^{C,0}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^i\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| = 0\right). \tag{50}$$

By Lemma 34 and Markov's inequality, there exists a universal constant $\tilde{C} > 0$ such that $\left|\mathcal{L}_s\left(\boldsymbol{F}_n\right)\right| \leq \tilde{C}\varepsilon^{-1}n^{1/2}(\log n)^{s-1}$ holds with probability higher than $1 - \varepsilon$. Therefore, we denote $\Omega_0 = \left\{\left|\mathcal{L}_s\left(\boldsymbol{F}_n\right)\right| \leq \tilde{C}\varepsilon^{-1}n^{1/2}(\log n)^{s-1}\right\}$, $\Omega_i = \left\{V^{C,0}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^i\right) \subset \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right\}$ for

$1 \leq i \leq K$, and $\Omega := \cap_{i=0}^{K} \Omega_i$, we now aim to bound the right-hand side of (50) on the event Ω:

$$\begin{aligned}
& \mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^{K} V_{\boldsymbol{G}_n}^{C,0}\left(\boldsymbol{T}^i\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| = 0\right) \\
= & \mathbb{P}\left(\left\{\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^{K} V_{\boldsymbol{G}_n}^{C,0}\left(\boldsymbol{T}^i\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| = 0\right\} \cap \Omega\right) + \mathbb{P}\left(\left\{\left|E_{\boldsymbol{R}_n}\left(V^{C,0}\left(\boldsymbol{G}_n\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| = 0\right\} \cap \Omega^c\right) \\
\geq & \mathbb{P}\left(\left\{\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^{K} \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| = 0\right\} \cap \Omega\right) \\
\geq & \mathbb{P}\left(\left\{\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^{K} \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| = 0\right\} \cap \Omega_0\right) - \mathbb{P}\left(\Omega^c \setminus \Omega_0^c\right). \\
\geq & \mathbb{P}\left(\left\{\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^{K} \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| = 0\right\} \cap \Omega_0\right) - \mathbb{P}\left(\Omega^c\right). \qquad (51)
\end{aligned}$$

We begin by bounding the first term

$$\mathbb{P}\left(\left\{\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^{K} \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| = 0\right\} \cap \Omega_0\right)$$

in (51). Note that the set $\cup_{i=1}^{K} \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right)$ and $\mathcal{L}_s\left(\boldsymbol{F}_n\right)$ is determined entirely by $\boldsymbol{F}_n$, which is independent of $\boldsymbol{R}_n$ by Definition 2. Therefore, the number of random edges induced by this set in $\boldsymbol{R}_n$ is stochastic dominant by $\mathrm{Bin}\left(L\left(\varepsilon, C\right) K \left|\mathcal{L}_s\left(\boldsymbol{F}_n\right)\right|, \theta\right)$ (conditioning on $\boldsymbol{F}_n$), since each pair of nodes in the selected set contributes an edge independently with probability θ. Therefore on the event Ω_0

$$\begin{aligned}
& \mathbb{P}\left(\left\{\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^{K} \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| = 0\right\} \cap \Omega_0\right) \\
= & \mathbb{P}(\Omega_0) - \mathbb{P}\left(\left\{\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^{K} \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| \geq 1\right\} \cap \Omega_0\right) \\
\geq & \mathbb{P}(\Omega_0) - \mathbb{E}\left(\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^{K} \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| \,\middle|\, \Omega_0\right) \\
\geq & \mathbb{P}(\Omega_0) - \tilde{C}\varepsilon^{-1} L\left(\varepsilon, C\right) K n^{1/2} \left(\log n\right)^{s-1} \theta \\
\geq & 1 - \varepsilon - O\left(\left(\log n\right)^{s-1} n^{-\delta}\right), \qquad (52)
\end{aligned}$$

where the bounds follow from the definition of Ω_0, Lemma 34, and Assumption 12.

Then we bound the second term $\mathbb{P}\left(\Omega^c\right)$. By Lemma 28 and Lemma 34, we have

$$\mathbb{P}\left(\Omega^c\right) = \mathbb{P}\left(\cup_{i=0}^{K} \Omega_i^c\right) \leq \sum_{i=0}^{K} \mathbb{P}\left(\Omega_i^c\right)$$

$$
\begin{aligned}
&= \sum_{i=1}^{K} \mathbb{P}\left(\left\{V_{\boldsymbol{G}_n}^{C,0}\left(\boldsymbol{T}^i\right) \subset \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right\}^c\right) + \mathbb{P}\left(\left|\mathcal{L}_s\left(\boldsymbol{F}_n\right)\right| > \tilde{C}\varepsilon^{-1} n^{1/2} (\log n)^{s-1}\right) \\
&\leq (K+1)\,\varepsilon + o(1). \qquad (53)
\end{aligned}
$$

Combining equations (50)–(53), we have shown that

$$
\mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(V^{C,0}\left(\boldsymbol{G}_n\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| = 0\right) \geq 1 - (K+2)\,\varepsilon - O\left((\log n)^{s-1} n^{-\delta}\right) - o\left(1\right).
$$

Since $\varepsilon > 0$ can be arbitrary small, we have

$$
\lim_{n\to\infty} \mathbb{P}\left(\left|E_{\boldsymbol{R}_n}\left(V^{C,0}\left(\boldsymbol{G}_n\right), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right| = 0\right) = 1.
$$

We now turn to the proof of (47). From the expression of $E^2_{\boldsymbol{R}_n}\left(V_{\boldsymbol{G}_n}^{C,0}\left(\boldsymbol{T}^j\right), \mathcal{L}_s\left(\boldsymbol{T}^i\right)\right)$ given in (46), and using the fact that the two sets $V_{\boldsymbol{G}_n}^{C,0}\left(\boldsymbol{T}^j\right)$ and $\mathcal{L}_s\left(\boldsymbol{T}^i\right)$ are disjoint for $i \neq j$, we obtain

$$
\begin{aligned}
\left|E^2_{\boldsymbol{R}_n}\left(V_{\boldsymbol{G}_n}^{C,0}\left(\boldsymbol{T}^j\right), \mathcal{L}_s\left(\boldsymbol{T}^i\right)\right)\right| &= \sum_{v\in V(\boldsymbol{G}_n)} \left|E_{\boldsymbol{R}_n}\left(V_{\boldsymbol{G}_n}^{C,0}\left(\boldsymbol{T}^j\right), v\right)\right| \cdot \left|E_{\boldsymbol{R}_n}\left(v, \mathcal{L}_s\left(\boldsymbol{T}^i\right)\right)\right| \\
&= \sum_{v\in V(\boldsymbol{G}_n)} \sum_{u_1 \in V_{\boldsymbol{G}_n}^{C,0}\left(\boldsymbol{T}^j\right)} \sum_{u_2 \in \mathcal{L}_s\left(\boldsymbol{T}^i\right)} \left|E_{\boldsymbol{R}_n}\left(u_1, v\right)\right| \cdot \left|E_{\boldsymbol{R}_n}\left(v, u_2\right)\right|.
\end{aligned}
$$

For any three different vertices v, u_1, u_2, the product $\left|E_{\boldsymbol{R}_n}\left(v, u_1\right)\right| \cdot \left|E_{\boldsymbol{R}_n}\left(v, u_2\right)\right| \sim \mathrm{Bin}\left(1, \theta^2\right)$.

Next, consider the size of $V_{\boldsymbol{G}_n}^{C,0}\left(\boldsymbol{T}^j\right)$ on the event Ω. By definition of Ω, it holds that $V_{\boldsymbol{G}_n}^{C,0}\left(\boldsymbol{T}^j\right) \subset \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^j\right)$. Hence,

$$
\mathbb{1}_\Omega \left|E^2_{\boldsymbol{R}_n}\big(V_{\boldsymbol{G}_n}^{C,0}\big(\boldsymbol{T}^j\big), \mathcal{L}_s\big(\boldsymbol{T}^i\big)\big)\right| \leq \left|E^2_{\boldsymbol{R}_n}\big(\pi_{1:L(\varepsilon,C)}\big(\boldsymbol{T}^j\big), \mathcal{L}_s\big(\boldsymbol{T}^i\big)\big)\right|.
$$

Conditioning only on $\boldsymbol{F}_n$, the sets $\pi_{1:L(\varepsilon,C)}(\boldsymbol{T}^j)$ and $\mathcal{L}_s(\boldsymbol{T}^i)$ are deterministic. Therefore, using the preceding expression for $\left|E^2_{\boldsymbol{R}_n}(\cdot,\cdot)\right|$, we obtain

$$
\begin{aligned}
&\mathbb{E}_{\boldsymbol{R}_n}\left(\frac{\mathbb{1}_\Omega \left|E^2_{\boldsymbol{R}_n}\big(V_{\boldsymbol{G}_n}^{C,0}\big(\boldsymbol{T}^j\big), \mathcal{L}_s\big(\boldsymbol{T}^i\big)\big)\right|}{\left|\mathcal{L}_s\big(\boldsymbol{F}_n\big)\right|} \,\Big|\, \boldsymbol{F}_n\right) \\
\leq\ &\mathbb{E}_{\boldsymbol{R}_n}\left(\frac{\left|E^2_{\boldsymbol{R}_n}\big(\pi_{1:L(\varepsilon,C)}\big(\boldsymbol{T}^j\big), \mathcal{L}_s\big(\boldsymbol{T}^i\big)\big)\right|}{\left|\mathcal{L}_s\big(\boldsymbol{F}_n\big)\right|} \,\Big|\, \boldsymbol{F}_n\right) \leq nL(\varepsilon,C)\theta^2 \frac{\left|\mathcal{L}_s\big(\boldsymbol{T}^i\big)\right|}{\left|\mathcal{L}_s\big(\boldsymbol{F}_n\big)\right|}.
\end{aligned}
$$

Summing the preceding bounds over all pairs $i, j \in [K]$ with $i \neq j$ gives

$$
\mathbb{E}_{\boldsymbol{R}_n}\left(\frac{\mathbb{1}_\Omega \left|\bigcup_{\substack{i,j\in[K]\\ i\neq j}} E^2_{\boldsymbol{R}_n}\big(V_{\boldsymbol{G}_n}^{C,0}\big(\boldsymbol{T}^j\big), \mathcal{L}_s\big(\boldsymbol{T}^i\big)\big)\right|}{\left|\mathcal{L}_s\big(\boldsymbol{F}_n\big)\right|} \,\Big|\, \boldsymbol{F}_n\right) \leq nKL(\varepsilon,C)\theta^2.
$$

Applying Markov's inequality conditional on $\boldsymbol{F}_n$, we obtain that, with probability at least $1-\varepsilon$,

$$\mathbb{1}_\Omega \frac{\left|\bigcup_{\substack{i,j\in[K]\\ i\neq j}} E^2_{\boldsymbol{R}_n}\left(V^{C,0}_{\boldsymbol{G}_n}\left(\boldsymbol{T}^j\right), \mathcal{L}_s\left(\boldsymbol{T}^i\right)\right)\right|}{\left|\mathcal{L}_s\left(\boldsymbol{F}_n\right)\right|} \leq \frac{nKL(\varepsilon, C)\theta^2}{\varepsilon} \leq C^0 n^{-2\delta},$$

where the last inequality follows from Assumption 12. Finally,

$$\mathbb{P}\left(\frac{\left|\bigcup_{\substack{i,j\in[K]\\ i\neq j}} E^2_{\boldsymbol{R}_n}\left(V^{C,0}_{\boldsymbol{G}_n}(\boldsymbol{T}^j), \mathcal{L}_s(\boldsymbol{T}^i)\right)\right|}{\left|\mathcal{L}_s(\boldsymbol{F}_n)\right|} > C^0 n^{-2\delta}\right) \leq \mathbb{P}\left(\Omega^c\right) + \varepsilon \leq (K+2)\varepsilon + o(1).$$

This completes the proof of (47).

Then we turn to the proof of (49). To this end, we introduce enlarged sets $\tilde{V}_i$ such that the $V_1^{C,0}(\boldsymbol{G}_n)$ in (48) is the subset of $\cup_{i=1}^K \tilde{V}_i$ on the event Ω:

$$\tilde{V}_i := \left\{ v \in V(\boldsymbol{G}_n) \,\middle|\, \mathrm{dist}_{\boldsymbol{T}^i}\left(v, \pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right) \leq 1 \right\}.$$

Then we define the "good event" $\tilde{\Omega}(t_1, t_2) := \left(\cap_{i=1}^K \bar{\Omega}_i(t_1)\right) \cap \left(\cap_{i=1}^K \breve{\Omega}_i(t_2)\right) \cap \Omega$, where

$$\bar{\Omega}_i(t_1) := \left\{ \deg_{\boldsymbol{T}^i}\left(\pi_{1:L(\varepsilon,C)}\left(\boldsymbol{T}^i\right)\right) \leq t_1\sqrt{n} \right\}$$

and

$$\breve{\Omega}_i(t_2) := \left\{ \frac{\left|E_{\boldsymbol{R}_n}\left(\tilde{V}_i, \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right|}{\left|\mathcal{L}_s\left(\boldsymbol{F}_n\right)\right|} \leq t_2 n^{-\delta} \right\}.$$

for any $t_1, t_2 > 0$.

We can observed that for fixed t_1, t_2, on the event $\tilde{\Omega}(t_1, t_2)$

$$\frac{\left|E_{\boldsymbol{R}_n}\left(V_1^C(\boldsymbol{G}_n), \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right|}{\left|\mathcal{L}_s\left(\boldsymbol{F}_n\right)\right|} \leq \frac{\left|E_{\boldsymbol{R}_n}\left(\cup_{i=1}^K \tilde{V}_i, \mathcal{L}_s\left(\boldsymbol{F}_n\right)\right)\right|}{\left|\mathcal{L}_s\left(\boldsymbol{F}_n\right)\right|} \leq K t_2 n^{-\delta}.$$

Then we claim that there exist $t_1, t_2 > 0$ depending only on ε, C, such that $\mathbb{P}\left(\tilde{\Omega}^c\right)$ is bounded by $(3K+1)\varepsilon + o(1)$ and hence (49) holds.

Since a bound for $\mathbb{P}(\Omega^c)$ has already been established in the proof of (45), we try to control $\mathbb{P}\left(\bar{\Omega}_i^c(t_1)\right)$ and $\mathbb{P}\left(\breve{\Omega}_i^c(t_2) \cap \bar{\Omega}_i(t_1)\right)$ separately in the following proof.

By Lemma 38, there exists some constant $t_1 := t_1(\varepsilon, C)$ such that $\mathbb{P}\left(\bar{\Omega}_i^c(t_1)\right) \leq \varepsilon$ for every $i \in [K]$. The next step is to bound $\mathbb{P}\left(\breve{\Omega}_i^c(t_2) \cap \bar{\Omega}_i(t_1)\right)$, We do so by showing that, on event $\bar{\Omega}_i(t_1)$, $\breve{\Omega}_i(t_2)$ stands with high probability for a suitable choice of t_2.

Note that the sets $\left\{\tilde{V}_i, i \in [K]\right\}$ and $\mathcal{L}_s(\boldsymbol{F}_n)$ is determined entirely by $\boldsymbol{F}_n$, which is independent of $\boldsymbol{R}_n$ by Definition 2. Therefore, the number of random edges induced by this set in $\boldsymbol{R}_n$, conditioning on $\boldsymbol{F}_n$ is stochastic dominated by $\text{Bin}\left(\left|\tilde{V}_i\right| \left|\mathcal{L}_s(\boldsymbol{F}_n)\right|, \theta\right)$. Combining the fact that, on the event $\bar{\Omega}_i$, $\left|\tilde{V}_i\right| \leq L(\varepsilon, C)(1 + t_1\sqrt{n}) \leq (1 + t_1)L(\varepsilon, C)\sqrt{n}$,

$$\mathbb{E}_{\boldsymbol{R}_n}\left(\frac{\left|E_{\boldsymbol{R}_n}\left(\tilde{V}_i, \mathcal{L}_s(\boldsymbol{F}_n)\right)\right|}{\left|\mathcal{L}_s(\boldsymbol{F}_n)\right|}\right) \leq \left|\tilde{V}_i\right|\theta \leq (1 + t_1)L(\varepsilon, C)\, n^{1/2}\theta.$$

By Markov inequality and Assumption 12, for any $\varepsilon > 0$, there exist and a universal constant $C^1 > 0$ such that the following holds with probability at least $1 - \varepsilon$:

$$\frac{\left|E_{\boldsymbol{R}_n}\left(\tilde{V}_i, \mathcal{L}_s(\boldsymbol{F}_n)\right)\right|}{\left|\mathcal{L}_s(\boldsymbol{F}_n)\right|} \leq \varepsilon^{-1}(1 + t_1)L(\varepsilon, C)\, n^{1/2}\theta \leq C^1\varepsilon^{-1}(1 + t_1)L(\varepsilon, C)\, n^{-\delta}.$$

Therefore, takes $t_2 := C^1\varepsilon^{-1}(1 + t_1)L(\varepsilon, C)$ the bound $\mathbb{P}\left(\breve{\Omega}_i^c(t_2) \cap \bar{\Omega}_i(t_1)\right) \leq \varepsilon$ is satisfied.

Set $\bar{\Omega}_i := \bar{\Omega}_i(t_1)$, $\breve{\Omega}_i := \breve{\Omega}_i(t_2)$, $\tilde{\Omega} := \tilde{\Omega}(t_1, t_2)$. Combining the preceding bounds with an union bound, we

$$\begin{aligned}
\mathbb{P}\left(\tilde{\Omega}^c\right) &= \mathbb{P}\left(\left(\left(\cap_{i=1}^K \bar{\Omega}_i\right) \cap \left(\cap_{i=1}^K \breve{\Omega}_i\right) \cap \Omega\right)^c\right) = \mathbb{P}\left(\left(\left(\cap_{i=1}^K \bar{\Omega}_i\right) \cap \left(\cap_{i=1}^K \left(\breve{\Omega}_i \cup \bar{\Omega}_i^c\right)\right) \cap \Omega\right)^c\right) \\
&\leq \sum_{i=1}^K \mathbb{P}\left(\bar{\Omega}_i^c\right) + \sum_{i=1}^K \mathbb{P}\left(\breve{\Omega}_i^c \cap \bar{\Omega}_i\right) + \mathbb{P}\left(\Omega^c\right) \leq K\varepsilon + K\varepsilon + (K + 1)\varepsilon + o(1) \leq (3K + 1)\varepsilon + o(1).
\end{aligned}$$

This completes the proof of (49).

■

F.7 Technical Lemmas

Lemma 36 (Theorem 2 in Rudas et al. (2007)). *For $\boldsymbol{T}_n \sim \text{APA}(\alpha, n)$ with $\alpha > -1$, as $n \to \infty$,*

$$\frac{\left|\left\{v : \deg_{\boldsymbol{T}_n}(v) = 1\right\}\right|}{n} \xrightarrow{P} p(1) > 0,$$

that is, the proportion of degree-1 vertices converges in probability to a positive constant $p(1)$ that depends on α.

Lemma 37 (Proposition 5.5 in Sénizergues (2021)). *Let $q > 2 + \alpha$. For $\boldsymbol{T}_n \sim \text{APA}(\alpha, n)$, for $\alpha > -1$, when n tends to infinity,*

$$n^{-\frac{1}{2+\alpha}}\left(\deg_{\boldsymbol{T}_n}\left(\pi_1(\boldsymbol{T}_n)\right), \deg_{\boldsymbol{T}_n}\left(\pi_2(\boldsymbol{T}_n)\right), \ldots, \deg_{\boldsymbol{T}_n}\left(\pi_n(\boldsymbol{T}_n)\right), 0, 0, \ldots\right) \xrightarrow{a.s.} \left(Y_{1,\alpha}, Y_{2,\alpha}, Y_{3,\alpha}, \ldots\right),$$

almost surely with respect to the ℓ_q metric where $(Y_{1,\alpha}, Y_{2,\alpha}, \ldots)$ is a random sequence satisfying $\sum_{j=1}^\infty \mathbb{E}Y_{j,\alpha}^q < \infty$ and each random variable $Y_{j,\alpha}$ has a density with respect to the Lebesgue measure.

Lemma 38. *For $\boldsymbol{T}_n \sim \mathrm{APA}(\alpha, n)$, for $\alpha > -1$, for any fixed $L \in \mathbb{N}$,*

$$\deg_{\boldsymbol{T}_n}\left(\pi_{1:L}\left(\boldsymbol{T}_n\right)\right) = O_p\left(n^{\frac{1}{2+\alpha}}\right).$$

Proof. Let $(Y_{1,\alpha}, Y_{2,\alpha}, \ldots)$ is a random sequence from Lemma 37, we first show that for any $j \in [L], \varepsilon > 0$, there exist a constant $C := C(\varepsilon, L)$ such that $\mathbb{P}\left(Y_{j,\alpha} < C\right) > 1 - \frac{\varepsilon}{L}$. By Lemma 37, each $Y_{j,\alpha}$ admits a density $q_j(\cdot)$ supported on $[0,\infty)$ with respect to the Lebesgue measure. Consequently, for any $\varepsilon > 0$, we can select $C_j := C_j(\varepsilon)$ such that $\mathbb{P}\left(Y_{j,\alpha} \geq C_j\right) \leq \frac{\varepsilon}{L}$, for any $j \in [L]$. We may then take $C(\varepsilon, L) = \max_{1\leq j\leq L} C_j(\varepsilon)$.

We now consider the distribution of $\deg_{\boldsymbol{T}_n}\left(\pi_j\left(\boldsymbol{T}_n\right)\right)$. By Lemma 37 and Portmanteau lemma, we have

$$\limsup_{n\to\infty} \mathbb{P}\left(n^{-\frac{1}{2+\alpha}} \deg_{\boldsymbol{T}_n}\left(\pi_j\left(\boldsymbol{T}_n\right)\right) \geq C\right) \leq \mathbb{P}\left(Y_{j,\alpha} \geq C\right) \leq \mathbb{P}\left(Y_{j,\alpha} \geq C_j\right) \leq \frac{\varepsilon}{L}. \tag{54}$$

Then we proceed to show that $\deg_{\boldsymbol{T}_n}\left(\pi_{1:L}\left(\boldsymbol{T}_n\right)\right) = O_p\left(n^{\frac{1}{2+\alpha}}\right)$:

$$\begin{aligned}
\liminf_{n\to\infty} \mathbb{P}\left(\deg_{\boldsymbol{T}_n}\left(\pi_{1:L}\left(\boldsymbol{T}_n\right)\right) < Cn^{\frac{1}{2+\alpha}}\right) &= \liminf_{n\to\infty} \mathbb{P}\left(\max_{1\leq j\leq L} \deg_{\boldsymbol{T}_n}\left(\pi_j\left(\boldsymbol{T}_n\right)\right) < Cn^{\frac{1}{2+\alpha}}\right) \\
&= 1 - \limsup_{n\to\infty} \mathbb{P}\left(\max_{1\leq j\leq L} \deg_{\boldsymbol{T}_n}\left(\pi_j\left(\boldsymbol{T}_n\right)\right) \geq Cn^{\frac{1}{2+\alpha}}\right) \\
&= 1 - \limsup_{n\to\infty} \mathbb{P}\left(\cup_{j=1}^{L}\left\{\deg_{\boldsymbol{T}_n}\left(\pi_j\left(\boldsymbol{T}_n\right)\right) \geq Cn^{\frac{1}{2+\alpha}}\right\}\right) \\
&\geq 1 - \sum_{j=1}^{L} \limsup_{n\to\infty} \mathbb{P}\left(\deg_{\boldsymbol{T}_n}\left(\pi_j\left(\boldsymbol{T}_n\right)\right) \geq Cn^{\frac{1}{2+\alpha}}\right) \\
&\geq 1 - \sum_{j=1}^{L} \frac{\varepsilon}{L} = 1 - \varepsilon,
\end{aligned}$$

where the first equality follows directly from the definition of $\deg_{\boldsymbol{T}_n}\left(\pi_{1:L}(\boldsymbol{T}_n)\right)$, while the final inequality is a consequence of (54). Therefore, for all sufficiently large n

$$\mathbb{P}\left(\deg_{\boldsymbol{T}_n}\left(\pi_{1:L}\left(\boldsymbol{T}_n\right)\right) < Cn^{\frac{1}{2+\alpha}}\right) \geq 1 - \varepsilon - o(1) \geq 1 - 2\varepsilon,$$

and the Lemma follows as desired. ∎

Lemma 39. *Let $q > (2+\alpha) \vee \frac{1}{\delta}$. For $\boldsymbol{G}_n \sim \mathrm{PF}(\alpha, \theta, \ell, \pi)$ with decomposition in (2), under Assumption 11, 12. For any $i \in [K]$, as $n \to \infty$,*

$$n_i^{-\frac{1}{2+\alpha}}\left(\deg_{\boldsymbol{G}_n}\left(\pi_1\left(\boldsymbol{T}^i\right)\right) - \deg_{\boldsymbol{T}^i}\left(\pi_1\left(\boldsymbol{T}^i\right)\right), \ldots, \deg_{\boldsymbol{G}_n}\left(\pi_{n_i}\left(\boldsymbol{T}^i\right)\right) - \deg_{\boldsymbol{T}^i}\left(\pi_{n_i}\left(\boldsymbol{T}^i\right)\right), 0, \ldots\right) \xrightarrow{d} (0, 0, \ldots),$$

in distribution with respect to the ℓ_q metric.

Proof. Indeed, we have

$$\mathbb{E}\left\|n_i^{-\frac{1}{2+\alpha}}\left(\deg_{\boldsymbol{G}_n}\left(\pi_1(\boldsymbol{T}^i)\right)-\deg_{\boldsymbol{T}^i}\left(\pi_1(\boldsymbol{T}^i)\right),\ldots,\deg_{\boldsymbol{G}_n}\left(\pi_{n_i}(\boldsymbol{T}^i)\right)-\deg_{\boldsymbol{T}^i}\left(\pi_{n_i}(\boldsymbol{T}^i)\right),0,\ldots\right)\right\|_q^q$$
$$\leq n_i^{-\frac{q}{2+\alpha}}\sum_{k=1}^{n_i}\mathbb{E}\deg_{\boldsymbol{R}_n}\left(\pi_1(\boldsymbol{T}^i)\right)^q \overset{(a)}{=} n_i^{1-\frac{q}{2+\alpha}}\mathbb{E}\left(\mathrm{Bin}(n-1,\theta)^q\right) \overset{(b)}{\leq} n_i^{1-\frac{q}{2+\alpha}}\left((2\theta n)^q+C_q\right)$$
$$\leq 2^q C_0^q n_i^{1-\frac{q}{2+\alpha}} n^{\frac{q}{2+\alpha}-q\delta}+C_q n_i^{1-\frac{q}{2+\alpha}} \leq 2^q C_0^q H^{\frac{q}{2+\alpha}-1}n^{1-q\delta}+C_q H^{\frac{q}{2+\alpha}-1}n^{1-\frac{q}{2+\alpha}}$$

where the equality (a) follows since $\deg_{\boldsymbol{R}_n}\left(\pi_j(\boldsymbol{T}^i)\right)$ is distributed as $\mathrm{Bin}(n-1,\theta)$ and where inequality (b) follows from Lemma 41.

Since $q>(2+\alpha)\vee 1/\delta$ by assumption, we have that, as $n\to\infty$,

$$\mathbb{E}\left\|n_i^{-\frac{1}{2+\alpha}}\left(\deg_{\boldsymbol{G}_n}\left(\pi_1(\boldsymbol{T}^i)\right)-\deg_{\boldsymbol{T}^i}\left(\pi_1(\boldsymbol{T}^i)\right),\ldots,\deg_{\boldsymbol{G}_n}\left(\pi_{n_i}(\boldsymbol{T}^i)\right)-\deg_{\boldsymbol{T}^i}\left(\pi_{n_i}(\boldsymbol{T}^i)\right),0,\ldots\right)\right\|_q^q\to 0$$

and the Lemma follows as desired. ∎

Lemma 40. *Let $q>(2+\alpha)\vee 1/\delta$. For $\boldsymbol{G}_n\sim\mathrm{PF}(\alpha,\theta,\ell,\pi)$ with decomposition in (2), under Assumption 11 and 12. For any $i\in[K]$, as $n\to\infty$,*

$$n_i^{-\frac{1}{2+\alpha}}\left(\deg_{\boldsymbol{G}_n}\left(\pi_1(\boldsymbol{T}^i)\right),\deg_{\boldsymbol{G}_n}\left(\pi_2(\boldsymbol{T}^i)\right),\ldots,\deg_{\boldsymbol{G}_n}\left(\pi_{n_i}(\boldsymbol{T}^i)\right),0,0,\ldots\right)\xrightarrow{d}\left(Y_{1,\alpha},Y_{2,\alpha},Y_{3,\alpha},\ldots\right),$$

in distribution with respect to the ℓ_q metric where $(Y_{1,\alpha},Y_{2,\alpha},\ldots)$ is a random sequence satisfying $\sum_{j=1}^{\infty}\mathbb{E}Y_{j,\alpha}^q<\infty$ and each random variable $Y_{j,\alpha}$ has a density with respect to the Lebesgue measure.

Proof. This lemma is a direct result of applying Slutsky Lemma with results with Lemma 37 and Lemma 39. ∎

Lemma 41 (Lemma S8 in Crane and Xu (2024))**.** *Let X be a random variable with $\mathrm{Bin}(n,\theta)$ distribution. For any $q\geq 1,\theta\in[0,1]$ and any $n\in\mathbb{N}$, we have that*

$$\mathbb{E}X^q\leq(2\theta n)^q+C_q,$$

where $C_q>0$ is a constant that depends only on q.

Proof. Write X as a random variable with the $\mathrm{Bin}(n,\theta)$ distribution. Then,

$$\mathbb{E}X^q=\int_0^{\infty}\mathbb{P}\left(X^q\geq t\right)dt\leq(2\theta n)^q+\int_{(2\theta n)^q}^{\infty}\mathbb{P}\left(X^q\geq t\right)dt \tag{55}$$

We note that $\mathrm{Var}\,X\leq\theta n$. By Bernstein's inequality, we have that for all $t\geq(2\theta n)^q$,

$$\mathbb{P}\left(X^q\geq t\right)=\mathbb{P}\left(X-\theta n\geq t^{1/q}-\theta n\right)\leq\exp\left(-\frac{1}{2}\frac{\left(t^{1/q}-\theta n\right)^2}{\left(t^{1/q}-\theta n\right)+\theta n}\right)\leq\exp\left(-\frac{1}{8}t^{1/q}\right)$$

Therefore, we may bound the second term of (55) as

$$\int_{(2\theta n)^q}^{\infty} \mathbb{P}\left(X^q \geq t\right) dt \leq \int_{(2\theta n)^q}^{\infty} e^{-\frac{t^{1/q}}{8}} dt \leq \int_0^{\infty} q s^{q-1} e^{-\frac{s}{8}} ds < \infty.$$

The lemma thus follows by writing $C_q := \int_0^{\infty} q s^{q-1} e^{-\frac{s}{8}} ds$. ∎

Lemma 42. *Let $\boldsymbol{g} = (V, E)$ be an arbitrary graph without self-loop and let $A, B \subset V(\boldsymbol{g})$. Then, we have that*

$$\sum_{v \in A} |E(v, B)| \leq 2|E(A, B)|.$$

Proof. Define

$$I := A \cap B, \quad A_1 = A \cap B^c = A \cap I^c, \quad B_1 = B \cap A^c = B \cap I^c,$$

and note that

$$E(A, B) = E(I, B) \sqcup E(A_1, B) = E(I, I) \sqcup E(I, B_1) \sqcup E(A_1, B).$$

Then, we have that

$$\begin{aligned}
\sum_{v \in A} |E(v, B)| &= \sum_{v \in I} |E(v, B)| + \sum_{v \in A_1} |E(v, B)| \\
&= \sum_{v \in I} |E(v, I)| + \sum_{v \in I} |E(v, B_1)| + \sum_{v \in A_1} |E(v, B)| \\
&= 2|E(I, I)| + |E(I, B_1)| + |E(A_1, B)| \leq 2|E(A, B)|.
\end{aligned}$$

The lemma thus follows as desired. ∎

Lemma 43. *For fixed integers $s \in \mathbb{N}$,*

$$\sum_{i=1}^{n-1} \frac{(\log i)^{s-1}}{2i - 1} \asymp (\log n)^s.$$

Proof. It suffices to consider the sum from $i = 2$ to $n - 1$, the possible $i = 1$ term is either zero or a fixed constant and is irrelevant for the asymptotic order. We first show that $\sum_{i=2}^{n-1} \frac{(\log i)^{s-1}}{2i-1} \leq \frac{(\log n)^s}{s}$, begin from the left hand side:

$$\begin{aligned}
\sum_{i=2}^{n-1} \frac{(\log i)^{s-1}}{2i - 1} &\leq \sum_{i=2}^{n-1} \left(\frac{1}{2i - 1} \int_i^{i+1} (\log t)^{s-1} \, dt \right) \leq \sum_{i=2}^{n-1} \int_i^{i+1} \frac{(\log t)^{s-1}}{t} dt = \int_2^n \frac{(\log t)^{s-1}}{t} dt \\
&= \int_{\log 2}^{\log n} y^{s-1} dy = \frac{(\log n)^s}{s} - \frac{(\log 2)^s}{s} \leq \frac{(\log n)^s}{s}.
\end{aligned}$$

In the first identity of the second line, we use a change of variables $y = \log t$.

Then we show that $\sum_{i=2}^{n-1} \frac{(\log i)^{s-1}}{2i-1} \geq \frac{(\log n)^s}{4s}$, begin from the left hand side:

$$\begin{aligned}\sum_{i=2}^{n-1} \frac{(\log i)^{s-1}}{2i-1} &\geq \sum_{i=2}^{n-1} \left(\frac{1}{2i-1} \int_{i-1}^{i} (\log t)^{s-1}\, dt \right) \geq \sum_{i=2}^{n-1} \int_{i-1}^{i} \frac{(\log t)^{s-1}}{3t} dt = \int_{1}^{n-1} \frac{(\log t)^{s-1}}{3t} dt \\ &= \int_{\log 1}^{\log(n-1)} \frac{1}{3} y^{s-1} dy = \frac{\big(\log(n-1)\big)^s}{3s} - \frac{(\log 1)^s}{3s} \geq \frac{(\log n)^s}{4s}.\end{aligned}$$

In the second identity of the second line, we use a change of variables $y = \log t$. The last inequality stands for all sufficiently large n. The Lemma thus follows as desired. ■

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